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INSTITUTE OF DISTANCE LEARNING

(QUANTITY SURVEYING & CONSTRUCTION ECONOMICS 1)
(CONSTRUCTION TECHNOLOGY & MANAGEMENT 1)

GE184: PRINCIPLES OF SURVEYING

[Credits: 3]

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Course Introduction

Course *GE185: Principles of Land Surveying* has been deigned for BSc. (Quantity Surveying & Construction Economics 1) and BSc. (Construction Technology & Management 1) programmes offered by Department Of Building Technology to provide you with theoretical principles and practical approaches in the tasks of land surveying.

Course objectives
After reading this course you should be able to:

1. *Define* and *describe* the various terms in Principles Of Land Surveying
2. *Solve* problems that involve land surveying calculations
3. *Identify* and *Explain* the errors of surveying
4. *Undertake* practical tasks of land surveying procedures: compass surveying, plane survey, geodetic surveying, chain surveying,

Course outline:

Unit 1	Session 1-1: Surveying
	Session 2-1: Errors in Surveying
	Session 3-1: Plan Scales
Unit 2	Session 1-2: Chain Surveying
	Session 2-2: Compass Surveying
Unit 3	Session 1-3: Levelling and Contouring
	Session 2-3: Theodolite Surveying
Unit 4	Session 1-4: Global Positioning Systems (GPS)
	Session 2-4: Areas and Volumes

Reading list/Textbooks/References

1) Bannister, A., Raymond, S., "Surveying" (London: Longman Group UK Ltd, 1992)
2) Irvine, W., "Surveying for Construction, 2nd Edition" (London: McGraw-Hill Book Company (UK) Limited, 1980)
3) Kavanagh, B. F., Bird, G. S. J. (1996), "Surveying: Principles and Applications"
4) Simpson, T.K. (2000) , "Principles of Surveying"
5) Uren, J. and Price, W. F., "Surveying for Engineers" 3rd Edition (London: Macmillan Press Ltd., 1994)
6) http://www.sonoma.edu/GIC/Geographica/MapInterp/Scale.html
7) http:// www.garmin.com/aboutGPS

Examinations
The following will contribute the respective percentages to your 100% Final mark for the course as follows.

1. Continuous assessments (Exercises, Assignments, Quizzes) = 30%
2. End of Semester Examinations = 70%

Study guide
GE183/4: Principles of Land Surveying involves is a 3-hour credit course. Practical work is 2 hours

Week	Unit/session	FFFS, Unit Assignments/Quizzes
1	1	
2	1	
3	2	
4	2	
5	3	
6	3	
7	4	
8	4	

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Unit 1

SURVEYING, ERRORS AND PLAN SCALES

Introduction

Surveying is an art of making measurements of the relative positions of natural and man made features on or near the surface of the earth and the representation of this information either graphically or numerically. Measurements taken may be linear, angular or a combination of the two in fixing a point n relation to a given line. Whatever the method used in fixing a point, however, errors will be inherent due to imperfection of equipment, measurement conditions and experience of the observer. Thus in surveying the true value of a measurement is never known; consequently true error is never determined. Standard observational procedures and mathematical models are used to minimise errors. Results of measurements are plotted as maps/plans.

Scale of a map/plan is the ratio between any distance on the map and the corresponding distance on the ground. Scales are necessary for the correct interpretation of maps or plans.

Learning objectives

This unit serves as basic introduction to land surveying. After studying this unit students should be able to:

- Differentiate between plane and geodetic surveying
- Discuss the classes of land surveying
- Describe and compare the different types of errors
- Distinguish between the different types of scales and convert from one scale to the other.
- Calculate plan distances and areas when given the scale and corresponding ground distances or areas and vice versa.

Unit content

Session 1-1: Surveying
1-1.1 What Is Surveying?
1-1.2 Purposes of Surveying
1-1.3 Types of Surveys
1-1.4 Classes of Surveys

Session 2-1: Errors in Surveying
2-1.1 The Problem
2-1.2 Sources of Errors
2-1.3 Mistakes (Gross Errors or Blunders)

2-1.4 Classification of Errors

Session 3-1: Plan Scales
3-1.1 Definition
3-1.2 Types of Map Scales
3-1.3 Conversion of Scales to Representative Fractions
3-1.4 Conversion of Areas by Representative Fractions

SESSION 1-1: SURVEYING

1-1.1 What is Surveying?

Surveying is an art of making measurements of the relative positions of natural and man made features on the earth's surface and the representation of this information either graphically or numerically. It is an art in which only the surveyor who possesses a thorough understanding of the surveying techniques will be able to determine the most efficient methods required to obtain the optimal results over a variety of surveying problems.

Surveying is one of the oldest and most important arts practiced by man, because from the earliest times it has been necessary to mark boundaries and divide the land. Much has changed since the earliest times, especially with the modern technological advancement in the fields of electronics and space travel, and while the basic principles remain almost the same, surveying, in a more general sense, can be regarded as that discipline which encompasses all methods for gathering and processing information about the physical earth and environment, and conventional methods are now supplemented by aerial and satellite surveying techniques.

1-1.2 Purposes of Surveying

Surveying is done to locate the positions of points on or near the surface of the earth. This involves measurements of distances and angles to determine horizontal positions of arbitrary points on the earth's surface, heights of arbitrary points above or below a reference datum and the configuration of the ground. It also involves setting- out distances, angles and grid lines to locate construction lines for buildings, bridges, highways and other engineering works and establishing the positions of boundary lines on the ground

1-1.3 Types of Surveys

With respect to the area of the earth's surface involved, surveying is divided principally into Plane and Geodetic Surveying.

1-1.3.1 Plane Surveying

It is that type of surveying in which relatively small areas of the earth (up to 250square km) are involved and the surface of the earth is considered to be a horizontal plane or flat surface. All height dimensions are referenced to the mean spherical surface of the earth (mean Sea Level).

1-1.3.2 Geodetic Surveying
It is that type of surveying in which large areas of the earth's surface (more than 250 square km) are involved and the surface of the earth is considered to be spherical for X and Y dimensions. All height dimensions are also referenced to the Mean Sea Level.

1-1.4 Classes of Surveys

The classes of land surveying are:

- **Topographical Surveys:** These are surveys conducted to determine the configuration of the ground as well as the location of the natural and man-made features of the earth including hills, valleys, railways etc.
- **Cadastral Surveys:** These are surveys conducted for legal purposes such as deed plans showing and defining legal property boundaries and the calculation of area(s) involved.
- **Hydrographic Surveys:** These are surveys conducted to determine the position of the survey vessel, depth of water and to investigate the nature of the sea bed.
- **Photogrammetric Surveying**: It is the science of making precise measurement and creating detailed maps from aerial images or photographs.
- **Mining Surveys:** These are surveys executed to establish location and boundaries of mining claims. It also involves the establishment of underground workings horizontally, vertically and lay out shaft connections.
- **Engineering Surveying:** Surveys executed to locate or lay out engineering or building works such as roads, railways, tunnels, dams etc.
- **Global Positioning System (GPS) Surveys:** Positioning in which the coordinates X Y and Z of survey stations are determined by the reception and analysis of NAVSTAR Satellite signals.



Self-Assessment

1) Distinguish between plane and geodetic surveying
2) List and discuss four main classes of land surveying

SESSION 2-1: ERRORS IN SURVEYING

2-1.1 The Problem

The basic task in land surveying is establishment of three-dimensional controls using linear and angular measurements and more recently through the use of satellites (GPS). Such measurements contain errors, since no measurement is free of errors. 'True value' of a measurement can therefore, never be found, even though such a value exist. The true value is determined statistically after repeated measurements. Since the true value of a measurement is not known it follows that exact error will also never be known.

2-1.2 Sources of Errors

There are basically three main sources of errors, namely natural, instrumental and personal.

- Natural Sources of error may be caused by adverse or variable weather conditions examples of which are; expansion or contraction of steel tapes due to temperature changes, refraction of light rays, variation in the speed of electromagnetic waves through the atmosphere, etc.
- Instrumental Errors are caused by imperfect construction and adjustment of equipment used in surveying
- Personal errors results from the inability of the individual to make exact observations due to limitations of human sight, touch and hearing.

2-1.3 Mistakes (Gross Errors or Blunders)

Mistakes also known as gross errors or blunders should not be classified as errors at all. They include result from fatigue or inexperience of the surveyor. Examples of mistakes include, omitting a whole tape length when measuring a distance, sighting the wrong target in the round of angles, reading 9 on a levelling staff as 6 or recording 67 instead of 76. Mistakes are the largest errors of the errors likely to arise and great care must be taken to eliminate them preferably by those who make them. All survey measurements are suspect until verified.

Mistakes can be detected and eliminated by repeated or check measurements. This also improves the precision.

2-1.4 Classification of errors

2-1.4.1 Systematic errors

- Systematic errors are attributable to known circumstances.
- The can be constant (having the same magnitude and sign) or variable throughout an operation
- They conform to mathematical and physical laws; thus appropriate corrections can be computed and applied as corrections to the measurements.
- Repeated observations will increase the effect of systematic errors.
- Systematic errors are the most difficult to deal with and therefore they require very careful consideration prior to, during and after the survey.
- Systematic errors are minimized by systematic observation and reduction techniques.

Examples:

-Refraction of light rays;
-Variation of the speed of light through the atmosphere;
-Expansion or contraction of steel tapes due to temperature variation.

2-1.4.2 Random Errors

Repeated observation is a typical feature of all measurement procedures in surveying. If a quantity is measured several times with the same instruments in the same way, it is highly unlikely that all the results will be identical, even after the effects of systematic errors have been removed. The true value of a quantity is never known, except in few isolated cases, for example, in measuring the angles of a triangle where the true value of the sum should be 180°. Generally, the mean of the repeated and 'corrected' measurements is accepted as the true value of the quantity.

Random Errors are associated with the skill and vigilance of the surveyor. They are introduced into each measurement mainly because no human can perform perfectly. Thus equipments and our observations are imperfect. They represent the residual error after all other errors have been eliminated. They are compensating and generally unavoidable. They

usually conform to the law of probability. Taking the mean of repeated observations minimizes their effect.

The characteristics of Random errors are as follows:

1. Small errors occur more frequently than large ones
2. Positive and negative errors of equal magnitude occur with equal frequency; that is they are equally probable.
3. Large errors are rare (infrequent) and are more likely to be mistakes or untreated systematic errors.

Certain terms, typically "*accurate"* and "*precise"* are often used indiscriminately, to describe the quality or reliability of the measurements or observations as if they were synonymous. It is important that you know the correct meanings of these terms so as to apply them correctly. Accuracy is the nearness of the measured value to the true value of a quantity. Since the true value is seldom known, accuracy is generally indeterminate in practice; and another concept used to assess the reliability of a measurement, or to decide which of two or more measurements is best, is precision. Precision, in this context, is an indication of the spread of the measured values of a quantity. Several values grouped closely together are said to constitute a more precise set of measurements than one with a broader spread of values.

Example

Two groups of students measured a line of 100m nominal length with a tape, and obtained the following results:-

Group A: 99.01, 98.99, 99.02 and 98.98m
Group B: 99.85, 99.90, 100.15 and 100.00m

From the above definitions, the results of Group A, being closely grouped together with a small spread, constitute a more precise measurement than those of Group B. However, the results of Group B are nearer to the nominal length of the line and should therefore be the more accurate of the two.



Self-Assessment

1. Discuss the following
   a. Gross errors
   b. Random Errors
   c. Systematic errors

2. How should each of he above errors be handled in land surveying?

SESSION 3-1: PLAN SCALES

3-1.1 Definition

'A map scale is the ratio between a distance on a map and the corresponding distance in the terrain.

It is also an expression of how much the area represented has been reduced on the map. Map scale is important for understanding maps both in paper and computer form, so it will pay you to understand the types and uses of scales.

Almost always, maps takes up less physical space than the area that they represent. This reduction in size is reflected by the scale of the map, and is usually represented as a ratio or a graphic scale. Scale is considered to be the single most critical mathematical feature of any map and, because of its importance, scale is often used as a primary means of categorising maps - from large-scale maps of 1: 10,000 to atlases of 1:1,000,000.

3-1.2 Types of Map Scales

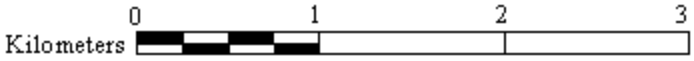
We can relate map and ground with three different types of scale (Source: http://www.sonoma.edu/GIC/Geographica/MapInterp/Scale.html)

1. **Verbal scale** expresses in words as the relationship between a map distance and a ground distance. Usually it is along the lines of:

   One inch represents 16 miles.

   Here it is implied that the one-inch is on the map, and that one-inch represents 16 miles on the ground. Verbal scales are commonly found on popular atlases and maps.

2. The second type of scale is a **graphic scale**, or **bar scale**. This shows directly on the map the corresponding ground distance. For example:



   Bar scales is probably the most common kind of scale found on maps, perhaps because their graphical nature makes them easily understood? Another great thing about bar scales is that *they remain correct if the map is reduced or enlarged* photographically. This is not true of the other two types of scales.

   The main drawback of bar scales is that they are usually short compared to the map itself, and hence measuring longer distances is difficult.

3. The third type of scale is a **representative fraction**, or ratio scale. Compared to the first two, it is the most abstract, but also the most versatile. A representative fraction, or RF, shows the relationship between one of any unit on the map and one of the same unit on the ground. RFs may be shown as an actual fraction, for example 1/24,000, but are usually written with a colon, as in 1:24,000. In this example, one unit of any length (one mm, one cm, one inch, one foot, etc.) on the map represents 24,000 of those same units on the ground (24,000 mm, 24,000 cm, 24,000", 24,000', etc.). The RF is versatile because you are not tied to any specific units. You may work in any unit you choose, either metric, English, or other.

   - The RF is a called a fraction because it is a fraction showing how much the real world is reduced to fit on the map. For example, a 1:100,000-scale map is 1/100,000th as large as the ground area shown on the map.

- The abstract nature of RF scales confuses many people. They ask, "one what is equal to 24,000 of what?" The key is that you can put in any unit you want instead of "what" in the question. That's what makes it the most flexible type of scale.
- RF scales are common among geographers as shorthand for types of maps. For instance, the common local-area quadrangle maps produced by the US Geological Survey, which each cover 7.5 minutes of latitude and longitude, are printed at 1:24,000. Hence these are often referred to as 1:24,000 maps, or even "24K quads." In a nutshell it is used in naming of maps.

3-1.3 Conversion of scales to representative fractions

Example 1:

If 1 centimetre on a map represents 10 metres on the ground, what will be the RF?

$$\frac{\text{Map units}}{\text{Ground units}} = \frac{\text{1 centimetre}}{\text{10 * 100 centimetres}} = 1:1000$$

Example 2

If the RF of a map is I: 120 how many units on the ground do 2 map units represent?

RF = 1: 120
i.e., 1 unit = 120 units
therefore 2 units = 2 x 120 unit~
= 240 units

Example 3

If the RF on a map is 1:1200 how far apart would two points be on the map if their actual ground distance is 360 metres?

The problem can be solved by using the following reasoning:

RF= 1:1200
i.e., 1 m on a map equal 1200 m on the ground. **x** metres on a map equal 360 m on the ground or

$$\frac{1}{1200} = \frac{\mathbf{x}}{360} \text{ metres}$$

$$\text{therefore } \mathbf{x} = \frac{360}{1200} \text{ metres}$$

= 0.3 metres = 300 mm

3-1.4 Conversion of areas by representative fractions

If the RF on a plan is a very large one, say ¼, the meaning assigned to the fraction as already discovered is that one unit on the plan represents 4 units on the ground. A square of 1 unit on the plan will therefore represent a ground area of (4 units x 4 units). From these facts emerges a simple formula:

Plan Scale = 1:4
Plan Area = 1 x 1 sq. units
therefore ground Area = (1 x 4) x (1 x 4) sq. units
= 1 x (4 x 4) sq. units
= 1 x 4^2 sq. units
i.e. Ground Area = Plan Area x 4^2
= Plan Area x (Scale factor) ^{2}

therefore Plan Area $= \frac{\text{Ground Area}}{4^2}$

therefore Plan Area $= \text{Ground Area} \times (RF)^2$

Example 4

The scale of a plan is 1:4. If a square on the plan measures 3 by 3 units what is the corresponding ground area?

Plan Scale = 1:4

Plan Area = 3 x 3 = 9 sq. units

Ground Area = (3 x 4) x (3 x 4) = 144 sq. units

By the formula:

Plan Area = Ground Area x $(RF)^2$

=> 9 = **x** * 1/16 sq. units

therefore **x** = 144 sq. units.

Example 5

An area was measured on a plan by, 250 mm x 175 mm. Calculate the ground area in metres if the scale is

(a) 1:2000

(b) 1: 500

Solution

(a) RF = 1: 2000

i.e., 1 mm (plan) = 2000 mm (ground)

therefore

1 mm^2 (plan) = 2000 x 2000 mm^2 (ground)

Plan Area = 175 x 250 mm^2

therefore

Ground Area = 175 x 250 x 2000 x 2000 mm^2

$= 175 \times 250 \times \frac{2000}{1000} \times \frac{2000}{1000}$ sq. m^2

$= 175{,}000\ m^2$

(b) By formula:

Plan Area = Ground Area x $(RF)^2$

therefore

Ground Area = (175 x 250) x 500^2 mm^2

$= 175 \times 250 \times \frac{500}{1000} \times \frac{500}{1000}\ m^2$

= 10937.5 m2

Example 6

A plot of land was surveyed and found to have an area of 2000 m^2. If it is plotted on a plan 1:500, what will be the plan area in mm^2?

Solution

RF = 1:500

i.e., 1 mm (plan) = 500 mm (ground)

therefore 1 mm^2 (plan) = 500 x 500 mm^2 (ground)

$$= \frac{500 \times 500}{1000 \times 1000} \text{ m}^2$$

$$= 0.25 \text{ m}^2$$

Ground Area = 2000 m^2

therefore Plan Area = $\frac{2000}{0.25}$ mm^2 = 8000 mm^2

Alternatively: Plan Area = Ground Area x $(RF)^2$

$$= 2000 \times \frac{1}{500} \times \frac{1}{500} \text{ m}^2$$

$$= 2000 \times \frac{1000}{500} \times \frac{1000}{500} \text{ m}^2$$

$$= 8000 \text{ mm}^2$$

Example 7

A plot of land is in the form of a rectangle in which the length is twice the breadth. When surveyed, it was found to have an area of 16,722.54 m2. Calculate the length of the sides as drawn on a plan whose scale is 1: 10 560.

Solution

Plan Area = Ground Area x RF^2

$$= 16722.54 \times \frac{1}{105602} \text{ m}^2$$

$$= \frac{16722.54 \times 1000 \times 1000}{10\,560 \times 10\,560} \text{ m}^2$$

$$= 149.96 \text{ mm}^2$$

Let dimensions be 2x by x

Then Area = 2 $\mathbf{x}^2$

therefore

$$2\,\mathbf{x}^2 = 149.96 \text{ mm}^2$$

$$\mathbf{x}^2 = 75.00 \text{ mm}^2$$

therefore x = 8.66 mm^2: therefore Plot measures 8.66 mm x 17.32 mm

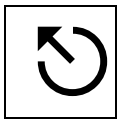


Self-Assessment

1. What are the merits and demerits of graphical scales?
2. A field distance of 200m was represented on a plan by 5cm. What is the ***scale*** of the plan?
3. A plot of land was surveyed and found to have an area of 10000 m^2. If it is plotted on a plan 1: 500, what will be the plan area in mm^2?

Learning Track Activities

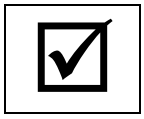
Summary



Unit Summary

In this unit

1. The Difference between plane and geodetic surveying has been discussed.
2. The main classes of land surveying have been highlighted.
3. A description of the different types of errors have been studied
4. Verbal, graphical and RF scales have been discussed
5. Calculations involving distances, areas and scales have been carried out.



Key terms

1. Plane Surveying
2. Geodetic surveying
3. Gross error
4. Random error
5. Systematic error
6. Map
7. Plan
8. scale
9. Representative fraction

Review Question

1. An area of ground lying between a straight path and a boundary fence has an area of 6250 square metres. The area as measured on an old map, whose scale has been obliterated, equals 1000 mm^2. What is the scale of the map?



Unit Assignment 1

- **Discussion Question:**
  - Distinguish between large scale and small scale maps
  - What is the difference between a map and a plan?
- **Reading:**

http://www.sonoma.edu/GIC/Geographica/MapInterp/Scale.html

Unit 2

CHAIN SURVEYING AND COMPASS SURVEYING

Introduction

This unit is intended to introduce you to the simplest and cheapest methods of surveying in terms of equipments and labour. Chain and Compass surveying are well adapted to small surveys on the accounts of their simplicity and most of the principles used here are also applicable to other surveying operations.

Chain surveying is the most basic method of surveying, relying entirely on accurate measurement of length. No angular measurements are made. The principle of trilateration is employed in chain surveying. Trilateration is a process in surveying where only linear measurements are made.

During Chain Surveying the area to be surveyed is divided into a series of well-conditioned triangles and each of the sides of all the triangles measured. Triangle is the simplest geometrical figure to plot, and therefore plans are easily plotted from chain surveys. This section also discusses how to deal with obstructions in the course of surveying.

Compass Surveys on the other hand relies on the magnetic fields of the earth. The earth has a magnetic field that enables a freely suspended magnetic needle to settle in the magnetic meridian. The magnetic property of the earth has enabled man to develop compass to measure bearings long before the advent of modern surveying equipment. A compass needle, suitable pivoted about its centre and properly balanced, so that it can oscillate freely without interference from any attractive forces, lines up with the magnetic meridian (north). Apart from the magnetic meridian other reference directions are discussed in this unit.

Learning objectives

At the end of the study of this unit students should be able to

- Carryout a survey of a small area using chain and compass surveying
- Discuss various obstructions and how they can be overcome in surveying

- List and differentiate the various reference directions
- Describe the equipment required to carryout chain and compass surveying
- Detect and eliminate local attraction from compass observation
- Plot and adjust a compass traverse graphically

Unit content

Session 1-2: Chain Surveying
1-2.1 Measurement of Distance
1-2.2 Setting Out Perpendiculars to a Chain Line
1-2.3 Location Methods
1-2.4 Field Problems (Obstacles)
1-2.5 Errors in Chain Surveying
1-2.6 Survey of Detail, Field Work and Plotting

Session 2-2: Compass Surveying
2-2.1 The Compass
2-2.2 Variation of the Compass
2-2.3 Reference Directions (Meridians)
2-2.4 Magnetic Bearing
2-2.5 Local Attraction
2-2.6 Traversing
2-2.7 Booking
2-2.8 Plotting
2-2.9 Advantages of Compass Surveying

SESSION 1-2: CHAIN SURVEYING

1-2.1 Measurement of Distance

Measurement of distances in surveying can be grouped under the following sub-sections.

1-2.1.1 Horizontal Distance

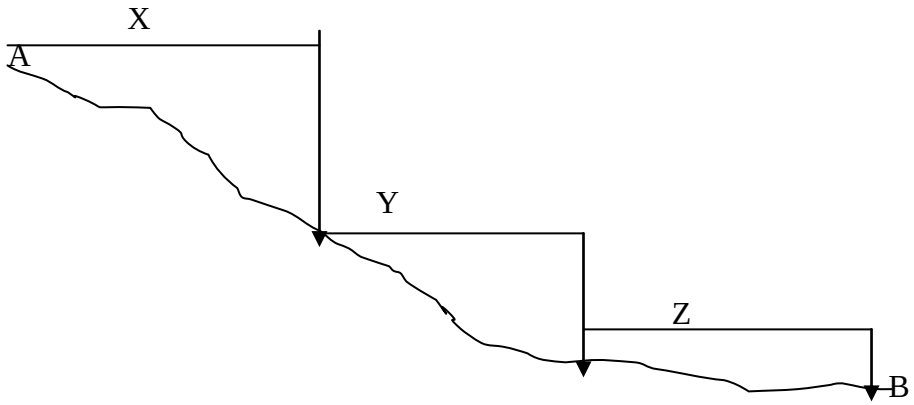
The procedure in measuring horizontal distances in surveying is:

- Step – chaining

This is an alternative method of obtaining horizontal measurements without using angle-measuring instruments. This is a field method where the horizontal distances are obtained directly.
Two men are required: the chainman/ Observer and an assistant. The chainman holds the chain or Lenin tape horizontally and aligns the assistant who holds the zero end of the chain or tape. A ranging pole or an arrow is dropped at the end of the tape.

The length of steps which can be adopted is limited by the gradient. At no time should the tape be above the surveyor's eye level, because plumbing becomes very difficult. As the gradient increases the length of step must therefore decrease.

When the observer has noted the length of the first step in his book, the second and third steps are measured and the process is repeated until the whole line has been measured (see Figure. 1). The sum of the steps will produce the required horizontal distance.



Total horizontal distance AB = X + Y + Z

Figure 1: Step Chaining

1-2.1.2 Measurement of Slope Distance

- **Abney level**

  This instrument consists of a square- section sighting tube 12.7cm long, fitted with a draw- tube extension, which extends to 17.8cm. The tube is fitted with a pin-hole sight and a horizontal cross wire at the viewing end. This completes the sighting arrangements.

  A highly polished mirror is fitted within the sighting tube at an angle of 45° to the line of sight allowing the observer to view simultaneously the spirit level via the mirror and some distant target against the crosshair.

  To measure an angle of inclination the instrument is placed to the eye so that the bubble is apparent in the mirror. The sighting tube is tilted to observe the forward

station and the slow motion screw controlling the spirit level is activated. The spirit level will now be in the centre of its run and the vernier will have been moved by the slow motion screw over the graduations giving the angle of inclination to 10 minutes of arc.

Examples of slope correction

1. Calculate the plan length for a measurement of 126.3metres along a gradient of 2° 34′.

   Solution
   Plan length = slope length X cosine inclination
   = 126.3 X Cosine 2° 34′
   = 126.3 x 0.9990
   = 126.174m

2. Calculate the plan length where a distance has been measured along a slope of 1 in 3 and found to be 149.5m.

   Solution
   Plan length = slope length X cosine inclination
   $= \frac{149.5 \times 3}{\sqrt{10}}$
   $= \frac{149.5 \times 3}{3.162}$
   = 141.840m

1-2.2 Setting out perpendiculars to a chain line

There are several small hand instruments that assist the surveyor in setting out right angles. They are Cross Staff, Optical Square and Prism Square.

1-2.2.1 Cross Staff

This instrument is probably the simplest and it consists of essentially two pieces of metal with upright sighting vanes crossing at right angles (Figure 2).

It is held on top of a ranging pole or a special rod called the Jacobs staff and planted at the point along the survey line at which the right angle is desired. Two slits are sighted along the survey line and the right angle is set out by sighting through the other pair of slits.

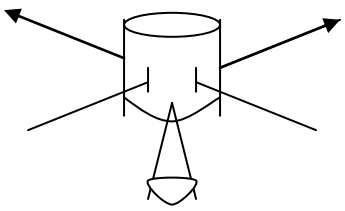


Figure 2: Cross staff

1-2.2.2 Prism Square

This instrument is made up of a pentagonal glass prism PQRST (Figure 3). The faces PT and RS, if produced contain 45°. The observer holds the prism at eye level and looks through the instrument to view directly at target 1. The incident rays carrying the image of target 2 are reflected to the observers' eye. When target 1 and the reflected image of target 2 coincide a right angle will have been formed.

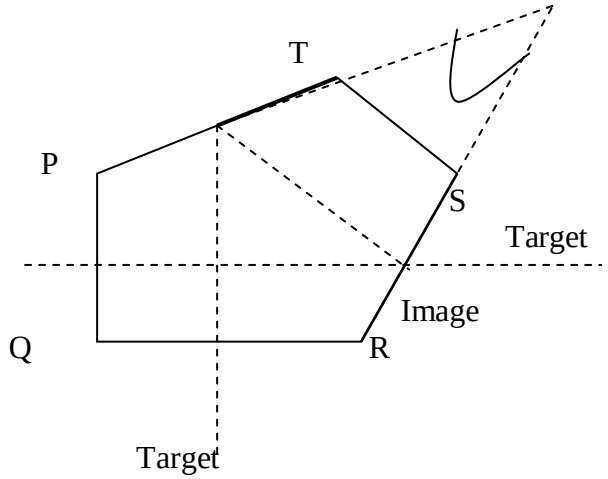


Figure 3: Pentagonal glass prism

1-2.3 Location methods

There are four location techniques which can be used to fix a point with respect to a survey line.

1-2.3.1 Right angle offset

Point P is fixed with respect to a survey line AB by the distance qp measured at right angle to AB from the point q. This is shown in Figure 4 below. The distances Aq and qB are also measured.

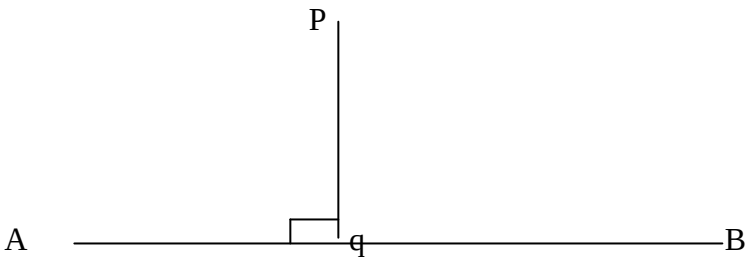


Figure 4: Right angle offset

1-2.3.2 Trilateration

The point P is located by the distances AP and BP which are measured respectively from' a' and 'b' as known points on the survey line AB. The distances Aa and bB are also measured.

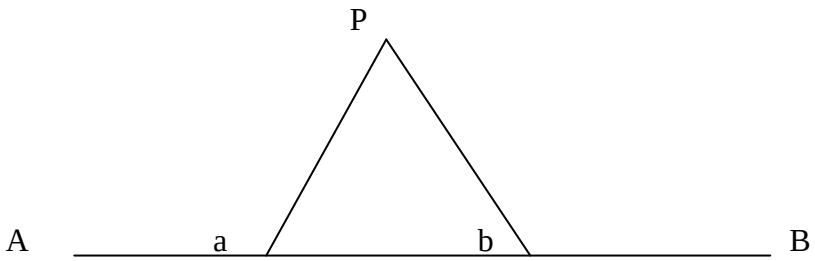


Figure 5: Trilateration

1-2.3.3 Triangulation

The point P is fixed with respect to the line AB by the intersection of two visual lines XP and YP which at known points X and Y make observed angles θ and α respectively with AB.

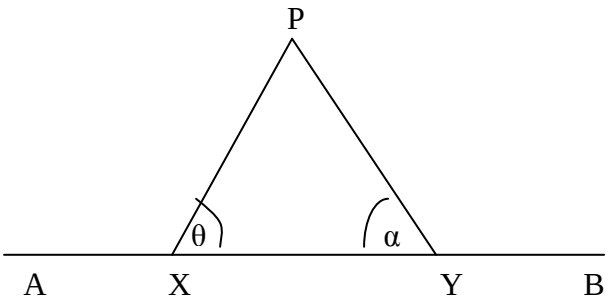


Figure 6: Triangulation

1-2.3.4 Traverse

The point P is fixed with reference to the survey line AB by the distance XP measured from a known point X on AB at a known angle β from that line.

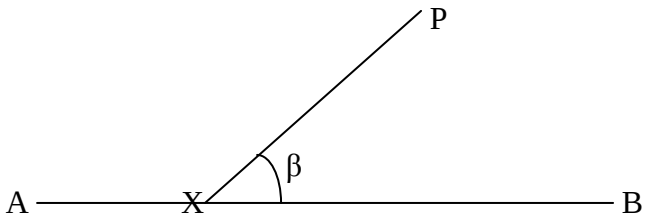


Figure 7: Traverse

1-2.4 Field Problems (Obstacles)

There are several obstacles in surveying. These either obstruct the direct measurement or ranging of any line. They are divided into the following:

i. Obstructions to measuring only.
ii. Obstructions to ranging only.
iii. Obstructions to measuring and ranging

1-2.4.1 Obstruction to measuring only

An example is a pond (Figure 8). If a pond lies directly on a survey line XY, it makes it impossible to measure the distance between the two stations directly. In other to overcome this problem, the following procedure is followed.

At a point A near the pond, a right angle is set out to point B by any of the previous described methods and the distance AB is measured. The distance from point B to point C on the survey line at the far end of the pond is measured. By Pythagoras theorem, the inaccessible distance AC is calculated.

$$\mathbf{AC = \sqrt{(BC^2 - AB^2)}}$$

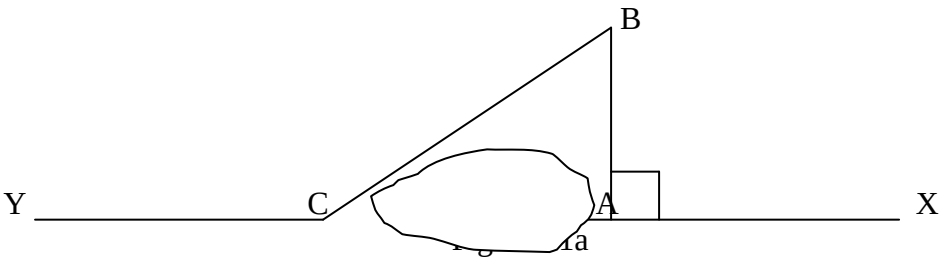


Figure 8: Example of obstruction to measuring only

Figure 9 shows an alternative method of obtaining the distance across the pond. At points A, B, C and D, right angles are set out and distances AB, BC, and CD measured. If all setting out is accurate AB will equal CD and the inaccessible distance AD will be equal CB.

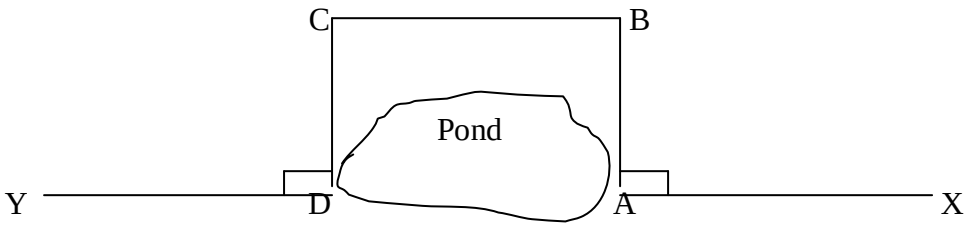
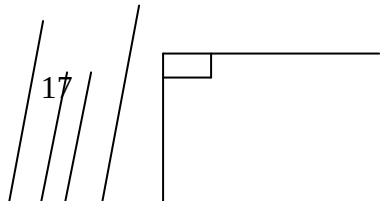


Figure 9: Example of obstruction to measuring only

- Obstructions cannot be measured around

The obstructions envisage under this heading are rivers and railway cuttings that exceed one tape length in width. Again many variations are possible. Figure 10 shows a survey line XY crossing a railway cutting; at point A, a right angle is set out to B and the distance AB is measured and bisected at C. At B another right angle is set out towards D and the point established where this line intersects the line EC produced. Lines BD and AE are parallel and triangles AEC and BCD are congruent. The inaccessible distance AE therefore equals BD.



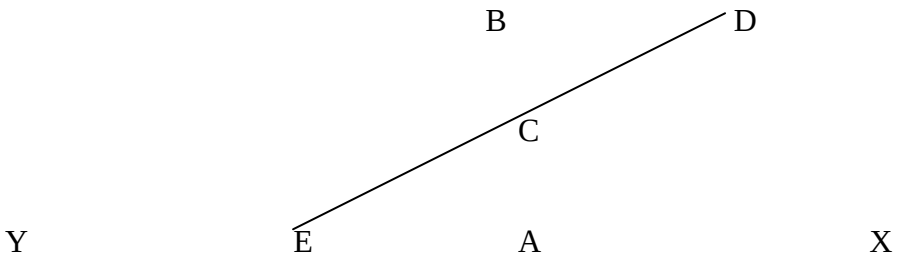


Figure 10: Obstructions cannot be measured around

1-2.4.2 Obstruction to ranging only (e.g. hill)

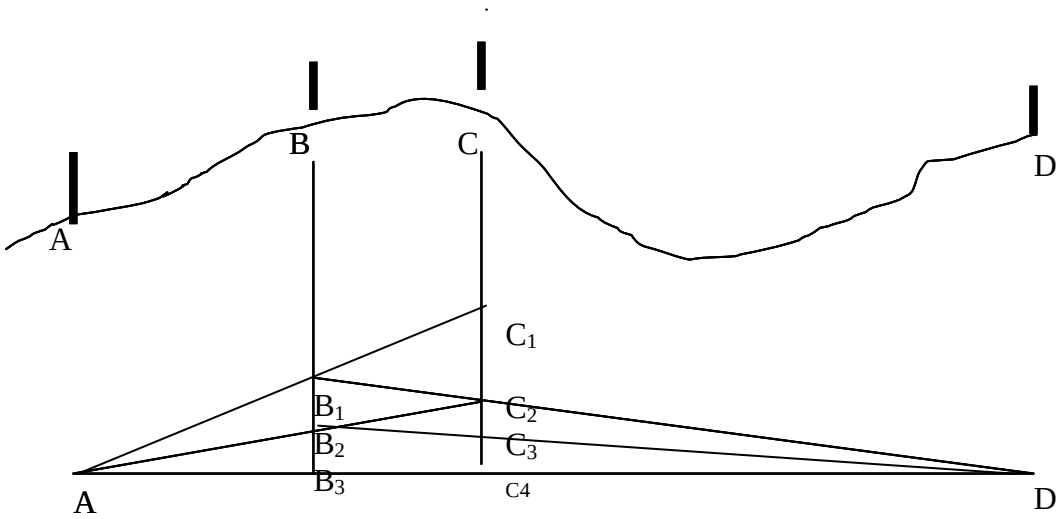


Figure 11: Obstructions to ranging only

a) Observers B and C each holding a ranging pole, place themselves on the ridge of the hill in line between A and D as nearly as possible so that one can see the other and the station beyond him

b) Observer C put B's pole in line with A at b.

c) Observer B at station b then looks to D and put C's pole in line with D at station c_1.

d) Observer C repeats his operation from c_1', putting B at b_1. In this manner, the two alternatively line each other gradually approaching the straight line between the stations A and D until at last they find themselves exactly in it at b_2 and c_2 as shown above.

1-2.4.3 Obstruction to ranging and chaining

Despite taking every precaution to avoid obstacles, the occasion arises where a building or wood lies on the survey line and the line can neither be measured nor ranged. Figure 12 shows a typical situation, where the survey line is interrupted by a grove.

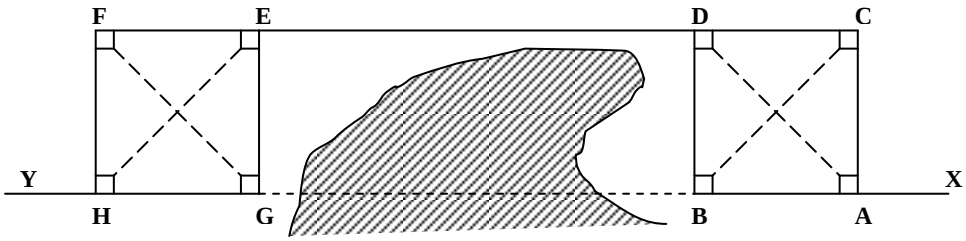


Figure 12: Obstruction to ranging and chaining

At points A and B, right Angles are set out to the points C and D respectively and AC made equal to BD. The length CD when check must equal the length AB. Line CD is then ranged forward to points E and F and the lengths DE and EF measured. Further, right angles are set out at E and F, the length EG and FH measured equal to AC and BD and points G and H establish. On checking GH, it should be found equal to EF. Points G and H are then on the line XAB and when the line GH is ranged forward it should point to Y. The inaccessible distance GB will equal ED.

Example 8 (see Figure 13)

A survey line XY is interrupted by a wide river and the distance across the river is required. On the near bank a right angle is set out at point A for a distance of 20 metres and a point B established. At point C, 40 metres back from A on the line AX, a second right angle is set out to D, on the same side of AX as point B. The point D is aligned with Y on the far bank and with B, already established. The line CD is found to measure 31metres. Calculate the length of the inaccessible portion AY.

Solution

Triangles YAB and YCD are similar.

Therefore $\frac{YA}{AB} = \frac{YA + AC}{CD}$

i.e., $\frac{YA}{20} = \frac{YA + 40}{31}$

Therefore $YA = 20(YA) + 800$

Therefore $YA = \frac{800}{11}$

$= \underline{72.72 \text{ metres}}$

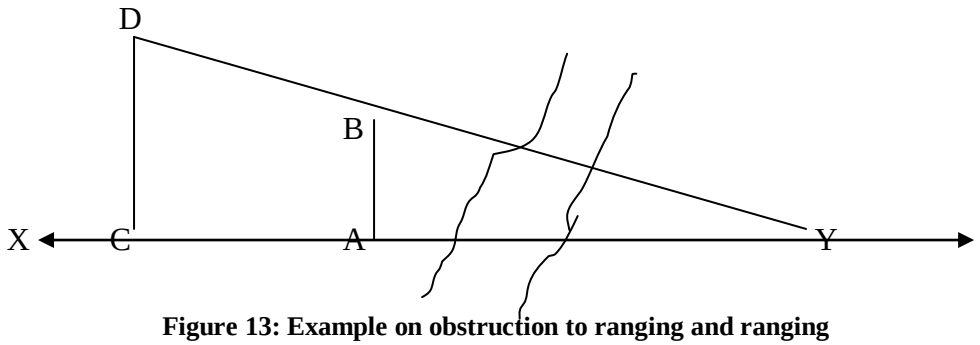


Figure 13: Example on obstruction to ranging and ranging

1-2.5 Errors in chain surveying

All measurements made with tapes and chains are subject to some form of error, no matter how carefully any line is measured. The error may be due to surveyor's carelessness or inexperience; it may be due to prevailing climatic conditions, or it may be inherent in the instrument being used. Errors can be divided into several classes:

1-2.5.1 Gross Errors

These are mistakes arising from inexperience, carelessness or fatigue in the observer. Examples are, miscounting of chain lengths, displacement of arrows from their true position and faulty booking of field information (6.32m instead of 16.32m). These can only be eliminated by proper and careful methods of observing booking.

1-2.5.2 Systematic/Cumulative/Constant errors

These are errors whose magnitude and algebraic sign can be determined. They always recur in the same instrument or operation. They include misalignment of the tape, error due to sag of the chain in stepping operations, the chain not being true to length and the chain not being pulled straight.

(a) Standardisation: C = (L- l) per tape/chain length. Where
C= correction
L= actual length of the tape
l = nominal length of tape.

(b) Temperature: C = L x C.Ex (Tm-Ts). Where
L= measured length
C.Ex= Co-efficient of expansion per degree centigrade.

Tm= Actual temperature at the time of measuring
Ts= Standard temperature.

c) Tension: C= L(Tf- Ts)/AE. Where,
Tf= Tension applied in the field.
Ts= Standard tension
A= Cross-sectional area of the tape (mm) ^{2}
E= Modulus of elasticity for the tape material
L= Recorded length.

1-2.5.3 Accidental/Human errors

These are caused by variations in the eye sight of persons using the instrument and sadden changes of temperature or wind and atmospheric disturbances. They are usually small and not cumulative.

1-2.6 Survey Of Detail, Field Work And Plotting

The survey of details and field work as well as the plot of the field work are illustrated in the next sub-sections

1-2.6.1 Chain surveying

The simplest form of carrying out work within plane surveying is known as Chain or linear surveying. This relies directly on the measurement of horizontal distance and the principle of trilateration. Angles are not measured except for occasional degree of slope. If fibred glass or steel tape is used the technique would still be termed 'Chain Surveying'.

1-2.6.2 Equipment and instruments used

The equipment necessary to enable a chain survey to be carried out comprises:

i. Chain (Engineers/ Gunter's), steel tape, linen tape, arrows, ranging poles, flags.
ii. Cross staff, optical square (prism), clinometers or Abney level.
iii. Field book, pencils and erasers.

A. Equipment

Equipments required are very simple. They are:-

1. A 30m or 50m tape, graduated in single meters, with decimetre and centimetre subdivisions. Ideally, light steel tape should be used for measuring the framework, but a linen tape, particularly the glass fibre synthetic type, properly handled, could produce fairly good results.
2. Ranging poles (also termed banderoles, in some textbooks)- Traditionally, these are wooden pieces of circular section 2m, 2.5m or 3m long, painted with characteristic red and white bands which are usually 0.5m long and tipped with a pointed steel shoe to enable them to be driven into the ground. They are used for marking, for ease of identification of any point which needs to be seen, and for line ranging, that is, establishing points on a given straight line. You may need a minimum of four ranging poles. (Note: There are now no the market, tubular aluminium, and even plastic, ranging poles).
3. Arrows – These are steel or aluminium skewers about 40cm long with red ribbons tied to circular rings at the top, enabling them to be more clearly and readily identified. Traditionally, the recommended number required is ten, but if the lines are not too long, we could do with a smaller number, say five or six.

B. Procedure in linear measurements

This consists of the following:-

- Standing at A, let your assistant take hold of the ring at the zero end of the tape and walk towards B while you slowly unroll the tape.
- (Generally we call the person who takes the lead towards B the 'leader' while the person at A becomes known as the 'follower').
- When the leader reaches B, instruct him to hold the tape firmly on the ground at B, Give your end of the tape one or two gentle shakes (vertical) to send 'ripples' or ' waves' down the tape to straighten it out between the two points, A and B.
- With the tape properly aligned and the zero graduation coinciding with the mark at B, alert the leader and give the tape a gentle pull to make it taut. [The zero graduation will shift from the mark at B, and as the leader attempts to restore the zero graduation to its position (on the mark at B) while you are striving to keep it taut, there will be some slight forward and backward movement of the tape).
- When the tape is steady, with the zero graduation coinciding with the mark at B, the leader signals, probably by saying the word "ON" in a prolonged manner, This is a signal that you should read the tape graduation against the mark at the A; which should be the required length of the line AB, since the reading on B is zero.

The foregoing simple procedure presupposes that length of the line AB is shorter than that of the tape and that the ground is ideally flat. In practice the contrary is generally the case; the lines are longer than the tape length, requiring the tape to be laid out more than once and hence calling for line ranging which is discussed below.

Very often our lines also pass over sloping grounds, and since the end result of our operation is the preparation of a plan or map, which is a horizontal projection of the features on the ground, it is imperative that we reduce all our measured lengths to their horizontal component.

There are two ways of dealing with grounds that slope in linear measurement. One way is to measure along the slope, if the ground surface allows for the laying of the tape parallel to the general ground slope. We would then determine the angle of slope with a hand-held instrument called the Abney Level (see 1 of 1-2.1.2) and then compute by trigonometry, the horizontal component H of the measured slope distance L. If **θ** is the angle of slope, then

Generally it is more convenient to compute a correction to be applied to the measured quantity, and the correction

$$H = L\cos\theta$$

$$= -(L - L\cos\theta)$$

$$= -L(1 - \cos\theta)$$

The other way is what we call step taping, which is cheaper because it does not require any additional equipment nor any trigonometrical calculation, and is generally preferred, especially where the ground surface is of variable slope, rough and not suitable for measuring along the slope. Step taping is also discussed in 1-2.1.2.

C. Line Ranging

Line ranging, as explained, involves establishing a series of points on a given straight line. We shall describe the procedure by an illustration:-

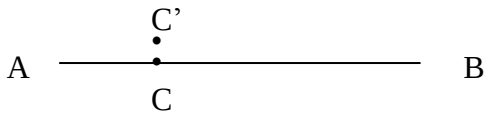


Figure 14: Line Ranging

Let A, B, be the two end points of the line. We are required to fix a point in line with A and B. The following are the steps:-

- Fix a ranging pole vertically at A and B respectively.
- Let the leader move the required distance – a tape length in linear measurement – to point C′, or somewhere around C.
- Standing close to the ranging pole at A, look along it to the pole at B (if you see the pole at B as well as that at C, it means C is not in line with A and B).
- Signal the leader to move his pole a little at time and in the appropriate direction, until his pole just blocks your view of the pole at B. The leader's pole is now deemed to be at C, and in line with A and B.

Note:

(i) The leader must always stand to the side of the line AB and in such a way that his body does not block your view of the ranging pole at B.

(ii) He must always hold the pole reasonably close to his body to ensure vertically of the pole to facilitate the operation.

- To avoid confusion, you must agree on a code of signals and ensure that you follow the code strictly, without ambiguity. For example: - Right (left) arm extended - continue to the right (left). Right (left) hand up and moved to the right (left) - plumb or straighten the pole by moving the top in the direction indicated.
- Left (right) hand down and moved to the left (right) - plumb or straighten the pole by shifting the bottom in the direction indicated.
- Both hands up, above the head – Correct

Signals should be made to be seen clearly. The arms should be extended clear of the body; signals to move to the left should not be made with the right hand, and vice versa. When considerable distances are involved, you should use a white handkerchief for some white paper to make the signals.

D. The use of the arrows in measuring long lines

The ten arrows are carried by the leader in addition to the ranging pole. The pole is fixed on line about 0.5m from the end graduation. When the tape is in steady condition for the measurement, the leader sticks an arrow in the ground against the end graduation.

After that the leader moves ahead with his pole, leaving the arrow still stuck in the ground. The follower then comes up. Replaces the arrow with his pole, and the measurement process is repeated. When moving forward, the follower picks the arrow. At the end of a line the number of arrows held by the follower represents the number of tape lengths measured, and to this is added the fractional bit to get the total length of the line. (Note: At the end of a line you must always check that the remaining arrows held

by the leader and those held by you, sum up to 10, or whatever number that was brought to the field.) This ensures that you have not overlooked any arrow placed in the ground by the leader, nor is any arrow misplaced. In a very long line, after you have collected the tenth arrow, you must send the ten arrows to the leader for the measurement process to be continued, but you must take note that you have already measured ten tape lengths.

1-2.6.3 Procedure in Chain Surveying

The procedures used in chain surveying are described below:

A. Reconnaissance survey

Existing maps/ plans of the area to be surveyed are consulted so as to enable the surveyor get the general impression of the area and also get clues to any problems that are likely to be encountered during the surveying process. The following factors are taken into consideration:

i. **Working from the whole to the part**

This is the fundamental rule of all survey operations. In surveying, the main frame must be set out first and the specific detail can be added on to the frame, not the reverse. There is far less error when a larger area is divided into smaller units than when smaller units are surveyed and an attempt is made to join them to form a larger area.

ii. **Selection of Survey stations**

A survey station is a point of importance at the beginning or end of a chain line, or at the junction of one line with another. It is usually marked by the insertion into the ground a peg and is normally intervisible. Stations should be placed at the corners of areas or at prominent points, so that the lines joining them are as close as possible to the boundaries of the site in order to keep offset measurements short.

iii. **The base line**

This is normally the longest of the chain lines forming the pattern of triangles. It should if possible be laid on level ground through the centre of the site and encompass the whole length of the area.

iv. **Trilateration**

Chain surveys are built up as a series of adjoining triangles known as the triangulation pattern. The Surveyor divides the area under survey into a number of triangles which when all the sides have been measured, can be plotted to scale in plan form. This frame work of triangles forms the skeleton upon which all the details such as boundaries, buildings, roads etc are added after further measurements have been taken. When all the sides are measured the term is trilateration.

v **Formation of well condition triangles**

All triangles should be well conditioned, that is they should not have angles less than 30° nor greater than 120°. The ideal figure is an equilateral triangle and every effort should be made to have triangles whose angles are all around 45° and 75°. The reason is simply that, it is easier to plot and more accurate.

vi **Offset area**

The area lying between the chain lines and the boundaries of the site is referred to as the offset area. Chain lines are placed as close as is practicable to the boundaries as well as offset measurements made short.

vii **Referencing the stations**

When the stations have to be used again it is necessary to be able to find them easily. If possible they should be placed near some permanent objects from which measurements may be taken to the survey stations. Each station should be referenced in case any check measurements are to be taken in cases of survey errors.

viii **Check or proof lines.**

When all three sides of a triangle have been measured, although the triangle is completely defined, it still requires to be proved. If one of the lines forming the sides is incorrectly measured, the check line will prove the accuracy of the work. Check line are not used for plotting purposes but are scaled from the drawing and compared with measurements taken in the field.

B. Conducting the Survey

Once the stations have been selected, the various lines are measured. The survey lines are as close as possible to the details which have to be surveyed. The details include the roadways, river fences buildings and any artificial object. Short distances, called offsets are measured to these points of detail from positions along the main survey lines. These latter points are called chainages, thus any point of detail must have at least two measurements to fix its position, namely a chainage and an offset (see Figure 7(a)).

Where ever possible, the offsets are measured at right angles to the survey lines. It may be necessary or desirable to fix certain points of detail by more than one offset, whereupon the estimation of right angles ceases. In Figure 7 (b) certain details are fixed from two chainage points by 'oblique' offsets. This is a more accurate method and is used to fix important details, like the corners of a house. It is necessary to use this method when the maximum allowable right angled offset is to be exceeded.

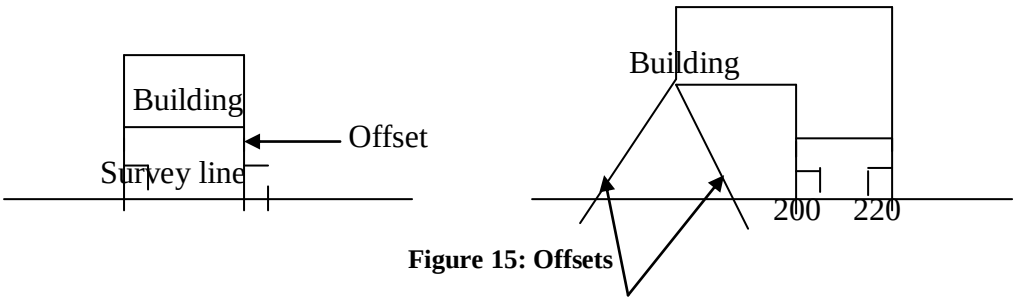


Figure 15: Offsets

C. Recording the Survey

One of the most important and probably most underrated points in linear survey is recording the information.

i. Referencing the survey

The first task in booking is to make a reference sketch of the survey as a whole. The sketch shows the main framework with the stations in their correct relationship.

As each line is measured, its length is written alongside the line, together with any gradient values. The measurements are always written in the direction of travel. From this reference sketch, the basic framework can be plotted.

ii. Booking the details

Usually two red lines, about 1.5cm apart are ruled at the centrally down the middle of the page to represent the survey line. The notes are recorded up the page, as in looking forward along the chain line to the next forward station. This method of upward booking should be characteristic of all forms of line notes. In keeping field notes, scale is relatively unimportant compared with neatness and clarity of interpretation, particularly in regard to offset detail.

Usually a page is allocated to a line regardless of its length, though obviously very long lines or lines with much detail will require two or more pages. Frequently the station chainages are inserted in rings. The crossing of a road or fence requires that the double red lines be imagine as a single line breaking the road or fence. An example is shown below:

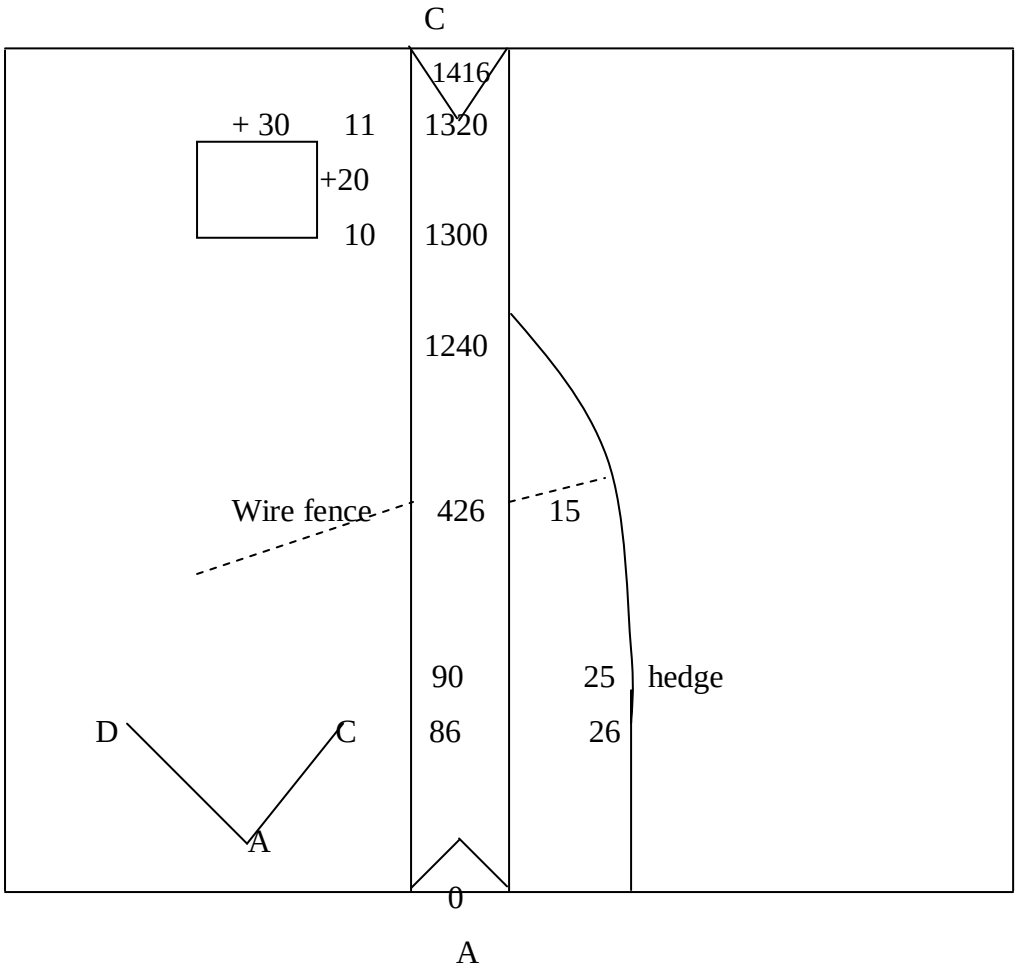


Figure 16: Field booking

In Figure 16, the details to be surveyed from the main survey lines include the hedge, wire fence, and building. These are surveyed by taking offsets at selected chainages along the main survey lines.

This method of booking is called the ‘double line’ method and all chainages are noted in the red lines. Virtually no attempt is made to draw to scale. The emphasis is placed rather on relative positioning of the features and on clear legible figuring.

D. **Plotting the Survey**

There are four principal steps in the routine of plotting maps and plans:

i. **Selecting the scale**

In selecting the scale, the objective of the survey and the extent of the surface to be represented must be borne in mind. Metric and Imperial units should not be mixed.

ii. **Placing the survey**

In placing the survey, the area should be viewed from the local aspect and approaches to the property should appear at the bottom with roads approximately parallel to the bottom edge of the paper regardless of the orientation of the north point, true or magnetic.

iii. **Constructing the triangles or skeleton**

In plotting the triangles, a survey line is selected as the base and this is drawn to scale in the most suitable position on the paper.

iv. **Inserting the detail**.

Details of surveys are inserted mostly by rectangular offsets being tied by means of ordinary compasses. The plotting, zero at the beginning station of the line, is placed carefully along the pencilled line and the points at which offsets are pricked off. Then short perpendiculars are erected to place the object at its correct position. All details should be carefully outlined in pencil, reducing the use of erases to a minimum.



Review Questions

1. (a) Describe four methods by which a point M can be fixed with respect to a line BC.

(b) Describe with the aid of suitable diagrams how you would solve the problem posed by the following:

i) An obstacle preventing ranging but not chaining in a course of a chain survey.

ii) An obstacle preventing chaining but not ranging in a course of a chain survey.

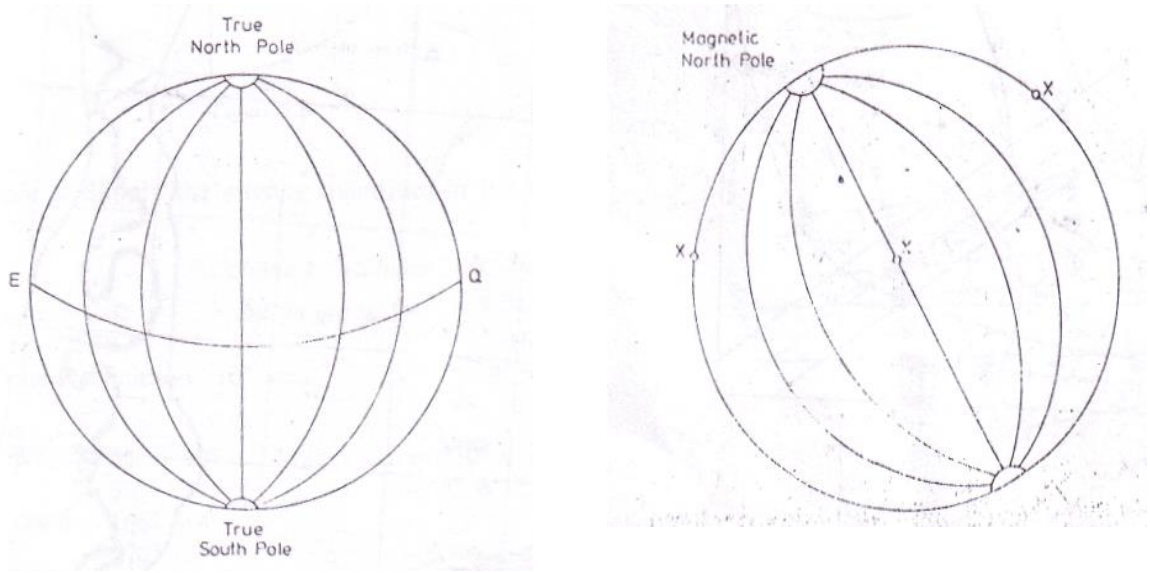
2. Write brief notes on the following
   a. Step Chaining
   b. Standardisation
   c. Horizontal distance
   d. Slope distance
   e. Plus measurement

SESSION 2-2: COMPASS SURVEYING

2-2.1 The compass

The earth has a magnetic field that enables a freely suspended magnetic needle to settle in the magnetic meridian. The scientific reason for this is beyond the scope of this course. The magnetic property of the earth has enabled man to develop compass to measure bearings long before the advent of modern surveying equipment.

The magnetic poles are not points but oval areas that are located not at the Geographic Poles (figure 17.a) but over a thousand kilometres from the poles. Magnetic Poles (figure 17.b) are not fixed they change from time to time. A compass needle lines up with the magnetic north in most places, meaning the needle points slightly east or west of astronomic (true) north depending on the locality.



(a) Geographic poles (b) Magnetic Poles

Figure 17: (a) Geographic poles, (b) Magnetic poles

Basically there are two main types of compasses namely, prismatic compass (Figure 18) and surveyors compass (Figure 19), and these are briefly discussed below.

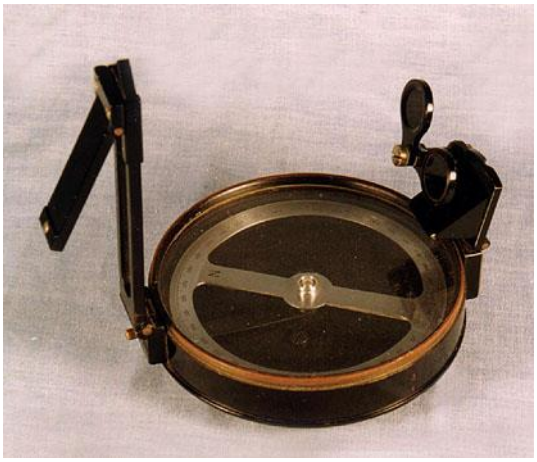


Figure 18: Prismatic compass

The **Prismatic Compass** is one of the two main kinds of magnetic compasses for measuring magnetic bearings, with the other being the Surveyor's Compass. The main difference between the two instruments is that the surveyor's compass is usually the larger and more accurate instrument, and is generally used on a stand or tripod. The prismatic compass on the other hand is often a small instrument which is held in the hand for observing, and is therefore employed on the rougher classes of work. The graduations on this prismatic compass are situated on a light Aluminium ring fastened to the needle, and the zero of the graduations coincides with the south point of the needle. The graduations therefore remain stationary with the needle, and the index turns with the sighting vanes. Since the circle is read at the observer's (rather than the target's) end, the graduations run clockwise from the south end of the needle (0º to 360º), whereas in the surveyor's compass, the graduations run anti-clockwise from north.

The prismatic attachment consists of a 45º reflecting prism with the eye and reading faces made slightly convex so as to magnify the image of the graduations. The prism is carried on a mounting which can be moved up and down between slides fixed on the outside of the case. The purpose of this up-and-down movement is to provide an adjustment for focusing. The image of the graduations is seen through a small circular aperture in the prism mounting, and immediately above this aperture is a small V cut on top of the mounting, over which the vertical wire in the front vane may be viewed. When the V cut, the vertical wire and the station whose bearing is required are viewed in one line; the bearing is directly read off the graduated arc at the point immediately underneath the vertical wire.

The oblong mirror located in front of the forward vane slides up and down the vane, and is hinged to fold flat over it or to rest inclined at any angle with it. This mirror is used for solar observations, or for viewing any very high object, and is not a normal fitting to a compass. The two circular discs in front of the back vane are dark glasses which can be swung in front of the vane when solar observations are being taken.



Figure 19: Surveyors Compass

While all compasses measure horizontal angles with reference to magnetic north, a surveyor's compass is equipped with vertical sights for taking aim at distant objects.

A vernier compass is a compass with a variation arc and a vernier mechanism. With it, a surveyor can compensate for magnetic variation, and thus run lines in relation to the true meridian, or retrace old lines (assuming that he knew the extent of magnetic variation at the time the line was originally run).

2-2.2 Variation of the Compass

The magnetic poles of the earth are constantly changing their position relative to the true north and south poles. There is a gradual unexplained shift in the magnetic field in one direction after which a gradual shift occurs in the other direction to complete a cycle.

Long term variation, annual variation, daily variation, magnetic storms and local attraction all affect the direction of the compass.

Secular variation: is the mean annual change in the value of the declination at a place.
Local Attraction: Apart from the earth's magnetic attraction other elements such as metal fences, underground pipes, passing cars iron ore deposits, high voltage power lines have distorting effect on compass readings these are called local attractions.

Local Attractions:

1. If there is local attraction at a station it affects all readings at the station equally.
2. Local attraction should be eliminated before a compass traverse is plotted.
3. Local attraction can be eliminated mathematically or by finding the included angle at a station.

2-2.3 Reference directions (Meridians)

In surveying the direction of a line is described by the horizontal angle that it makes with a reference direction also called meridians. A line on the mean surface of the earth joining the north and south poles is called a meridian. There are three types of directions namely:

- Astronomic, Geographic or True meridian is the direction of a line passing through the astronomic north or south poles and the observer's position. More precisely, it is the line formed by the intersection of the earth surface of a plane that includes the earth's axis of rotation.

- Magnetic Meridians are meridians that are parallel to the direction taken by freely moving magnetized needles as in a compass at the observer's position. Whereas true meridians are fixed, magnetic meridians vary with time and location. The magnetic meridian does not coincide with the true meridian, except in certain localities, and the horizontal angle between the two directions is termed the *magnetic declination* or the **'***declination of the needle***.'**

  The amount and direction of the declination is different at different places on the earth's surface: in some places the needle points west, in others east, of the true meridian; it even varies at the same place with time. Its value at any place may be determined by making observations for the true meridian, or may be interpolated from published **isogonics charts** - charts showing places of equal magnetic declination.

  [Isogonics lines, or isogons, are imaginary lines through points, at which the magnetic declinations are equal at a given time. Lines passing through places of zero declination are termed agonic lines].

- Grid Meridians are lines that are parallel to a grid referenced meridian. The national grid is a symmetrical network of parallel lines crossing at right angles like a sheet of graph paper.

For Ghana our meridian of origin is 01º 00 West of Greenwich and latitude of origin is 04º 40 North.

On the globe all lines of latitudes are parallel and run truly east west. Similarly all longitudes meet at the poles. However the north –south lines on the grid are parallel to 01º 00 west longitude and cannot point to true north. The difference between the grid north and the true north is the convergence.

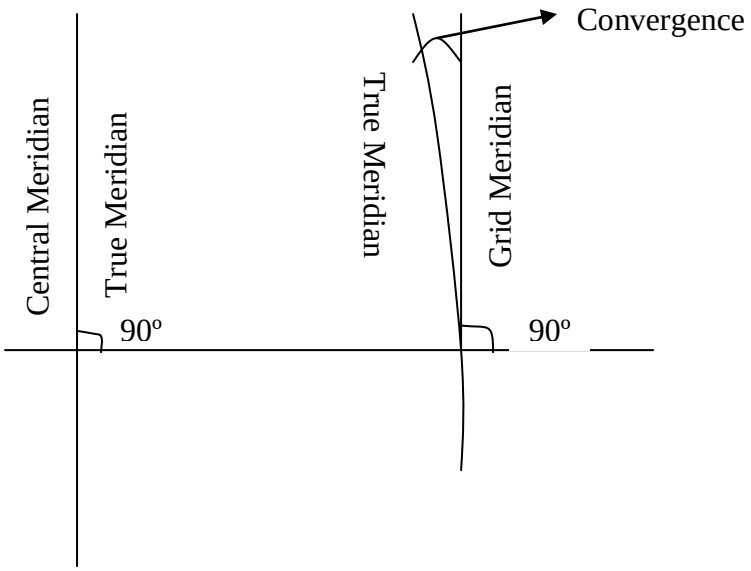


Figure 20: Relationship between true and grid bearing

- Assumed Meridian is an arbitrary direction taken for convenience. This is only used for small scale survey of limited importance. Here meridians are assumed and the survey is referenced to the assumed meridian.

2-2.4 Magnetic Bearing

The magnetic needle in all freely suspended compasses will settle in the magnetic meridian that is its north seeking will point to magnetic north pole. In the figure below if a compass were held at X, the magnetic needle will settle pointing to the magnetic North.

If a protractor were laid below the needle and oriented with the zero, pointing to the magnetic meridian, then the direction of any line radiating from X could be determined relative to the magnetic meridian. In the diagram below line XA makes an angle of 40º with the magnetic meridian. Bearings of lines XB and XC are respectively 160º and 290º. It is evident from the diagram that the maximum angle will be 360º. The 290º formed by XN and the line XC is the whole circle magnetic bearing of the line XC and is defined as the angle measured in a clockwise direction from the magnetic meridian to line XC.

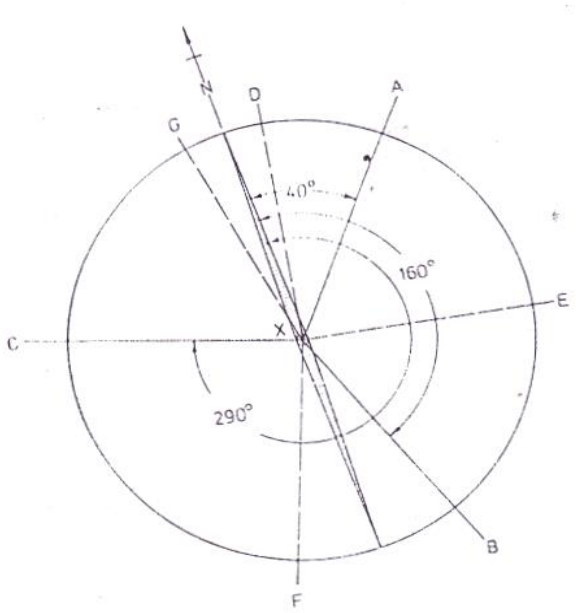


Figure 21: Magnetic Bearing

2-2.4.1 Whole-circle Bearings (WCB)

To establish the direction of a line between two points on the ground, its bearing has to be determined. The whole-circle bearing (WCB) of a line is measured in a clockwise direction in the range $0^{o^{\circ}}$ to $360^{o^{\circ}}$ from a specified reference or north direction. Examples of whole-circle bearings are given in **Figure 22.**

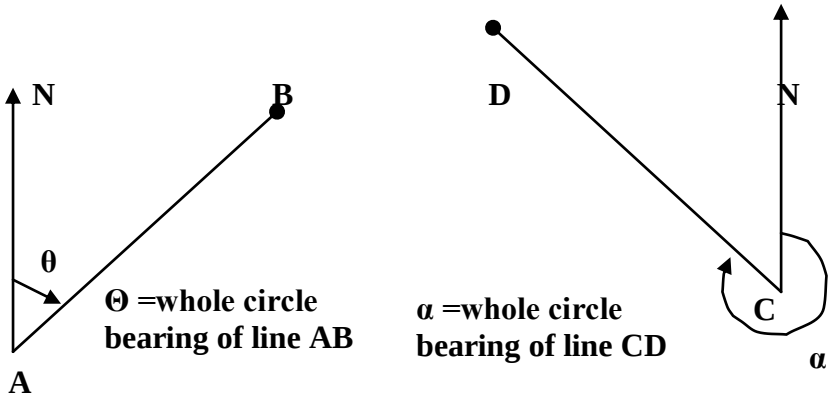


Figure 22: Whole-circle Bearings

- Whole circle bearing (WCB): this is the standard way of defining a bearing in survey practice. The whole circle bearing of a line AB is defined as the clockwise angle from 0º to 360 º at A between the reference meridian and line AB.

2-2.4.2 Quadrant bearing (reduced bearing)

If the cardinal points of a compass are drawn and labelled north, east, south and west moving clockwise, the whole 360º circle would have been divided into four quadrants of 90º (figure 23). The quadrants are known as northeast, southeast, southwest and northwest. ***The quadrant bearing of any line is the angle it makes with the north-south axis.***

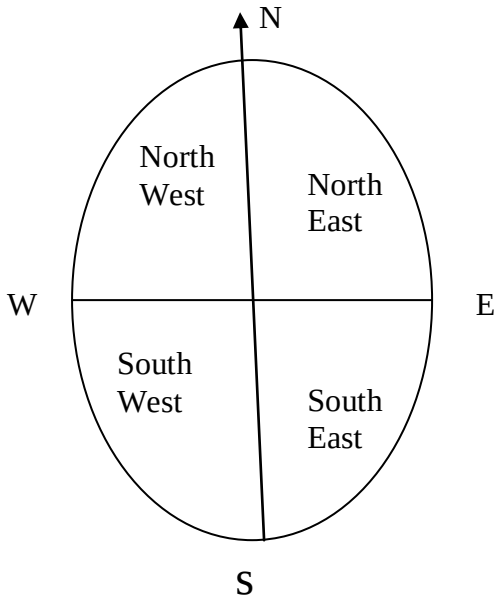


Figure 23: Quadrant bearing

It is defined as the angle lying between 0 º and 90 º between the direction to the North or south and the direction of the line. The North or South is assigned 0º when measuring the direction.

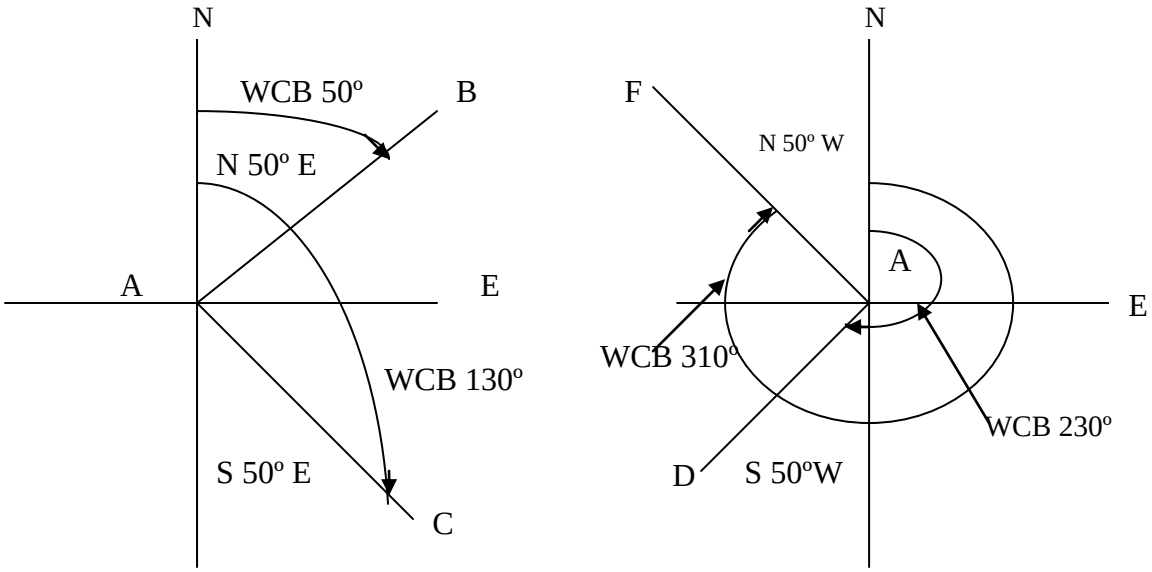


Figure 24: Whole circle bearings and quadrant bearings

Conversion of whole circle bearing (WCB) into quadrant bearing by the following rules:

- If the whole circle lies between 0º and 90º, the quadrant bearing has the same numerical value in the North east quadrant.
- If the WCB lies between 90º and 180º, the quadrant bearing is (180 – WCB) and lies in the south east quadrant.
- If the WCB lies between 180º and 270º, the quadrant bearing is (WCB - 180) and lies in the south west quadrant.

- If the WCB lies between 270º and 360º, the quadrant bearing is (360 - WCB) and lies in the North West quadrant.

Examples:

1. The magnetic bearing of a property line was S43º 30´ E. The magnetic declination at the place of the survey was 3º 15´ W of true North. What is the true bearing of the line?

Solution:
It is important to make a sketch to aid in the solution of the problem.

Let TN and MN represent true north and magnetic North respectively (Fig 25). Similarly TS and MS represent the south, and let the property line be represented by OA.

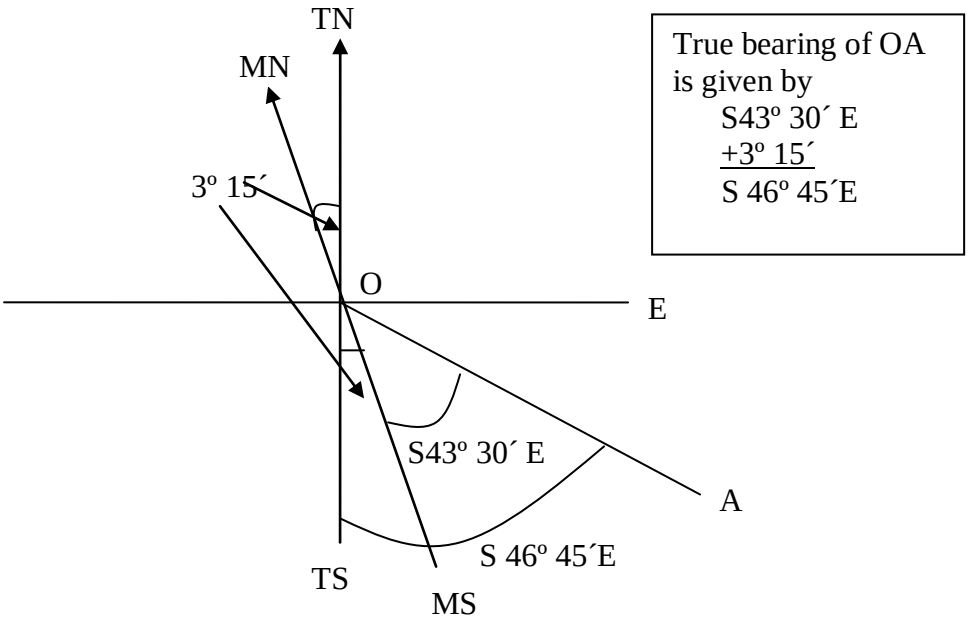


Figure 25: Example 1 (Solution diagram)

2. The magnetic WCB of a line AB read in 1900 was 026º 15´, the declination at that time was 7º 15´W, and in 2006 the declination is 4º 30´E. What is the magnetic bearing of AB in 2006?

Solution
This question can be broken into two parts, first find the true bearing of line AB at the initial time of survey 1900 (Fig. 26). Since the true bearing of a line does not change, use the true bearing and the current magnetic declination to find the magnetic bearing for 2006 (Fig.26)

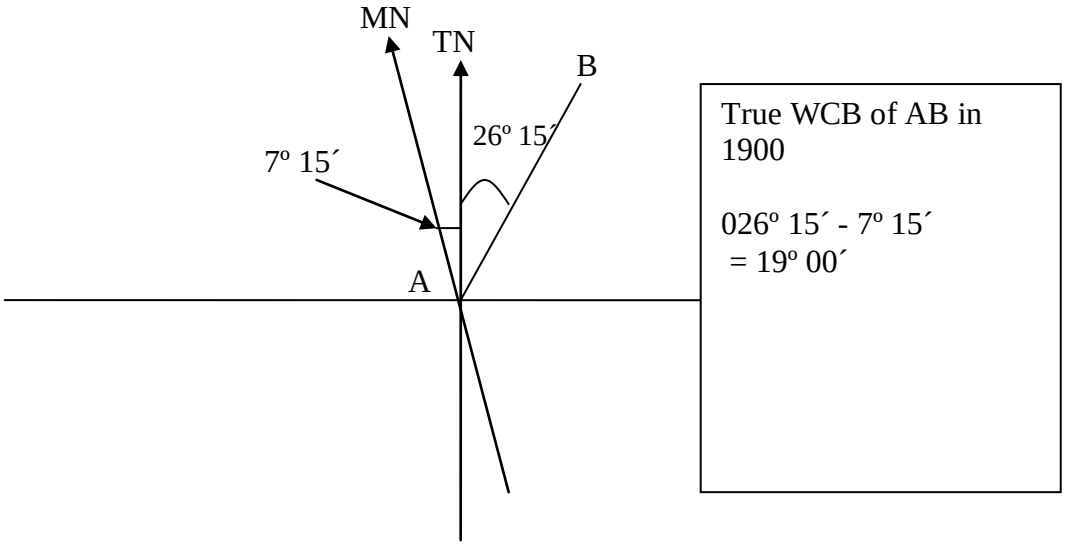


Figure 26: Example 2 (Solution diagram 1)

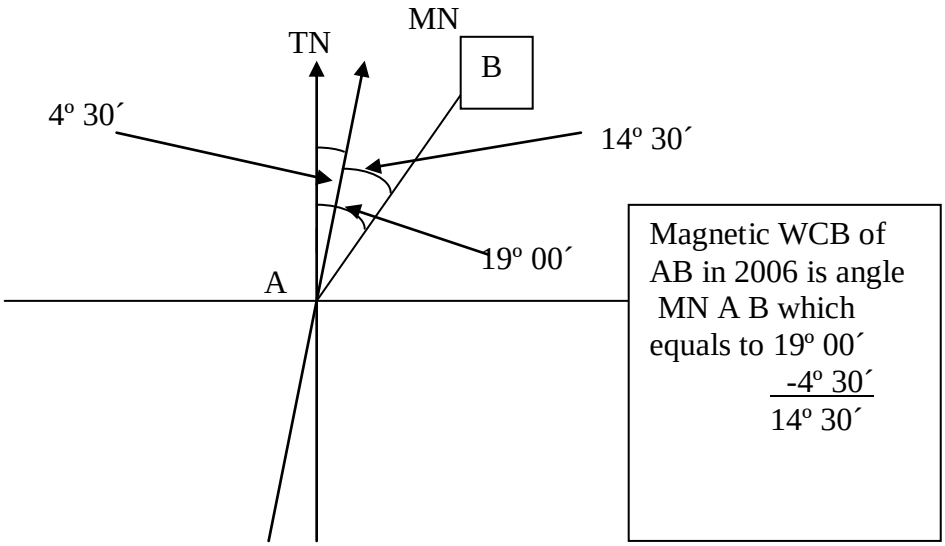


Figure 27: Example 2 (Solution diagram 2)

2-2.4.3 Forward and back bearing

Every line has two bearings, the forward and back bearings. The forward and back bearing of a line differ by 180 º if there is no local attraction in the case of compass bearing.

The bearing of a line, designated by the stations between which it lies, is taken as that from the first station mentioned. Thus, when we talk of the bearing of a line OA, it is the direction of the line from O to A. The direction from A to O is termed the back, or reverse, bearing. In Figure 28, OA is the line under discussion. NOS and N'AS' are north-south grid lines drawn through O and A respectively. The bearing of OA, that is the direction from O to A with reference to the grid line meridian NOS, is the clockwise angle NOA=α. The bearing of AO, i.e., the back bearing of OA, is

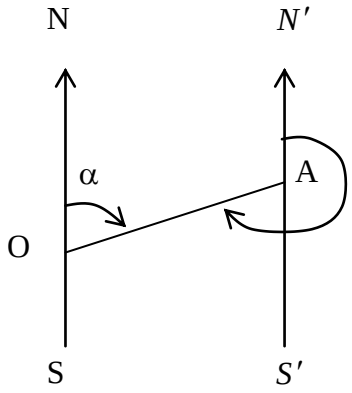


Figure 28: Forward and back bearing

From Figure 28, the clockwise angle $N'\hat{A}O = N'\hat{A}S' + O\hat{A}S'$. Since $N'\hat{A}S'$ is a straight line, $N'\hat{A}S' = 180^{\circ}$, and $O\hat{A}S' = N\hat{O}A = \alpha$ (alternate angles, since NOS and $N'\hat{A}S'$, both being grid lines, are parallel lines and OA cuts them). Therefore the back bearing A0 equals ($180^{\circ} + \alpha$), Fig.2.4 i.e. the bearing of OA, plus 180°.

Thus in the whole circle bearing system, the back bearing of a line is obtained by applying 180°, adding where the forward bearing is less than 180°, and subtracting, where it is more than 180°. In the quadrantal system it is only necessary to change the cardinal points by substituting N for S and E for W, and vice versa.

2-2.5 Local Attraction

The free flowing, well balanced magnetic needle will not come to rest in the magnetic meridian, if there is any material such as mineral deposits, particularly magnetite (Fe_3O_4), in the locality. The proximity of steel structures, rails, current-carrying electric cables, iron fences and pipes, lamp posts, machetes and felling axes and even, chains and bunches of keys in your pockets may produce undesirable interference.

Thus, a bearing read from a compass under the influence of such attractive forces in the vicinity, may originate from a direction other than that of the magnetic meridian, and the error in the bearing is proportional to the magnitude of the attractive force. *Local attraction* is the term given to both the cause and effect of this error.

- **Detection of local attraction**
  Observe the bearing of each line in both directions. If the forward and back bearings differ by 180°, the inference is that there is no local attraction at either of the stations and that in the absence of other instrumental errors, the observation is correct so far as the reading is concerned.

  If the two bearings do not agree within permissible error of reading, check the forward bearing, since the discrepancy may have been caused by avoidable attractions such as chains, bunches of keys, machetes, felling axes, etc carried by persons around. If the discrepancy still exists, the presence of local attraction, at one or both stations, is suggested. (NB. An allowable error of between 1° - 2° may be a guide).

- **Elimination of effects of local attraction**

There are two ways of eliminating the effects of local attraction:-

- As stated above, the magnetic bearing of a line read from a compass under the influence of local attraction originates from a direction other than that of the magnetic meridian passing through the point.
- Since all bearings read from the station will originate from the same direction, the error will be the same for each bearing read at the station, and therefore subtracting one erroneous bearing from the other eliminates the error and gives the correct included angles between the various directions at the station.

To obtain the corrected bearings, select a line whose bearing is unaffected by local attraction, i.e. a line whose forward and back bearing differ by exactly 180°. To the bearing of this line, apply the deduced included angles to obtain the corrected bearings of all the other lines, as follows:

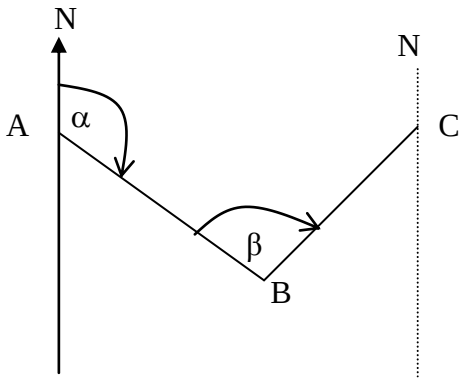


Figure 29: Relationship between back and forward bearings and included angle

Let α be the bearing of the line AB, figure 29 free from local attraction. Let β be the deduced included angle at B. Then the bearing of BC is the back bearing of AB plus the deduced included angle at B, i.e. bearing of

$$\mathbf{BC = (\alpha + 180^{o}) + \beta.}$$

Subtract 360° if the sum exceeds 360°.

In the second method, the magnitude of the effect of local attraction at each station is determined and a corresponding correction is applied to each bearing read at the station.

Table 1: Removing local attraction from an observation

Line	Observed Bearing	Correction	Corrected Bearing
AE	319 º 00´	-2º 30´	316 º 30´
AB	72º 45´	-2º 30´	70º 15´
BA	252º 00´	-1º 45´	250º 15´
BC	349º 00´	-1º 45´	347º 15´
CB	167º 15´	0º	167º 15´
CD	298º 30´	0º	298º 30´
DC	118º 30´	0º	118º 30´
DE	229º 00´	0º	229º 00´
ED	48º 00´	+1°	49º 00´
EA	135º 30´	+1°	136º 30´

In the above, Table 1, the forward and back bearings observed round a figure ABCDEA are entered in the 'observed bearing' column. As in the first method, we examine the figures to find a line whose bearing is unaffected by local attraction. We find that the bearing of the line CD and its back bearing differ by exactly 180 °. We take it that there is no local attraction at the two stations, C and D and therefore all bearings read from C and D must be correct. Hence the correction to the bearing of CD, CB, DC and DE is zero. If the bearing of CB is correct, then we would expect the bearing of BC to be (167º 15´+ 180°), i.e. 347º 15´, but the observed bearing is 349°,

This figure, which indicates the presence of some local attraction at station B, exceeds the expected figure by 1º 45´, i.e. the error in the bearing is +1º 45´, and the correction is -1º 45´. We therefore subtract 1º 45´from all bearings read from station B. Hence the observed bearing BA gets a correction of -1º 45´and the corrected bearing BA is 250º 15´.

Similarly, if there were no local attraction at station A, we would expect the bearing AB to be (250º 15´ - 180º), i.e. 70º15´. The observed bearing is 72º 45´, indicating an error of +2º 30´due to local attraction. The correction to all bearings read form the station A is - 2º 30´; hence the corrected bearing of AE is 316 º 30´. Following these deductions, we find that the bearings read at station E are in error by -1º, due to local attraction and the correction to bearings EA and ED is +1º respectively. In this way the effect of local attraction is removed and we obtain the corrected bearings.

2-2.6 Traversing

The cardinal principle of 'working from whole to part' demands that for any survey project we establish a series of control points to cover the entire area of the survey. The relative positions of these points, constitute a framework of reference points, are determined and used to serve as a base for detail survey or setting out works. The measurements are made to such a degree of accuracy that the measurements for the location of detail or setting out can be made to a lower degree of accuracy, with the certainty that the errors in these measurements will be contained by the control framework.

One of the methods of establishing such control frameworks is traversing. In traversing, a series of control points (stations), each being intervisible with the adjacent stations, are carefully selected and fixed. The lines joining these points (stations) are known as traverse lines and the measurements required are:-

- the angles between successive lines (or the bearing of each line), and
- the length of each traverse line, or leg.

A traverse may be described by the shape of the figure that the lines form. If the figure formed by the traverse lines closes at a station, i.e. if they form a polygon, or if the traverse starts and ends at points of known positions (coordinates) the traverse is described as a *closed traverse*, the former being distinguished as a *closed loop traverse* while the latter is a *closed line traverse*. A traverse which starts on a known point, say A, and ends at a point E, which is not already coordinated, is known as an *unclosed, or hanging, traverse***.**

Each type has its particular uses, but a closed traverse is the more satisfactory figure, since it is the easier one to apply corrections for the errors which invariably occur in the measurements. The closed line traverse however has the advantage over a closed loop traverse in that errors in the finishing coordinates and bearings as well as in the scale of the distance measurements are revealed. A closed traverse may be used for the frame work of surveys for township, housing or factory sites and for the determination of the perimeters of lakes, etc. An unclosed traverse can be carried out when the survey is comparatively long and narrow, such as that required for a trunk sewer, pipeline, main trunk roads or rail construction. (However, where the length is great, consideration should be given to the possibility of tying in to national survey pillars).

2-2.7 Booking

Below, in Figure 30, is a sample booking sheet for compass surveying. All procedures concerning booking of survey records such as no erasure, no copying etc., must be observed. Booking of details in compass surveying is the same as that of chain surveying.

Note! It is crucial in surveying that a sketch of the area accompanies the booking.

Job Title…….. Job No…….. Compass No. ……..
Name of Surveyor:……. Date…….. Location. …….

Line	**Ob. Bearing**		**Correction**		**Corrected Bearing**		**Distance**	**Remarks**
	FB	**BB**	**FB**	**BB**	**FB**	**BB**		

Figure 30: a sample booking sheet for compass surveying

2-2.8 Plotting

Compass traverses, like theodolite traverses, can, after the elimination of local attraction from the bearings, be computed and adjusted, and the coordinates used in plotting the survey. However, in general, for the accuracy of compass observations and the types of survey on which they are employed, the survey is plotted direct form the field measurements, after eliminating local attraction. Since the measurements are not without random errors, it generally happens that a plot of a closed loop compass traverse will not end on the starting point.

The following procedure could be followed in plotting your compass survey:

- Select a suitable scale and spot to start your plotting (forward bearing and distance of 1st line)
- Draw and label magnetic North and true north at starting point.
- Place a circular protractor with its centre at starting point and its zero line up with the reference direction and then mark on the paper at the protractor's edge the required bearing of the line.

- Remove the protractor, draw line and scale off the required distance to fix next station.
- Draw a magnetic North or south at B parallel to that at A and repeat the whole process.
- In most cases in a loop compass traverse, there is error of closure. Thus there is a small gap between the starting and end points. Since this represent the same point, this error must be corrected.

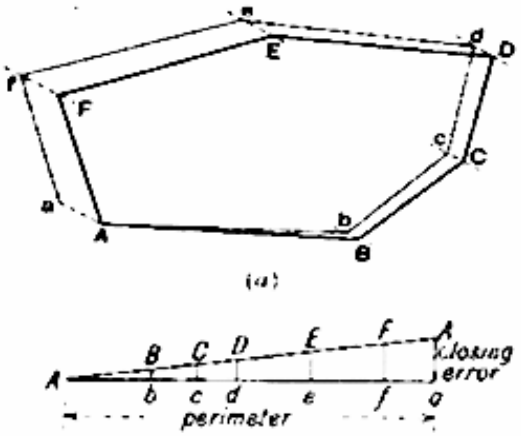


Figure 31: (a), graphical adjustment of a compass traverse (b) error diagram

For example, in Figure 31 (a), closed loop traverse ABCDEFA plots as abcdefa. Aa then represents the closing error. This can be adjusted graphically according to the Bowditch rule as follows:-

Set out, to scale, a line Aa, Figure 31(b), equal to the total length of the traverse, and mark off the distances Ab, bc,, cd, ef, fa equal to the lengths of the different legs. At a, draw aA´

perperndicular to Aa, and equal to th closing error. Join AA´ and at b, c, d, e and f draw perpendiculars bB, cC, dD, eE and fF to meet the line AA´ in B C D E and F respectively. These perpendiculars then represents proportions of the closing error up to that point, and the graphical adjustment is as follows: Draw lines through b,c,d,e, and f parallel to Aa, Figure 31 (a) and along these lines set out distances bB, cC, dD, eE, and fF equal to the corresponding perpendiculars above, and ABCDEFA then represents the adjusted plot of the traverse.

2-2.9 Advantages of Compass surveying

- Compass measures the bearing with respect to the magnetic meridian at each station and not the angle. No backsight need be taken.
- Small obstacles can be passed merely by placing the instrument beyond the obstacle as nearly on line as can be estimated.
- When blunder is caused in determining the direction of a traverse cause it affects the direction of none of the following courses.
- Local attraction affect all readings taken from a station equally
- It is easier and cheaper to use



Self-Assessment

1. Convert the following quadrant bearings into whole circle bearing
   a. N 4º 30´W
   b. S 60º 30´E
   c. S79º 10´W
   d. N 68º 15´E
2. The following whole circle bearings are known in a certain area.
   a. 164º 30´ b. 350º 50´ c. 220º 10´ d. 64º 15´

At a certain time the magnetic declination was found to be 10º 45´ west. Calculate the magnetic whole circle bearings of the lines and convert them to magnetic quadrant bearings.

Learning Track Activities

Select a suitable area (there should be at least one building, road or footpath, electricity pole etc). Survey, detail and plot the area using a suitable scale, using Chain surveying
And compass surveying methods.

Summary

Unit Summary

This unit has discussed simple mapping techniques using the principles of Chain and compass surveying. Among other things this unit students;

- Studied how to plan, carryout and plot a survey of small areas using chain and compass surveying
- Differentiated between horizontal and slope distance measurements and how to convert one to the other.
- Discussed various obstructions and how they can be overcomed during surveying operations
- Studied various reference directions and their differences
- Described the equipment required to carryout chain and compass surveying
- Detected and eliminated local attraction from compass observations
- Plotted and adjusted a compass traverse graphically

Key terms

a. Step chaining
b. Slope distance
c. Horizontal distance
d. Standardisation
e. Sag
f. Abney level
g. Cross staff
h. Reference meridian

Discussion questions

Discuss briefly the following terms:

a. Magnetic declination
b. Local Attraction
c. Magnetic variation
d. Grid Meridian
e. Astronomic meridian

Unit Assignment – 2

1. Calculate the back bearing corresponding to the following whole circle bearings.

a. 001º 30´ b. 179º 30´ c. 181º 00´ d. 359º 15´ e. 091º 10´

Convert the back bearing into true quadrant bearing given that the magnetic declination is 9º 45´east.

2. Distinguish between a closed loop traverse and a closed line traverse. What are their relative merits?

3. The following compass traverse was carried out in the vicinity of overhead power lines

Line	**Whole Circle bearing**		**Distance (m)**
	Forward	**Back**	
PQ	30º 00´	205º 00´	32
QR	97º 00	272º 00	75
RS	164º 00	346º 00	33
ST	235º 00	55º 00	21
TU	261º 00	81º 00	88

Calculate the correct bearing and plot the survey to a scale of 1:500.

Unit 3

LEVELLING, CONTOURING AND THEODOLITE SURVEYING

Introduction

In unit two, we discussed how to determine the relative positions of points, however, since the earth surface is undulating; the definition of these points would not be complete without a height dimension. Whenever two or more points are not at the same height, they are said to have a "difference of level". This difference can be measured in many ways; this unit discusses how to determine the difference of level using levelling instruments. Levelling is the procedure used to determine the relative heights of points above or below a plane of reference called a datum. In unit one it was learnt that the end result of a survey is the production of a map or plan which is the horizontal projection of the area under survey. Representation of the vertical dimension on the plan is therefore not easy task; but one way of doing this is by the use of contours which would be discussed in this unit. Contour lines are lines drawn on a plan showing points of the same elevation above or below a datum. Contours, thus satisfies the requirement of the vertical dimension being represented in such a manner that the form of the ground can easily be interpreted. This unit concludes with a brief introduction of theodolite surveying which is a more accurate way of mapping an area.

Learning objectives

This unit seeks to equip the student to be able to determine relative heights of positions and to represent such heights on a map or plan in the form of contours. Specifically at the end of the study, students should be able to

- Use the levelling instrument or theodolite to collect field data
- Record, reduce and apply corrections to corrected field data
- Determine height differences between remote points
- Generate contours from reduced field data
- Differentiate between direct and indirect contouring
- Deduce includes angles and bearings from theodolite data

Unit content

Session 3-1: Levelling and Contouring
3-1.1 Levelling
3-1.2 Contouring
3-1.3 Sectioning

Session 3-2: Theodolite Surveying
3-2.1 Description of Theodolite
3-2.2 Setting Up the Theodolite
3-2.3 Measurement of Horizontal and Vertical Angles (Uren & Price, 1994)
3-2.4 Booking and Calculating Angles
3-2.5 Plotting and Adjustment of Theodolite Surveys

SESSION 1-3: LEVELLING AND CONTOURING

3-1.1 Levelling

Levelling is the means by which a comparison of the heights of points on the earth's surface may be made. In other words, levelling is the procedure used to determine differences in elevation between points that are remote from each other.

3-1.1.1 Basic definitions

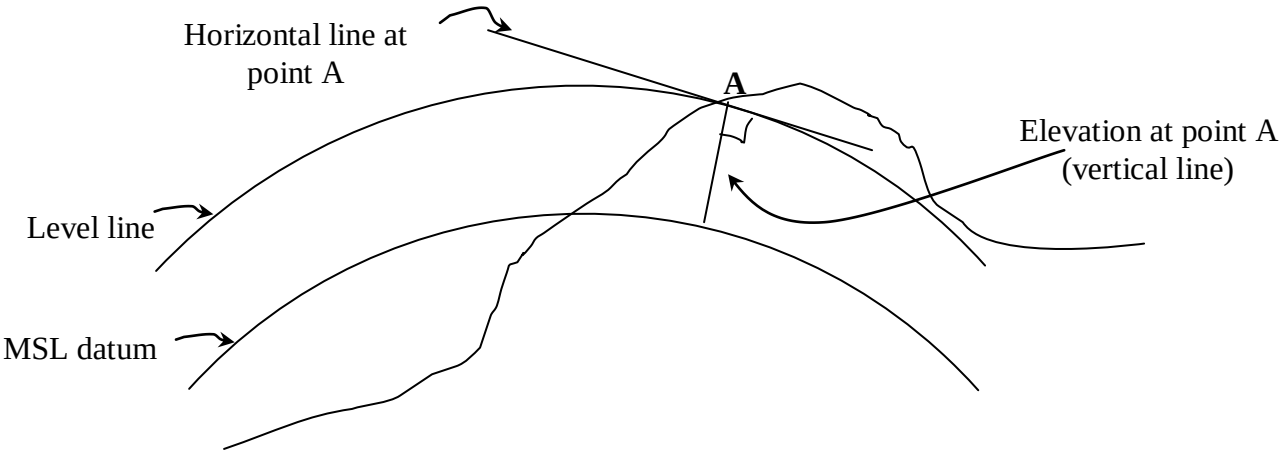


Figure 32: Levelling concepts

- **Vertical line**

A *vertical line* is a line from the surface of the earth to the earth's centre. It is also referred to as a *plumb line* or a *line of gravity* because the vertical line is parallel to the direction of gravity.

At a particular point, say **A** it is the direction assumed by the plumb-bob string if the plumb-bob is allowed to swing freely. Due to the earth's curvature, plumb-bob lines at points some distance apart are not parallel, but in plane surveying they are assumed to be parallel. See Figure 32.

- **Horizontal line**

  A horizontal line is a straight line perpendicular to a vertical line. See Figure 32.

- **Level line**

  A level line is a curved line in a level surface. See Figure 32.

- **Level surface**

  A level surface is a curved surface parallel to the mean surface of the earth. That is, a level surface is a surface of constant elevation that is perpendicular to a plumb line at every point. A level surface is best visualized as the surface of a large body of water at rest. See Figure 32.

- **Elevation**

  An elevation is a vertical distance above or below a reference datum. In surveying, the reference datum that is universally employed in levelling is that of the mean sea level (MSL)

3-1.1.2 Curvature and refraction

It is essential to understand the nature of the earth's curvature and atmospheric refraction as they affect levelling operations.

The definition of a level surface indicates that it is parallel to the curvature of the earth. A line of constant elevation, termed a *level line*, is likewise a curved line and is everywhere normal to the plumb line. However, a *horizontal line of sight* through the surveyor's telescope is perpendicular to the plumb line only at the point of observation.

- **Curvature**

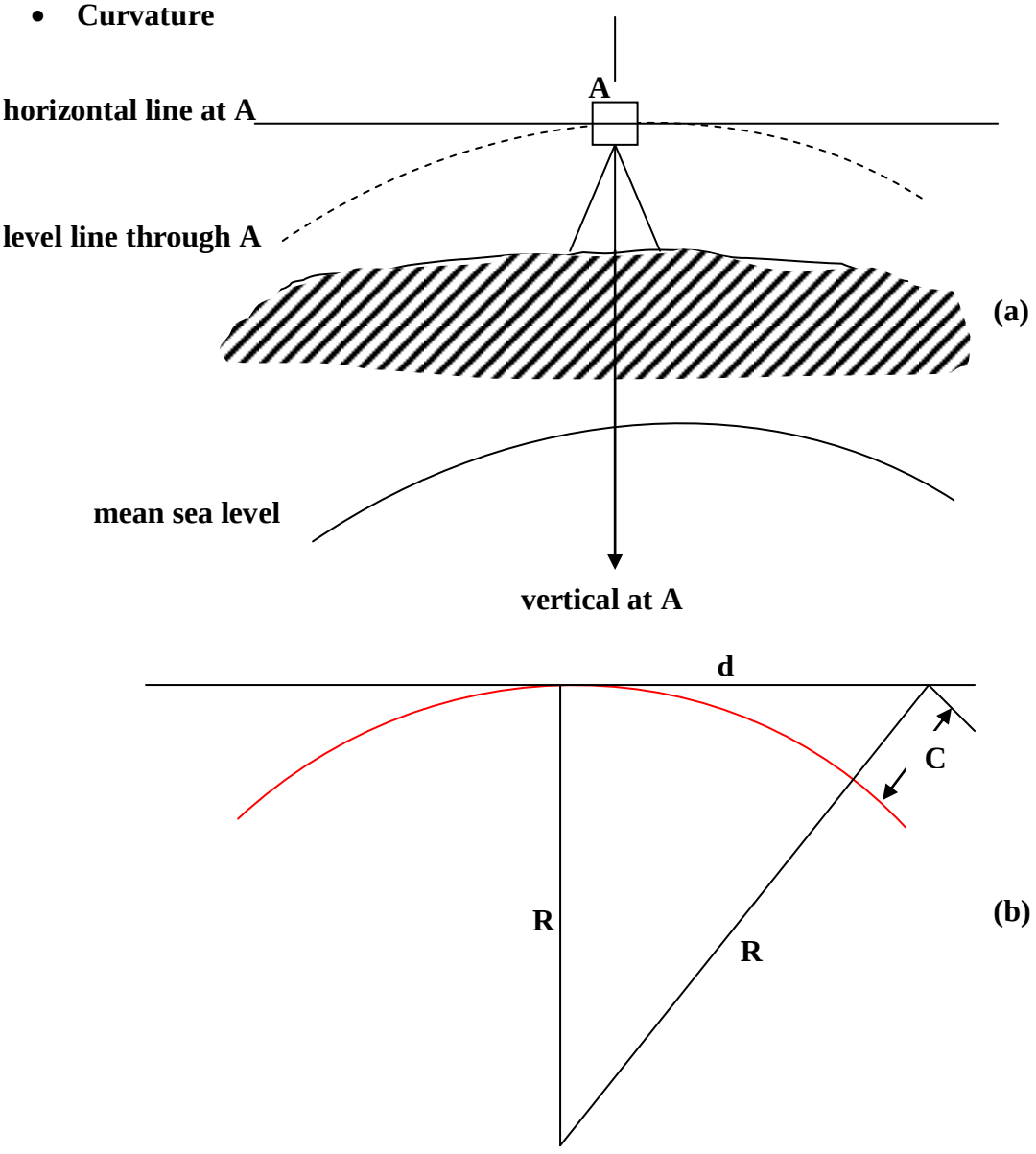


Figure 33: Earth's curvature, (a) and (b)

Figure 33 shows a section passing through the earth's centre. It can be seen that the level and horizontal lines through the instrument diverge. This divergence between the level line and horizontal line over a specified distance is known as the curvature, **C**. This is caused by level

lines following the curvature of the earth which is defined by the mean sea level are observed along the horizontal line instead of the level line.

- **Refraction**

  When considering the divergence of between level and horizontal line, one must also account for the fact that all sight lines are refracted downward by the earth's atmospheric conditions. That is, the effect of atmospheric refraction on a line of sight is to bend it towards the earth's surface causing staff readings to be too low.

Note: the combined effects of curvature and refraction are negligible when undertaking levelling where the sighting length is less than 120 metres. However, if longer sight length must be used, the effects of curvature and refraction will cancel out if the sight lengths are equal.

3-1.1.3 Description of the level instruments

The basic principle of a level is as follows:

The basic instrument used to measure differences of elevation is the level. Many design types exist, however, a level essentially consists of a telescope for sighting and levelling device for maintaining the line of sight in a horizontal position. A level is constructed such that the line of sight of the telescope is perpendicular to its vertical axis.

After a level has been properly set up on a tripod, its vertical axis should coincide with the direction of gravity and the line of sight should be horizontal. As the telescope is rotated about its vertical axis the line of sight moves in the horizontal plane.

Full description of the level instruments will be done during field practice. However, the basic difference in the various instruments will be discussed.

Some terms associated with the level instruments are:

- A level instrument set at a point or station establishes a horizontal line of sight. The telescope provides the accurate line of sight.
- A *bubble tube* is used in many levels to establish horizontal line
- A *circular bubble vial* is used in many modern levels to approximately set a horizontal plane.
- *Three 'foot screws'* are used to level the bubble in the circular bubble vial.
- *Line of collimation* or *line of sight*: the object lens, focusing lens, diaphragm and eyepiece are all mounted on the same optical axis and the imaginary line passing through the centre of the cross hairs and the optical centre of the object lens is called the line of collimation or the line of sight. See Figure 34.

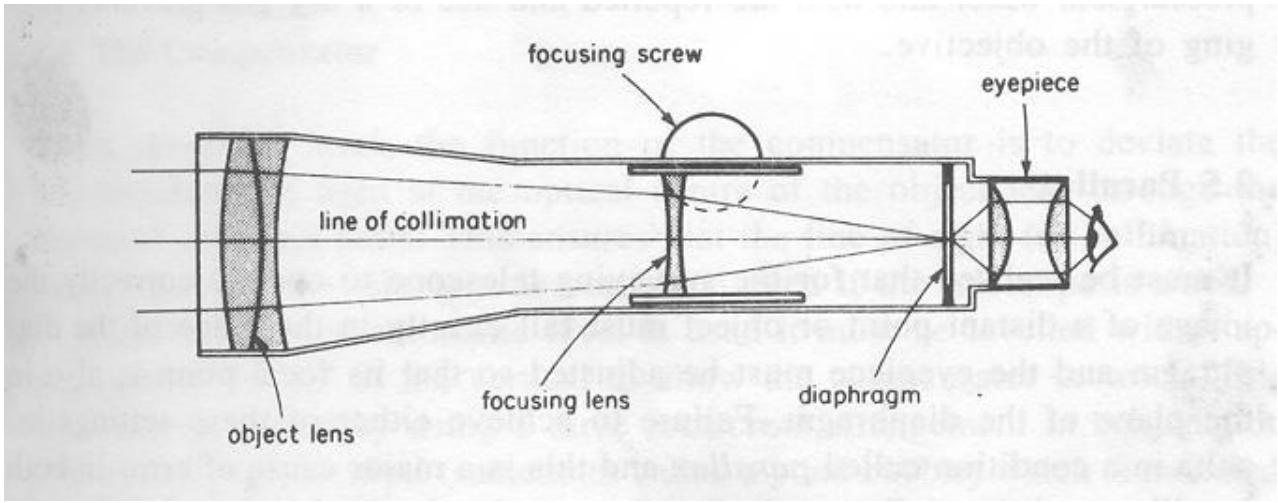


Figure 34: The object lens, focusing lens, diaphragm and eyepiece

- **Dumpy level**

  A dumpy level has the telescope rigidly attached to a horizontal bar that houses the level tube.

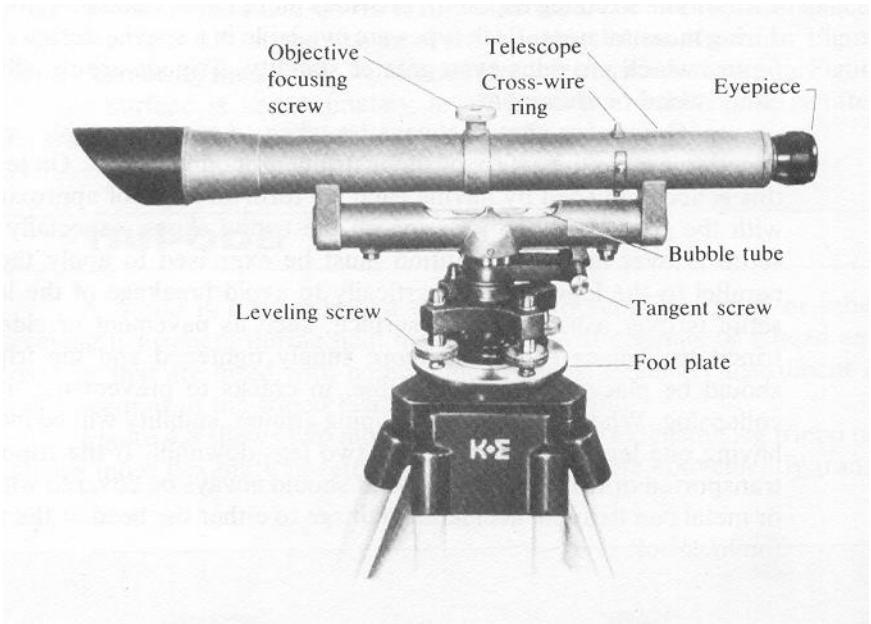


Figure 35: Dumpy level

The dumpy level spindle revolves in the socket of a levelling head controlled in position by four levelling screws. The levelling head is screwed to a tripod. At the lower end of the spindle is a ball-and-socket joint, which permits a flexible connection between the instrument proper and the foot plate. Hence when the foot screws are manipulated, the level is moved about this point as a centre. Figure 35 shows a dumpy level with an internal focusing telescope providing an erect image. The telescope is rigidly attached to a horizontal bar that houses the level tube. The line of sight of the instrument is the line fixed by the intersection of the cross-wires and the centre of the objective lens. When the instrument is in proper adjustment, the line of sight is parallel to the axis of the level tube.

- **Tilting level**

  A tilting level has a telescope that can be tilted about a horizontal axis. This design enables the operator to quickly and accurately centre the bubble and thus bring the line of sight into a horizontal plane.

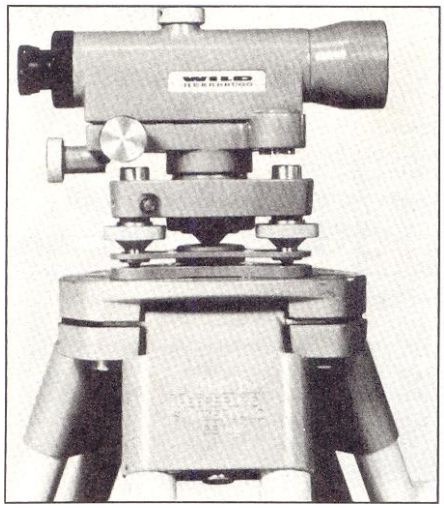


Figure 36: Tilting level

Figure 36 shows a level instrument of this type. A tilting screw is provided to raise or lower the eyepiece end of the telescope. The bubble tube is enclosed in a metal cover. A split image of the two opposite ends of the bubble is viewed optically through a small microscope situated next to the eyepiece of the telescope. When the bubble is off-centred, a split image of the two ends of the bubble is seen through the viewing microscope. As the bubble moves in its tube, the images of the two parts of the bubble move in opposite directions. When the bubble is correctly centred, the two images coincide to form a continuous U-shaped curve. This arrangement permits the accurate centring of the bubble from the normal observing position of the operator. There is no need to walk about the level in order to view the bubble.

Such a level instrument is first approximately levelled by means of a bull's-eye level. The split bubble is then brought into coincidence by turning the tilting screw. Once coincidence is achieved, the line of sight should be in a horizontal position.

The critical difference between a dumpy and tilting level:

- The essential difference between the levels is that the telescope of the dumpy level is rigidly attached to the vertical spindle and can be rotated only in azimuth (i.e. horizontally). The telescope of the tilting level is supported on a central pivot and can therefore be moved both in azimuth and altitude (i.e. vertically). The latter movement is very limited but the fact that it is possible gives the tilting level a great advantage over the dumpy.

- **Automatic level**

  The distinctive feature of this type of level is an internal compensator that automatically makes horizontal the line of sight and maintains it in that position through the application of the force of gravity. As soon as the instrument is levelled by means of a circular bubble, the movable component of the compensator swings free to a position that makes the line of sight horizontal. For the best performance of the compensator, which operates within a limited angular range from the horizontal, it is always essential to accurately centre the circular bubble and to maintain it in good adjustment.



Figure 37: Automatic level

Automatic or self-levelling level instruments are accurate, fast, and easy to use. Their employment substantially increases the productivity of a levelling party, and they have been found to be acceptable for the highest grades of levelling work. See Figure 37.

- **Tripod**

  A tripod is a three-legged stand used to support a level or other surveying instrument during field measurements. It consists of a head and three legs that are fitted with pointed metal shoes. The survey instrument is usually secured to the tripod head by a thread bolt. The tripod comes in two models: the extension leg tripod and fixed leg tripod. The latter is becoming obsolete. See Figure 38.

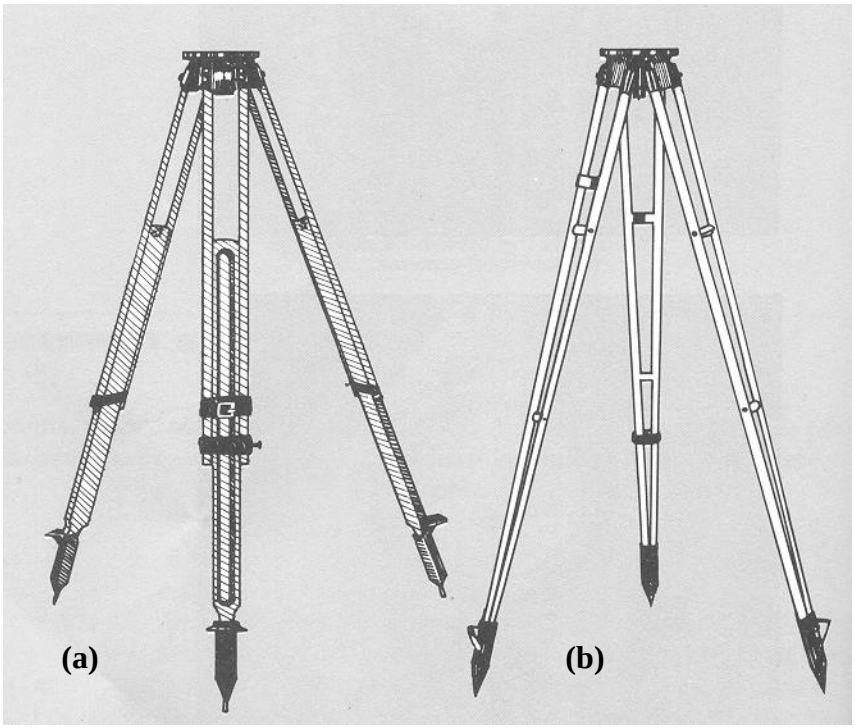


Figure 38: Tripod: (a) Extension leg tripod and (b) Fixed leg tripod

- **Levelling staff**

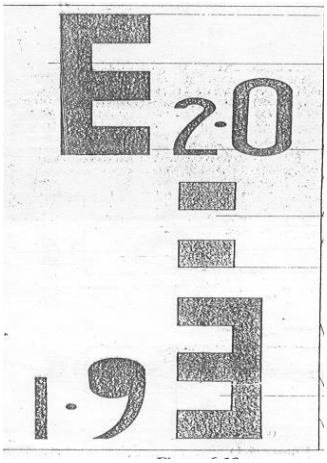


Figure 39:

The levelling staff enables distances to be measured vertically above or blow points on which it is held relative to a line of collimation used. Many a type of staff are in current use and these can have lengths between 2 and 4 m. 4m is the normal length for a staff. Different colours are used to show the graduation marks in alternate metres. The most common colours are: black and red on white background. The levelling staff can be read correctly to 0.01m, with estimation to 0.001m

3-1.1.4 Fieldwork in levelling

- **Making a reading**

The staff holder holds the staff on the mark, say **A**, and ensures that it is held perfectly vertically facing towards the instrument. The observer directs the telescope towards the staff and using the focusing screw on the side of the telescope brings the staff clearly into focus. The reading on the staff is then taken. The staff holder is then directed to the next staff station, say **B**, where the reading procedure repeated. See figure 40.

- **Levelling between two points**

This is the basis for all levelling work no matter how complex the particular levelling may be.

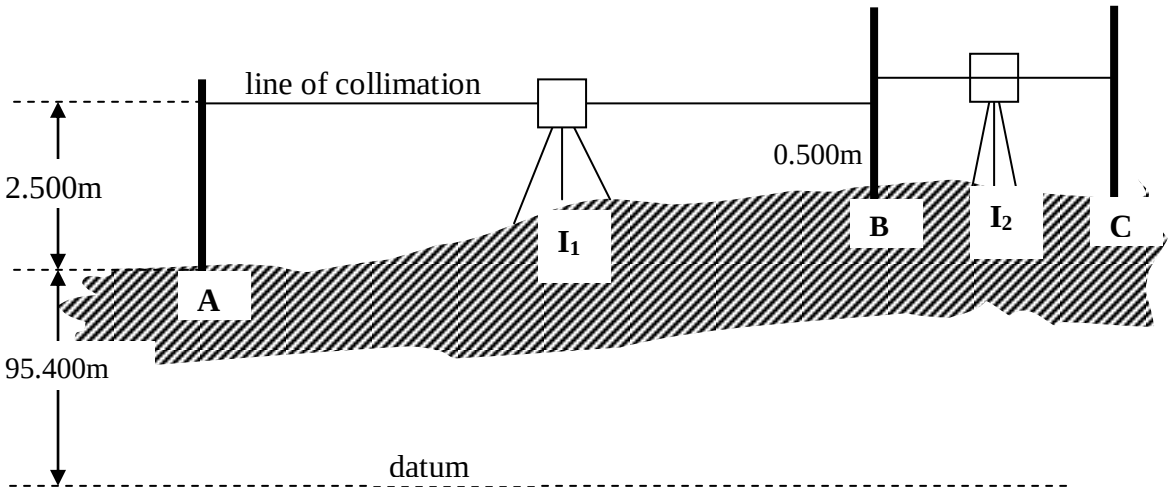


Figure 40: Principles of levelling

Referring to figure 40, the direction of levelling is from A to B and to C:

- **A** is known as the *back sight* **(BS)** and **B** is known as the *fore sight* **(FS)** if level instrument is at $\mathbf{I_1}$
- **B** is known as the *back sight* and **C** is known as the *fore sight* if level instrument is at $\mathbf{I_2}$**.** On and on it goes

- Point **B** is called a *change point* (CP) since at point **B** both back sight and fore sight readings were taken.
- In practice, a BS is the first reading taken after the instrument has been set up and is always to a point of known or calculated reduced level. Conversely, a FS is the last reading taken before the instrument is moved. Any readings taken between the BS and FS from the same instrument position are known as *intermediate sights* (IS).
- 2.500m and 0.500m are the respective readings observed on the staff at stations **A** and **B.** In all cases, the back-station is observed on a known reduced level (RL).

Back-sight (BS) = 2.500m
Foresight (FS) = 0.500m
If RL at A = RL_A = 95.400m
Then RL at B = RL_B = 95.400 + (2.500 - 0.500)
= 97.400m

In general, the reduced level is given by:

$$\mathbf{RL_B = RL_A + (BS - FS)}$$

1-3.1.5 Field booking and reduction of levels

Figure 41 will be used in demonstrating the field bookings and the reduction of the levels. Two methods of booking and reduction of levels will be discussed, i.e. the rise and fall method and the height of collimation method.

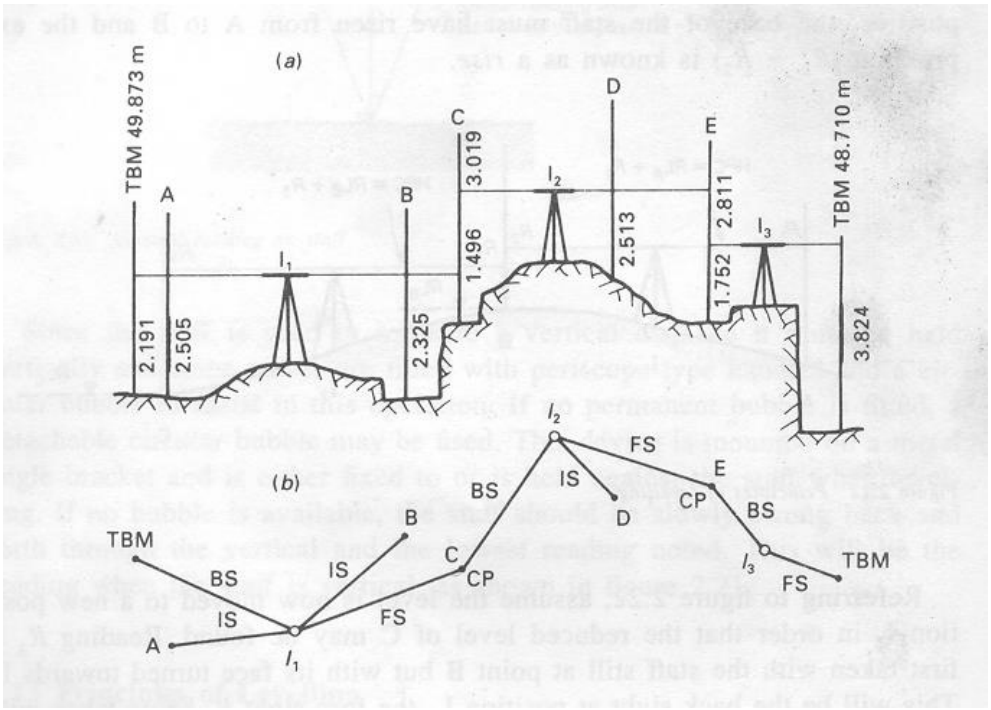


Figure 41: Levelling and booking sequence

- **Rise and fall method**

Table 2: Rise and fall method

BS	IS	FS	Rise	Fall	Initial RL	Adjustment	Adjusted RL	Remarks

2.191					49.873	0	49.873	TBM 49.873
	2.505			0.314	49.559	+0.002	49.561	A
	2.325		0.180		49.739	+0.002	49.741	B
3.019		1.496	0.829		50.568	+0.002	50.570	C(CP)
	2.513		0.506		51.074	+0.004	51.078	D
1.752		2.811		0.298	50.776	+0.004	50.780	E(CP)
		3.824		2.072	48.704	+0.006	48.710	TBM 48.710
6.962		8.131	1.515	2.684	48.704			
8.131			2.684		49.873			
-1.169			-1.169		-1.169			

Checks:

Checks to be performed for the rise and fall method are as follows:

- $\sum$ (BS) - $\sum$ (FS) = $\sum$(Rises) -$\sum$(Falls)
  = Last RL – First RL

Adjustment:

Adjustments to be performed for the rise and fall method are as follows:

- The difference between the calculated and known values of the RL of the final Bench Mark (BM) is -0.006m. This is known as the *misclosure.*
- Since the misclosure is -0.006m, it implies that the total adjustment is +0.006m. this must be distributed to all the RL's
- Since there three stations, +0.002m (+0.006/2) is added to the reduced levels found from each instrument position. These cumulative adjustment are done in Table 2.

- **Height of Collimation method.**
  The basic idea here is to determine the height of collimation (HPC) for all the instrument stations (change points). This is then used in calculating the RL's of the various stations.

Table 3: Height of Collimation method.

BS	**IS**	**FS**	**HPC**	**Initial RL**	**Adjustment**	**Adjusted RL**	**Remarks**
2.191			52.064	49.873	0	49.873	TBM 49.873
	2.505			49.559	+0.002	49.561	A
	2.325			49.739	+0.002	49.741	B
3.019		1.496	53.587	50.568	+0.002	50.570	C(CP)
	2.513			51.074	+0.004	51.078	D
1.752		2.811	52.528	50.776	+0.004	50.780	E(CP)
		3.824		48.704	+0.006	48.710	TBM 48.710
6.962	7.343	8.131	106.115	300.420			
8.131				7.343 + 8.131 + 300.420 = 315.894			
-1.169				(52.064 x 3) + (53.587 x 2) + 52.528 = 315.894			

6.962	48.704
8.131	49.873
-1.169	-1.169

Calculation for HPC's and RL's:
Calculations to be performed for the height of collimation method are as follows:

- For instrument station I_1, HPC = 49.873 + 2.191 = 52.064m
- RL of A = 52.064 – 2.505 = 49.559m
- RL of B = 52.064 – 2.325 = 49.739m
- RL of A = 52.064 – 1.496 = 50.568m
- For instrument station I_2, HPC = 50.568 + 3.019 = 53.587m
- The same procedure is continued for the other stations.

Checks:
Checks to be performed for the height of collimation method are as follows:

- $\sum$ (BS) - $\sum$ (FS) = Last RL – First RL
- $\sum$ (IS) + $\sum$ (FS) + $\sum$(RL's except first) = $\sum$(each HPC x number of applications)

The critical difference between rise and fall and height of collimation methods:

- The rise and fall method is quicker to reduce where a lot of back sights and fore sights have been taken and very few intermediate sights taken. For this reason, the rise and fall method tends to be used when establishing control when no intermediate sights would normally be taken.
- The collimation method is quicker to reduce where a lot of intermediate sights have been taken since fewer calculations are required and it is a good method to use when setting out levels where, usually, many readings are taken from each instrument position. A disadvantage of this method is that the check can be lengthy.

1-3.1.6 Precision of levelling

For levelling, the allowable misclosure is given by;

$$\textit{Allowable misclosure} = \pm 5\sqrt{n} \text{ mm}$$, where n is the number of instrument positions.

When the actual and allowable misclosures are compared and it is found that the actual value is greater than the allowable value, the levelling should be repeated. If, however, the actual value is less than the allowable value, the misclosure should be distributed equally between the instruments positions as already described.

1-3.1.7 Inverted staff

Frequently the reduced levels of points above the height of instrument are required. For example: the soffit of a bridge or underpass; the underside of a canopy; the levels of roof, caves, buildings etc. Generally, these points will be above the line of collimation. To obtain the reduced levels of such points, the staff is held upside down in an inverted position with its base on the elevated point.

When booking an inverted staff reading it is entered in the levelling table with a minus sign. The calculation proceeds in the normal way, taking the minus sign into account.

1-3.1.8 Errors in levelling

Errors in levelling can be classified under the following headings:

1. **Gross errors:** These are mistakes arising as a result of carelessness, inexperience or fatigue. Examples of these in levelling are:
    a. **Wrong staff reading.** For example reading the wrong metre value
    b. **Wrong booking.** For example, interchanging figures
    c. **Omission or wrong entry.** For example, writing staff reading in the wrong column.
    d. **Spirit level not centred.** For example, if the staff is read without centring the bubble.

    These errors can be eliminated by double levelling. Also readings should be re-mentioned back to the observer by the recorder.

2. **Constant errors:** these are instrumental errors and will always be of the same sign. Examples of these in levelling are:
    a. **Non-verticality of the staff**: this error can be eliminated by fitting the staff with a circular spirit level. The staff holder then ensures that the bubble is always centred when the staff is being read. An alternative method is for the staff holder to swing the staff slowly backwards and forwards across the vertical position during observation. The observer then reads the lowest reading.
    b. **Collimation error in the instrument:** arises if the line of collimation is not perfectly horizontal although the bubble of the spirit level is central. Such error is eliminated by making back sights and fore sights equal in length.
    c. Parallax: parallax is detected by moving the eye to different parts of the eyepiece when viewing distant object. If different parts of the object appear against the cross hairs then the telescope is has not been properly focussed and parallax is present.

    To remove parallax, the eyepiece is first adjusted while moving viewing a light background, for example the sky or a booking sheet, until the cross hairs appear in sharp focus. The distant point at which readings are required is now sighted and brought into focus and is viewed while moving the eye. If the object and cross hairs do not move relative to each other then parallax has been eliminated; if there exit apparent movement between then the procedures should be repeated.

3. **Random errors:** these are errors that are due to physical and climatic conditions. These resulting errors are small and are compensatory. Examples of these in levelling are:
    a. **Effect of wind and temperature**:
    b. **Soft and hard ground**:
    c. **Change point**: Firm spot must be chosen. For precise levelling, a base plate is used.
    d. **Human deficiencies**: errors arise in estimating the mm readings, particularly when visibility is bad or sights are long.

1-3.1.9 Uses of levelling

Levelling is principally needed in:

- Contouring
- Sectioning (*longitudinal* and *cross sections*) and
- Setting out

Contouring is discussed under 1-3.2, and sectioning is discussed under 1-3.3. Setting out is not discussed.

1-3.2 Contouring

Contour is by far the most extensively used method of showing relief features on a flat sheet of paper. This is because both the height and shape of the land surface can be specifically shown.

> *Contours are lines drawn on maps to join places of equal heights above or below sea level.* For example, on topographic map, a 500ft contour line means the line joins points on the land that are 500ft above sea level.

The level of the sea is taken as a stating point in all height measurements. The reason being that when the sea is calm has perfectly level surface. This level surface is known as mean or average sea level.

When planning a housing estate the builder will always try to take advantage of the natural slope of the ground. Houses built on different levels always give the estate a more interesting and pleasant appearance. Contouring is a method of depicting the undulation of the surface (i.e. the surface relief).

1-3.2.1 Contour line

Definition: A contour line is a line drawn on a plan joining all points of the same height above or below a datum. See figure 42.

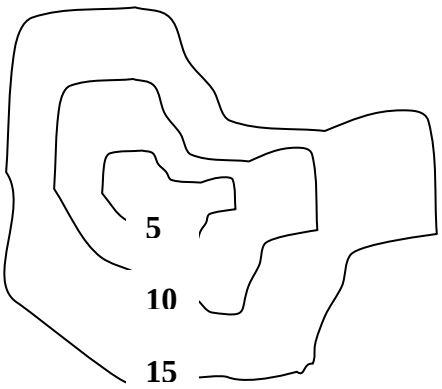


Figure 42: Contour line

1-3.2.2 Characteristics of contour lines

Contour lines are continuous lines and cannot meet or cross any other contour line nor can any line split or join any other line, except in cases of cliff.

Characteristically, contours are indicated by brown curved lines on maps. Every fifth contour line is thickened to facilitate easy reading. The interval between two contours, i.e. *vertical interval (VI)*, is constant on the same map.

Widely spaced contours indicate a gentle slope. Contours which are closer together indicate steep slopes and contours which are evenly spaced indicate uniform slope.

1-3.2.3 Contour Interval

The difference in height between successive contours is known as the *contour* or *vertical interval* and this interval dictates the accuracy to which the ground is represented. The value chosen for any application depends on:

- the intended use of the plan
- the scale of the plan
- the costs involved
- the nature of the terrain

The above and other dependants in the selection of contour interval are explained below:

1. **Scale of the map:** The scale of the map when large needs a small contour interval. When the scale of the map is small the contour interval should be large.
2. **Purpose of map (Map use requirements):** Land use requires large contour interval; dams require small contour interval; soil requires large contour interval because much information detail is not needed.
3. **Available information:** More information requires large contour interval and less information requires small contour interval.
4. **Nature of terrain/ or type of area:** Flat areas requires large contour interval; steep and valley areas require small contour interval.
5. **Survey process applied:** When levelling is used, the spot height interpolation is done manually which demands large contour interval. With Photogrammmetric methods, contour intervals can be automated hence small contour interval. But manually will demand large contour interval.
6. **Economics:** Smaller contour interval takes more time hence the cost is high; large contour interval takes less time hence the cost is less.

Generally, a small vertical interval of up to 1 m is required for engineering projects, for large-scale plans and for surveys on fairly even sites. In hilly or broken terrain and at small scales, a wider vertical interval is used. Very often, a compromise has to be reached on the value chosen since a smaller interval requires more fieldwork time, thus increasing the cost of the survey.

1-3.2.4 Gradients

The gradient of the ground between two points is given by:

$$\text{Gradient} = \frac{\text{Vertical Interval}}{\text{Horizontal Equivalent}}$$

For example, using Figure 43, the gradient of the ground between the points A and B is:

$$= \frac{AB}{BC} = \frac{\text{Vertical Interval}}{\text{Horizontal Equivalent}}$$

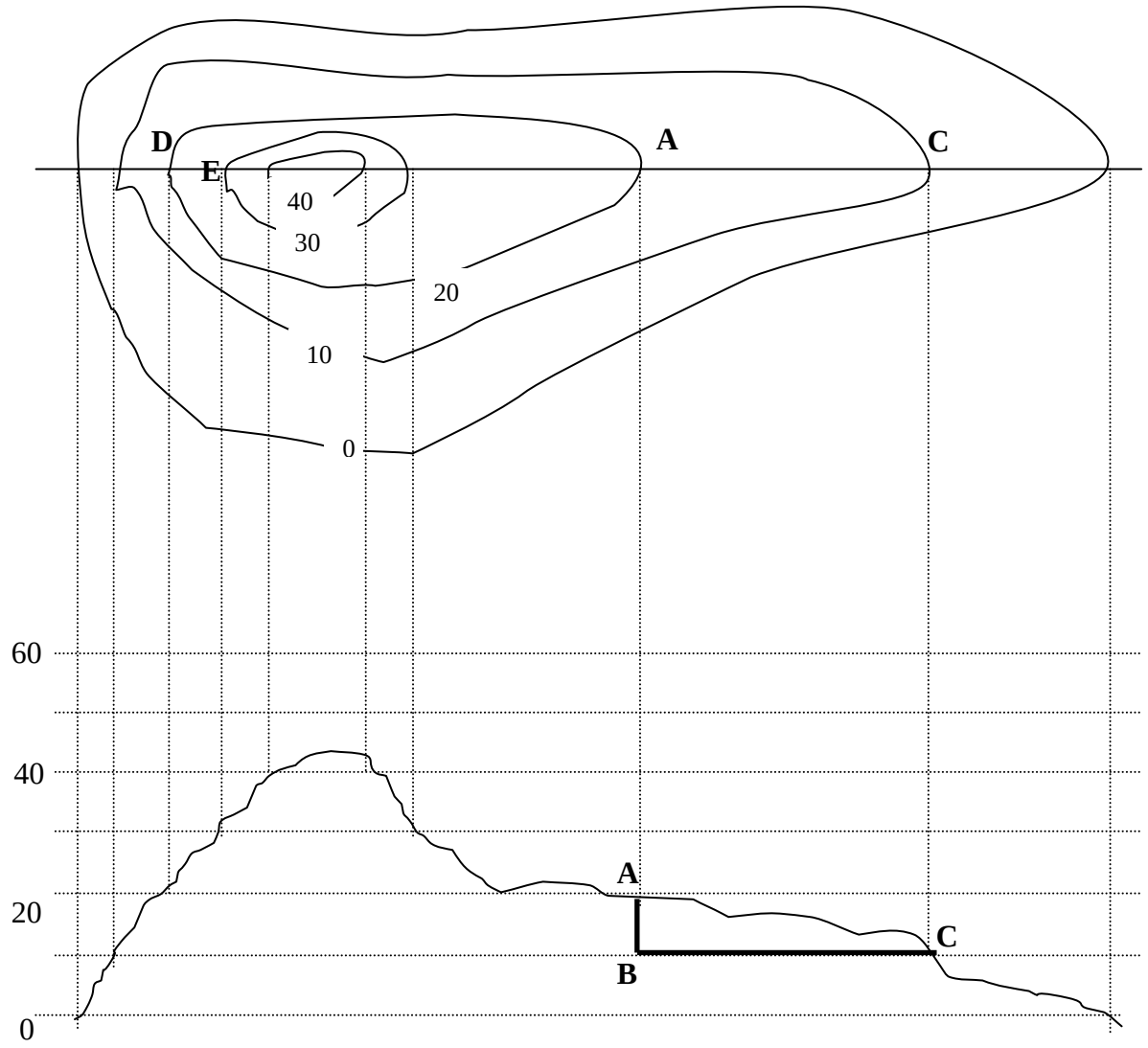


Figure 43: Gradient and profile

1-3.2.5 Contours: Merits and Demerits

Merits

1. They show accurately the heights of places with reference of a nearest contour.
2. Contours show the form or shape of the land. Important features such as ridges, valleys and spurs can clearly be shown.
3. Contours may combine effectively with other methods like trigonometrical points and spot heights to show relief.
4. Cultural or human features like roads, settlements, towns etc. as well as physical features like vegetation, streams etc. can be superimposed on contoured map. This may further facilitate the study of the relationships between physical and human features.

Demerits

1. The exact heights between two contours cannot be determined accurately. It is often done by approximations.

2. The process of contouring itself is laborious. Contouring as a process is not only time-consuming but also costly to undertake.
3. Contouring alone may not be effective in depicting certain features like cliffs or scarps, which require other symbols like bench mark.

1-3.2.6 Contours: Plotting

Contours can be plotted directly or indirectly.

- **Direct contouring**:

  In this method, the position of particular contour is located on the ground and marked using a level instrument. Using this method, the contour lines are physically followed on the ground. Two distinct operations can be used in this method:

  a. **Levelling**
     The height of collimation (= reduced level at observed station + backsight reading at observed station) of the levelling instrument is first of all determined. Staff-man holds the staff facing the instrument and backs slowly until prompted by observer for a particular reading on the staff. Pegs are positioned are the various prompted points which will indicate a particular contour line.
  b. **Survey of the pegs**
     Here the plan positions of the pegs are established to allow plotting to take place. On a smaller site, chain lines and offset can be used to survey these pegs. On a larger site, compass, Theodolite or tacheometric traverse can be employed. The plan positions of the contours are then plotted directly onto the site plan and smooth curves drawn through them.

- **Indirect contouring**:

  When using this method, no attempt is made to follow the contour lines. Instead, a series of spot levels is taken and the contour positions are interpolated. Three distinct operations are involves:

  a. **Setting out grid**
     In this method, the area to be contoured is carefully divided into a series of lines that form squares. The interval of the grid lines depends on the contour interval.

     The area to be contoured is divided into a series of lines forming squares and ground levels are taken at the intersection of the grid lines. The sides of the squares can vary from 5 to 30 m, the actual figure depending on the accuracy required and on the nature of the ground surface. The more irregular the ground surface the greater the concentration of grid points.

  b. **Levelling**

     Levels are taken at the intersections of the grid lines. The reduced level of every point on the grid is obtained.

     To obtain the ground level at each grid point the person holding the staff lines the staff in with the two ranging rods in each direction that intersect at the point being levelled, and a reading is taken. The procedure is repeated at all grid points.

  c. **Interpolating the contours**

This is done *mathematically* or g*raphically*. With the mathematical method, the positions of the contours are interpolated mathematically from the reduced level by simple proportions. The height difference between each spot height is calculated and used with the horizontal distance between them to calculate the position on the line joining the spot heights at which the required contour is located

1-3.2.7 Use of contour maps

Contour maps are used in obtaining *sections* (i.e. cross sections and longitudinal sections). For example, it is possible to use contours to obtain sectional information for use in the initial planning of such projects as roads, pipelines, earthworks and reservoirs. These sections are used in earthworks.

1-3.3 Sectioning

Sectioning is usually undertaken for construction work such as road works, railways and pipelines. Two types of section are often necessary and these are *longitudinal* and *cross sections.*

1-3.3.1 Longitudinal sections

In surveying, a longitudinal section (or profile) is taken along the complete length of the proposed centre line of the construction showing the existing ground level. Levelling can be used to measure heights at points on the centre line so that the profile can be plotted.
Generally, this type of section provides data for determining the most economic formation level, this being the level to which existing ground is formed by construction methods. The fieldwork in longitudinal sectioning normally involves two operations.

- Firstly, the centre line of the section must be set out on the ground and marked with pegs. For most works, this is done by theodolite and some form of distance measurement so that pegs are placed at regular intervals (frequently 20 m) along the centre line.
- Secondly as soon as the centre line has been established levelling can commence.

For longitudinal sections, it is usually sufficiently accurate to record readings to the nearest 0.01 m. Levels are taken at the following points, the object being to survey the ground profile as accurately as possible:

1. At the top and ground level of each centre line peg noting the through chainage of the peg.
2. At points on the centre line at which the ground slope changes.
3. Where features cross the centre line, such as fences, hedges, roads, pavements, ditches and so on. At points where, for example, roads or pavements cross the centre line, levels should be taken at the top and bottom of kerbs. At ditches and streams, the levels at the top and bottom of any banks as well as bed levels are required.
4. Where necessary, inverted staff readings to underpasses and bridge soffits would be taken.

In order to be able to plot levels obtained in addition to those taken at the centre line pegs; the position of each extra point on the centre line must be known. These distances are recorded by measurement with a tape, the tape being positioned horizontally between appropriate centre line pegs. The method of booking longitudinal sections should always be by the height of collimation method since many intermediate sights will be taken. Distances denoting chainage should be recorded for each level and most commercially available level books have

a special column for this purpose. Careful booking is required to ensure that each level is entered in the level book with the correct chainage. Good use should be made of the 'remarks' column in this type of levelling so that each point can be clearly identified when plotting.

1-3.3.2 Cross sections

A longitudinal section provides information only along the centre line of a proposed project. For works such as sewers or pipelines, which usually are only of a narrow extent in the form of a trench cut along the surveyed centre line, a longitudinal section provides sufficient data for the construction to be planned and carried out. However, in the construction of other projects such as roads and railways, existing ground level information at right angles to the centre line is required. Taking cross sections provides this. These are sections taken at right angles to the centre line such that information is obtained over the full width of the proposed construction.

For the best possible accuracy in sectioning a cross-section should be taken at every point levelled on the longitudinal section. Since this would involve a considerable amount of fieldwork, this rule is generally not observed and cross-sections are, instead, taken at regular intervals along the centre line usually where pegs have been established. A right angle is set out at each cross-section either by eye for short lengths or by theodolite for long distances or where greater accuracy is needed. A ranging rod is placed on either side of the centre line to mark each cross-section.

The longitudinal section and the cross-sections are usually levelled in the same operation. Starting at a temporal benchmark (TBM), levels are taken at each centre line peg and at intervals along each cross-section. These intervals may be regular, for example, 6m, 20 m, 30 m on either side of the centre line peg or, where the ground is undulating, levels should be taken at all changes of slope such that a good representation of existing ground level is obtained over the full width of the construction. The process is continued taking both longitudinal and cross-section levels in the one operation and the levelling is finally closed on another known point. Such a line of levels can be very long and can involve many staff readings and it is possible for errors to occur at stages in the procedure. The result is that if a large miss-closure is found all the levelling will have to be repeated, often a soul destroying task. Therefore, to provide regular checks on the levelling it is good practice to include points of known height such as traverse stations at regular intervals in the line of levels and then, if a large discrepancy is found.

1-3.3.3 Drawing cross-sections and profiles

A map view looks at the surface of the earth from overhead. Contour maps use the contour lines to represent the third dimension of elevation. Another view of the earth is a profile view. A profile or cross section shows a cut through the earth. The top line on the cross section or profile represents the surface of the earth. We can apply this alteration of perspective to contour maps. From the information provided by the contour map, we can produce a cut across this surface into the earth and, thus, show a side view like a silhouette or skyline. Specifically, this illustration is called a *topographic profile* (Figure 44).

A topographic profile is a diagram that shows the change of elevation of the land surface along a given line. As indicated above, it represents graphically the skyline viewed from a distance. The *vertical* scale is the scale used to plot the elevation. It is usually larger than the horizontal or map scale, *exaggerated*, in order to emphasize the difference in the relief. The *maximum relief* is the difference in elevation between the highest and lowest points.

Following are the steps for drawing a topographic profile:

- Lay the edge of a strip of paper along the line between the starting and ending points for the profile.
- Mark on the edge of the strip the EXACT places where each CONTOUR, STREAM, and HILLTOP crosses this line.
- Label these marks with the elevation and correct identification.
- Mark any important other features such as bottoms of depressions or landmarks to be included.
- If a graph is not provided, construct the horizontal line for your profile of the SAME LENGTH as your profile (unless a different horizontal scale is to be used for the profile.) Generally, the same horizontal scale is used. Prepare the VERTICAL SCALE by lightly drawing lines parallel to your horizontal base line on the proper scale for each of the elevations to be represented. Label these lines with the correct elevations starting one or two intervals below the lowest elevation that will be plotted (lowest elevation on the profile). Thus, the side represents a kind of graphic scale.
- Place the edge of the strip of paper with the labelled contour lines at the bottom of the profile base line and project each contour and feature to the horizontal line of the same elevation. Put a small dot at the intersection of these two lines.
- Connect all of the points with a smooth line being careful to show all hilltops at the proper height and all valleys and depressions at their correct approximate values.

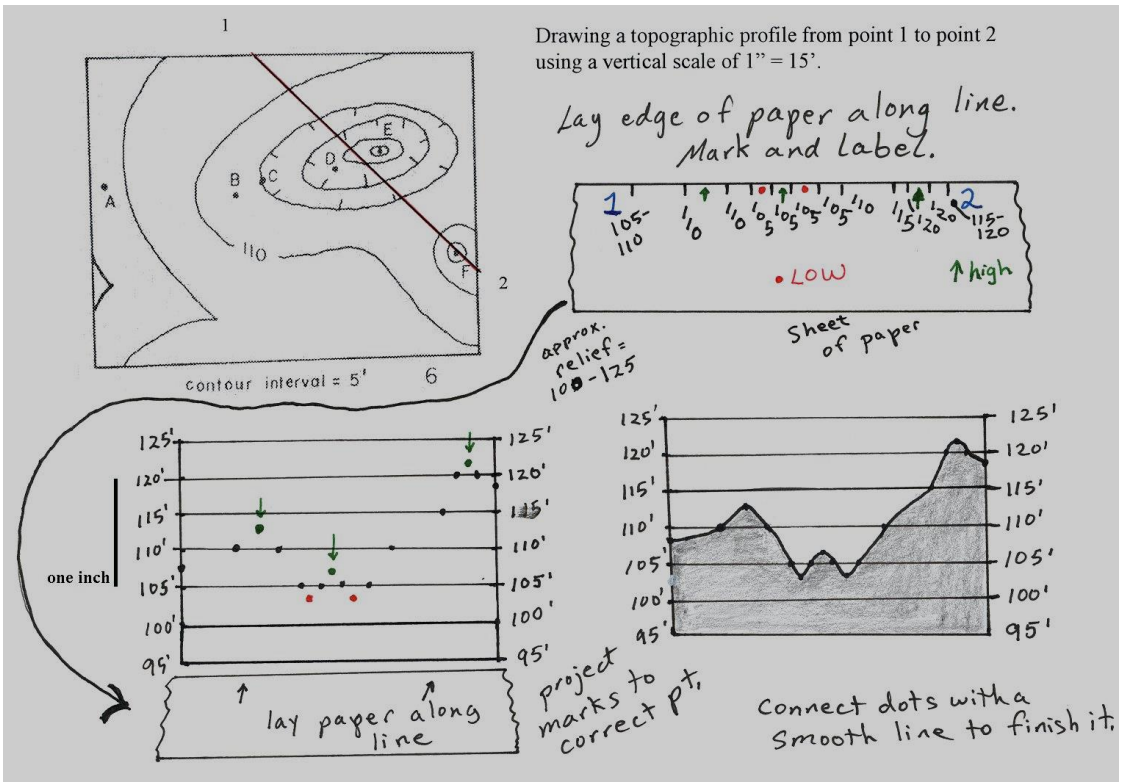


Figure 44: Drawing a topographic profile

Self-Assessment

1. Explain the following expressions with respect to the level instrument:
   i) Line of sight

ii) Line of Collimation
iii) Collimation error

How would you ascertain that your level instrument has no collimation error?

2. Write short notes about the following:
   - A level surface
   - The mean sea level as a reference surface
   - Height of collimation
   - Turning point

3. a) Define the following terms:
   i) Foresight
   ii) Backsight

   b) Explain why it is desirable that foresight and backsight distances be equal

   c) In levelling from a bench mark 106.010m above datum and closing onto anothe benchmark 100.770m above datum, staff readings were taken in the following order.

   1.134, 2.502, 1.497, 2.540, 2.219, 0.271
   1.301, 0.695 1.190 1.130 2.813

   The instrument was moved immediately after taking the 2nd, 5th and 7th reading Show how these readings should be booked and reduced by the 'rise and fall' metho with all the arithmetical checks and the closing error.

SESSION 2-3: THEODOLITE SURVEYING

2-3.1 Description of Theodolite

Theodolites are precision instruments used extensively in construction work for measuring angles in the horizontal and vertical planes. Many different theodolites are available for measuring angles and they are classified according to the smallest reading that can be taken with the instrument. For example, a 1" theodolite is one which can be read to 1" directly without estimation.

The components of the theodolite are:

- Vertical circle:
- Horizontal circle
- Telescope
- Eyepiece:
- Circle reading eyepiece.
- Reflecting mirror for the reading system
- Standard:
- Plate level
- Circular bubble
- Upper plate (or alidade): it carries the horizontal circle reading system.
- Lower plate: this carries the horizontal circle.
- Lower plate tangent screw
- Lower plate clamp

- Tribrach: it is the body of the instrument carrying all other parts. It has a hollow, slightly conically shaped socket into which fits the remainder of the instrument.
- Trivet: it forms the base of the instrument and in order to be able to attach the theodolite to the tripod. The trivet also caries the feet of three *threaded levelling footscrews.*
- Levelling footscrews
- Optical plummet
- Telescope clamp
- Upper plate tangent screw and clamp

2-3.2 Setting Up The Theodolite

The sequence of operations required to prepare the theodolite instrument is as follows:

- Setting the tripod
- Mounting the instrument
- Centring: that is, centring the theodolite over the ground mark
- Levelling: that is, centralising the bubble by levelling the plate spirit level.
- Removing parallax: parallax is removed using the same principle described under levelling.

2-3.3 Measurement of Horizontal and Vertical Angles

Horizontal angles are measured with respect to the horizontal plane and vertical angles are measure with respect to the vertical plane. (Figure 45).

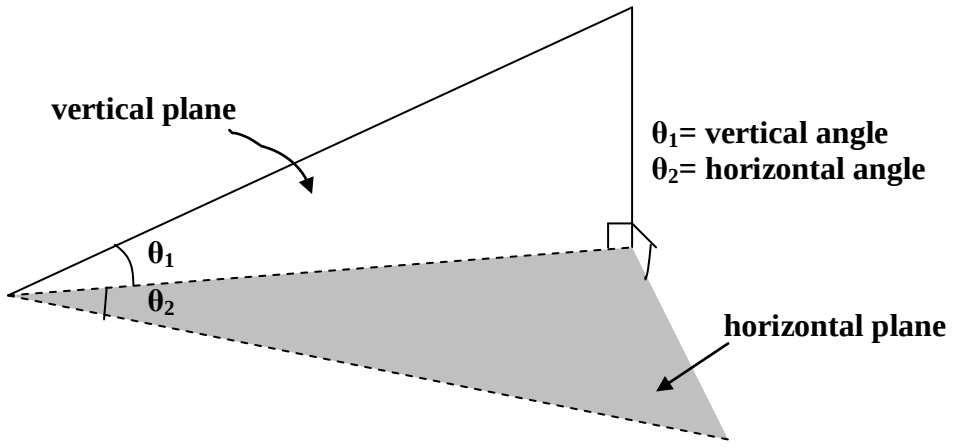


Figure 45: Horizontal and vertical angles

When measuring an angle whilst the theodolite is set over a survey mark and properly levelled, the theodolite can be used in two positions (or faces):

- Face Left (L)
- Face Right (R)

The theodolite can be swung in two directions:

- Swing Left (L) or anticlockwise
- Swing Right (R) or clockwise

Combining the face and swing we can have:

- LL meaning face left, swing left
- RR meaning face right, swing right

Measurements of the angles are illustrated in the field bookings in Table 4.

2-3.4 Booking And Calculating Angles

The horizontal angle booking for the theodolite is illustrated in Table 4. An example of the field diagram is also demonstrated in Figure 46.

Table 4: Horizontal angle booking

Instrument Station	Face & Swing	Back Station	Fore Station	Horizontal circle reading	Included Angle	Mean Included Angle
B	LL	A		00º00'00"		65º25'10"
	LL		C	65º25'00"	65º25'00"	
	RR		C	245º25'20"	65º25'20"	
	RR	A		180º00'00"		
B	LL	A		20º00'00"		65º25'10"
	LL		C	85º25'00"	65º25'00"	
	RR		C	265º25'20"	65º25'20"	
	RR	A		200º00'00"		

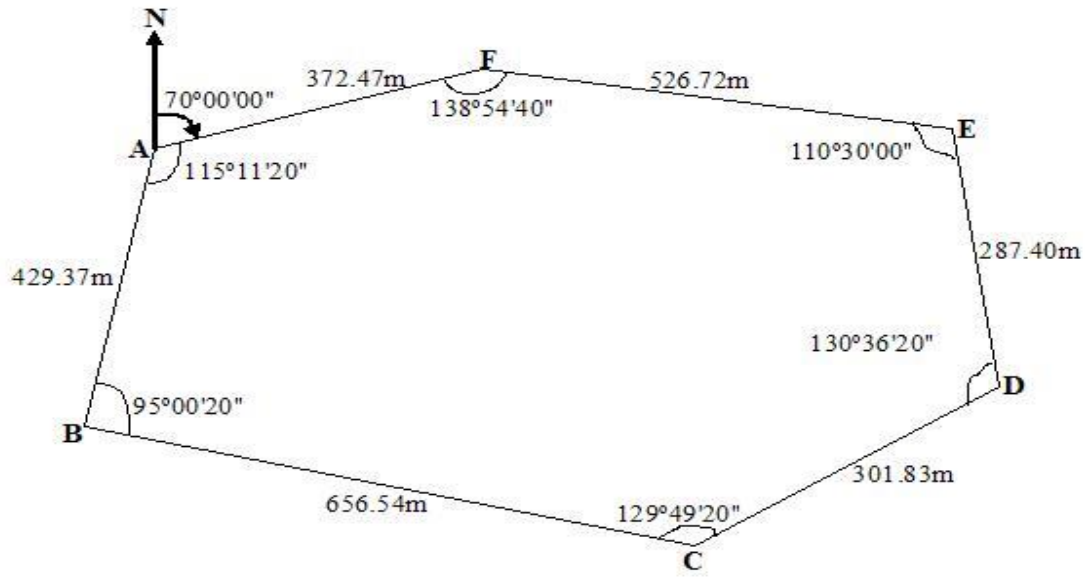
The horizontal angle and vertical angle bookings for the theodolite are illustrated in Table 5.

Table 5: Horizontal angle and vertical angle bookings

Instrument Station	Face & Swing	Back Station	Fore Station	Horizontal circle reading	Included Angle	Mean Included Angle	Vertical Angle	Remarks
B	LL	A		00º00'00"		65º25'10"	4º22'15"	
	LL		C	65º25'00"	65º25'00"			
	RR		C	245º25'20"	65º25'20"			
	RR	A		180º00'00"				
B	LL	A		20º00'00"		65º25'10"	6º23'42"	
	LL		C	85º25'00"	65º25'00"			
	RR		C	265º25'20"	65º25'20"			
	RR	A		200º00'00"				

2-3.5 Plotting and Adjustment of Theodolite Surveys

An example will be used in illustrating the adjustment of the theodolite surveys. With this example, consideration is based on abstraction of angles and lengths being entered on to a traverse diagram.



Fi

gure 46: Traverse abstract

- **Step 1: Angular misclosure**

  The first step is to determine the angular misclosure of traverse. That is, to check that the observed theodolite angles sum to their required value. Checks will be considered for both polygon and link theodolite traverses.

  - **Check for polygon traverse:**

    The observed angles of a polygon traverse can be either internal or external angles.

    For internal angles: Sum of internal angles = (2*n* – 4) x 90º

    For external angles: Sum of external angles = (2*n* + 4) x 90º

    Where *n* is the number of angles measured.

  - **Check for link traverse:**

    Sum of angles = (final forward bearing – initial back bearing) + *m* x 180º

    Where, *m* depends on the nature of the traverse. *m* = (*n* – 1) or (*n* + 1), where n is the number of angles measured between the initial back bearing and final forward bearing.

  Considering the above example we will have the sum of angles as shown in Table 6:

Table 6: Sum of angles

Station	Observed angle
A	75º11’20’’
B	95º00’20’’
C	129º49’20’’
D	130º36’20’’

E	110º30’00’’
F	138º54’40’’
Sum	720º02’00’’

Required sum = (2*n* – 4) x 90º = (2 x 6 – 4) x 90º = 720º00’00’’

Hence misclosure = 720º02’00’’ - 720º00’00’’ = + 20’00’’

- **Step 2: Adjustment of angles**

  The angles are adjusted by distributing the misclosure over all the measured angles as in Table 7. For the above the adjustment per angle will be – (20’00’’)/6, since six (6) internal angles were measured:

Table 7: Adjustment of angles

Station	Observed angle	Adjustment	Adjusted angle
A	115º11’20’’	-20’’	115º11’00’’
B	95º00’20’’	-20’’	95º00’00’’
C	129º49’20’’	-20’’	129º49’00’’
D	130º36’20’’	-20’’	130º36’00’’
E	110º30’00’’	-20’’	110º29’40’’
F	138º54’40’’	-20’’	138º54’20’’
Sum	720º02’00’’	-20’00’’	720º00’00’’

- **Step 3: Conversion of angles to whole circle bearings**

  The conversion of the angles to bearings for the theodolite surveys is illustrated in figure 47. This is done because the angles measured by the theodolite are not bearings.

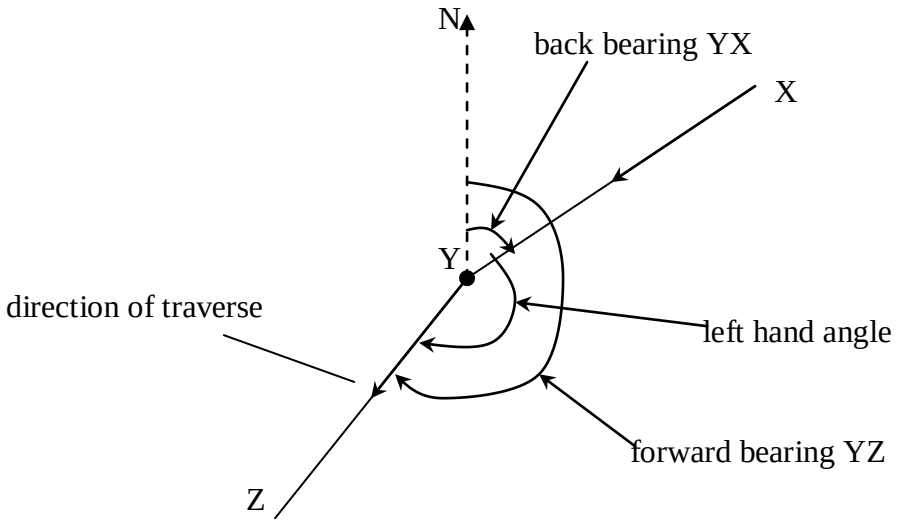


Figure 47: Whole circle bearing calculation (Source: Uren and Price 1994)

From figure 47:
Forward bearing = back bearing + left-hand angle.

Considering the above example:

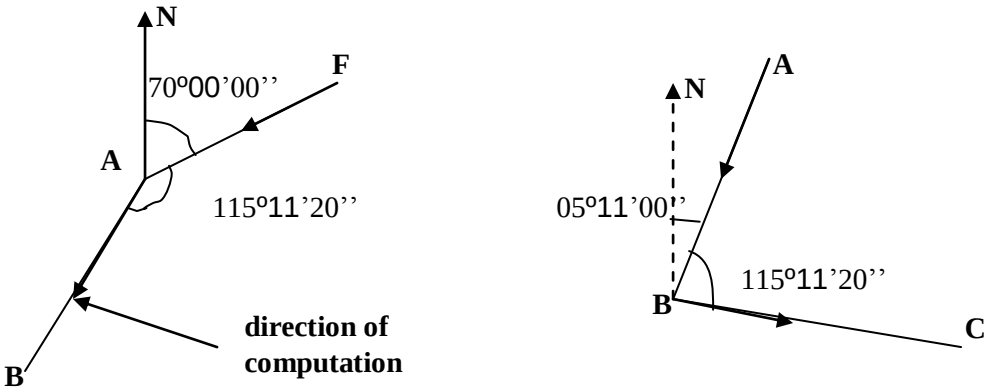


Figure 48: Conversion of angles to whole circle bearings

At station A:

forward bearing AB = back bearing AF + left-hand angle at A
= 70º00’00’’ (given) + 115º11’00’’
= 185º11’00’’

At station B:

forward bearing BC = back bearing BA + left-hand angle at B

but back bearing BA = forward bearing AB ± 180º
= 185º11’00’’ ± 180º
= 365º11’00’’ or 05º11’00’’
= 05º11’00’’ (to keep bearing in the range 0 to 360º)

hence forward bearing BC = 05º11’00’’ + 95º00’00’’
= 100º11’00’’

The bearings of all the lines can be computed in the same manner.

- **Step 4:** Determine the **partial coordinates**
- **Step 5:** Determine the **total coordinates**
- **Step 6:** Conduct **arithmetic check**
- **Step 7: Adjustment** of the survey
- **Step 8:** Determine the **closing error**
- **Step 9: Plot** the survey

Steps 4 to 9 are the same as the compass survey as discussed under 2-4.

Self-Assessment

1. The following staff readings were taken with a tilting level instrument on a line of levels in whic the instrument was moved forward after the 2nd 4th and 11th readings. The first reading was take on a bench mark whose elevation is 190.415m.

On a sample page of a level book, enter and reduce these readings, applying all the arithmetic check Reduce using both methods of reduction.

(1) 1.611	(2) 0.662	(3) 2.418	(4) 1.296	(5) 1.796
(6) 0.854	(7) 1.095	(8) 1.369	(9) 1.608	(10) 2.070
(11) 2.372	(12) 0.422	(13) 1.916		

2. Several angles have been measured at a station 'A'. The following data refer to the angles

Instrument Station	Target station	Face Left	Face Right	Mean Angle	Included Angle
A	B	00º 24' 40''	180º 24' 40''		
	C	65º 36' 00''	245º 36' 20''		
	D	132º 10' 20''	312º 10' 40''		
	E	247º 08' 00''	67º 08' 20''		

Calculate the mean observed angles BAC, CAD, DAE

Learning Track Activities

Summary

Unit Summary

In this unit has discussed the use of levelling for the determination of heights and also theodolite was used in angular measurements from which bearings were extracted. Basically in this unit;

- Levelling instrument or theodolite were used to collect field data
- Recording of levelling notes and theodolite traversing were discussed
- Field data were reduced and corrections where necessary were applied c
- Height differences between remote points were determined
- Contours were generate from reduced field data
- Direct and indirect contouring were discussed
- Angles and bearings were deduced from theodolite field data

Key terms

a. Level surface
b. Level line
c. Horizontal line
d. Vertical line
e. Line of collimation
f. Backsight
g. Foresight
h. Intermediary (intersight) sight
i. Change point
j. Datum
k. Contour interval
l. Spot height

Review Question

1. Make a list of errors which may arise in measuring heights using a levelling instrument(tilting level) use the following headings;
    a. Gross error
    b. Systematic error
    c. Random error

2. Describe briefly any method of measuring horizontal readings such that errors are minimised.

1.

Unit Assignment – 3

1. AB and FG are two survey lines with bearings fixed from a primary traverse as 39° 40′ 20″ and 36° 18′ 30″ respectively. These bearings are to be unaltered. A secondary traverse run between the above stations produced the following results

Angle	Mean observed value
ABC	179° 59′ 40″
BCD	210° 05′ 40″
CDE	149° 44′ 40″
DEF	177° 46′ 40″
EFG	179° 02′ 20″

Calculate the corrected bearing of each line.

Reading: a. http://www.civl.port.ac.uk/survey/

b. Irvine,, W. (1980). " Surveying for Construction". Second Edition. McGraw-Hill Book Company (UK) Limited.

c. Uren, J. and Price, W. F. (1994) Surveying for Engineers. Third edition. Macmillan Press ltd., London, UK.

Unit 4

GLOBAL POSITIONING SYSTEMS, AREAS AND VOLUMES

Introduction

Traditional Surveying has been greatly transformed by the introduction of Global positioning system in the mid 1980s. The Global Positioning System (GPS) is a location system based on a constellation of about 24 satellites orbiting the earth at altitudes of approximately 11,000 miles. GPS was developed by the United States Department of Defense (DOD), for its tremendous application as a military locating utility. However, over the past several years, GPS has proven to be a useful tool in non-military mapping applications as well.
GPS satellites are orbited high enough to avoid the problems associated with land based systems, yet can provide accurate positioning 24 hours a day, anywhere in the world. Uncorrected positions determined from GPS satellite signals produce accuracies in the range of 50 to 100 meters. When using a technique called differential correction, users can get positions accurate to within 5 meters or less. This unit introduces the reader to some basic information about GPS which is increasingly becoming a major source of data acquisition in Land surveying.

In unit one it was made clear that the end result of land surveying data capturing is to produce a map or a plan. However, plans intended for planning purposes or land ownership should also show the extent of land in question. The last session of this unit therefore deals with calculation of areas and volumes.

Learning objectives

This unit introduces readers to basic concepts for GPS data capturing and use in surveying. It also discusses the calculation from field notes, from plans and using coordinates. At the end of the study of this unit, students should;

- Understand the concept of GPS data capturing
- Know some factors that affect the accuracy of GPS data
- Differentiate different positioning systems
- Calculate areas from field notes, plans or coordinates
- Compare the accuracies of different methods of area computation

Unit content

Session 1-4: Global Positioning Systems (GPS)
1-4.1 Geopositioning – Basic Concepts
1-4.2 GPS – Components and Basic Facts
1-4.3 Components of a GPS
1-4.4 GPS Positioning Types
1-4.5 Accuracy of GPS
1-4.6 Different types of answers given by a GPS
1-4.7 Factors that Affect GPS
1-4.8 GPS Applications
1-4.9 Future of GPS Technology

Session 2-4: Computation of Areas
2-4.1 Areas from Field Notes
2-4.2 Areas enclosed by Irregular Sides
2-4.3 Measurement of Area by Planimeter
2-4.4 Calculation of Volumes

SESSION 1-4: GLOBAL POSITIONING SYSTEMS (GPS)

The Global Positioning System (GPS) is a burgeoning technology, which provides unequalled accuracy and flexibility of positioning for navigation, surveying and GIS data capture. The GPS NAVSTAR (Navigation Satellite timing and Ranging Global Positioning System) is a satellite-based navigation, timing and positioning system. The GPS provides continuous three-dimensional (X, Y, Z) positioning 24 hrs a day throughout the world. The technology seems to be beneficiary to the GPS user community in terms of obtaining accurate data up to about 100 meters for navigation, metre-level for mapping, and down to millimetre level for geodetic positioning. The GPS technology has tremendous amount of applications in GIS data collection, surveying, and mapping.

1-4.1 Geopositioning – Basic Concepts

By positioning we understand the determination of stationary or moving objects. These can be determined as follows:

1. In relation to a well-defined coordinate system, usually by three coordinate values and
2. In relation to other point, taking one point as the origin of a local coordinate system.

The first mode of positioning is known as point positioning, the second as relative positioning. If the object to be positioned is stationary, we term it as static positioning. When the object is moving, we call it kinematic positioning. Usually, the static positioning is used in surveying and the kinematic position in navigation.

1-4.2 GPS – Components and Basic Facts

The GPS uses satellites and computers to compute positions anywhere on earth. The GPS is based on satellite ranging. That means the position on the earth is determined by measuring the distance from a group of satellites in space. The basic principle behind GPS are really simple, even though the system employs some of the most high-tech equipment ever developed. In order to understand GPS basics, the system can be categorised into **FIVE logical Steps**

1. Triangulation from the satellite is the basis of the system.
2. To triangulate, the GPS measures the distance using the travel time of the radio message.
3. To measure travel time, the GPS need a very accurate clock.
4. Once the distance to a satellite is known, then we need to know where the satellite is in space.
5. As the GPS signal travels through the ionosphere and the earth's atmosphere, the signal is delayed.

To compute a positions in three dimensions. We need to have four satellite measurements. The GPS uses a trigonometric approach to calculate the positions, The GPS satellites are so high up that their orbits are very predictable and each of the satellites is equipped with a very accurate atomic clock.

1-4.3 Components of a GPS

The GPS is divided into three major components

- The Control Segment
- The Space Segments
- The User Segment

1-4.3.1 The Control Segment

The Control Segment consists of five monitoring stations (Colorado Springs, Ascesion Island, Diego Garcia, Hawaii, and Kwajalein Island). Three of the stations (Ascension, Diego Garcia, and Kwajalein) serve as uplink installations, capable of transmitting data to the satellites, including new ephemerides (satellite positions as a function of time), clock corrections, and other broadcast message data, while Colorado Springs serves as the master control station. The Control Segment is the sole responsibility of the DoD (Department of Defence) who undertakes construction, launching, maintenance, and virtually constant performance monitoring of all GPS satellites.

The DOD monitoring stations track all GPS signals for use in controlling the satellites and predicting their orbits. Meteorological data also are collected at the monitoring stations, permitting the most accurate evaluation of tropospheric delays of GPS signals. Satellite tracking data from the monitoring stations are transmitted to the master control station for processing. This processing involves the computation of satellite ephemerides and satellite clock corrections. The master station controls orbital corrections, when any satellite strays too far from its assigned position, and necessary repositioning to compensate for unhealthy (not fully functioning) satellites.

1-4.3.2 The Space Segment

The Space Segment consists of the Constellation of NAVASTAR earth orbiting satellites. The current Defence Department plan calls for a full constellation of 24 Block II satellites (21 operational and 3 in-orbit spares). The satellites are arrayed in 6 orbital planes, inclined 55 degrees to the equator. They orbit at altitudes of about 12000, miles each, with orbital periods of 12 sidereal hours (i.e., determined by or from the stars), or approximately one half of the earth's periods, approximately 12 hours of 3-D position fixes. The next block of satellites is called Block IIR, and they will provide improved reliability and have a capacity of ranging between satellites, which will increase the orbital accuracy. Each satellite contains four precise atomic clocks (Rubidium and Cesium standards) and has a microprocessor on board for limited self-monitoring and data processing. The satellites are equipped with thrusters which can be used to maintain or modify their orbits.

1-4.3.3 The User Segment

The user segment is a total user and supplier community, both civilian and military. The User Segment consists of all earth-based GPS receivers. Receivers vary greatly in size and complexity, though the basic design is rather simple. The typical receiver is composed of an antenna and preamplifier, radio signal microprocessor, control and display device, data recording unit, and power supply. The GPS receiver decodes the timing signals from the 'visible' satellites (four or more) and, having calculated their distances, computes its own latitude, longitude, elevation, and time. This is a continuous process and generally the position is updated on a second-by-second basis, output to the receiver display device and, if the receiver display device provides data capture capabilities, stored by the receiver-logging unit.

1-4.4 GPS Positioning Types

The GPS positioning types are described in the next sub-sections:

1-4.4.1 Absolute Positioning

This mode of positioning relies upon a single receiver station. It is also referred to as 'stand-alone' GPS, because, unlike differential positioning, ranging is carried out strictly between the satellite and the receiver station, not on a ground-based reference station that assists with the computation of error corrections. As a result, the positions derived in absolute mode are subject to the unmitigated errors inherent in satellite positioning. Overall accuracy of absolute positioning is considered to be no greater than 50 meters at best by Ackroyd and Lorimer and to be + 100 meter accuracy by the U.S. Army Corps of Engineers.

1-4.4.2 Differential Positioning

Relative or Differential GPS carries the triangulation principles one step further, with a second receiver at a known reference point. To further facilitate determination of a point's position, relative to the known earth surface point, this configuration demands collection of an error-correcting message from the reference receiver.

Differential-mode positioning relies upon an established control point. The reference station is placed on the control point, a triangulated position, the control point coordinate. This allows for a correction factor to be calculated and applied to other roving GPS units used in the same area and in the same time series. Inaccuracies in the control point's coordinate are directly additive to errors inherent in the satellite positioning process. Error corrections derived by the reference station vary rapidly, as the factors propagating position errors are not static over time. This error correction allows for a considerable amount of error of error to be negated, potentially as much as 90 percent

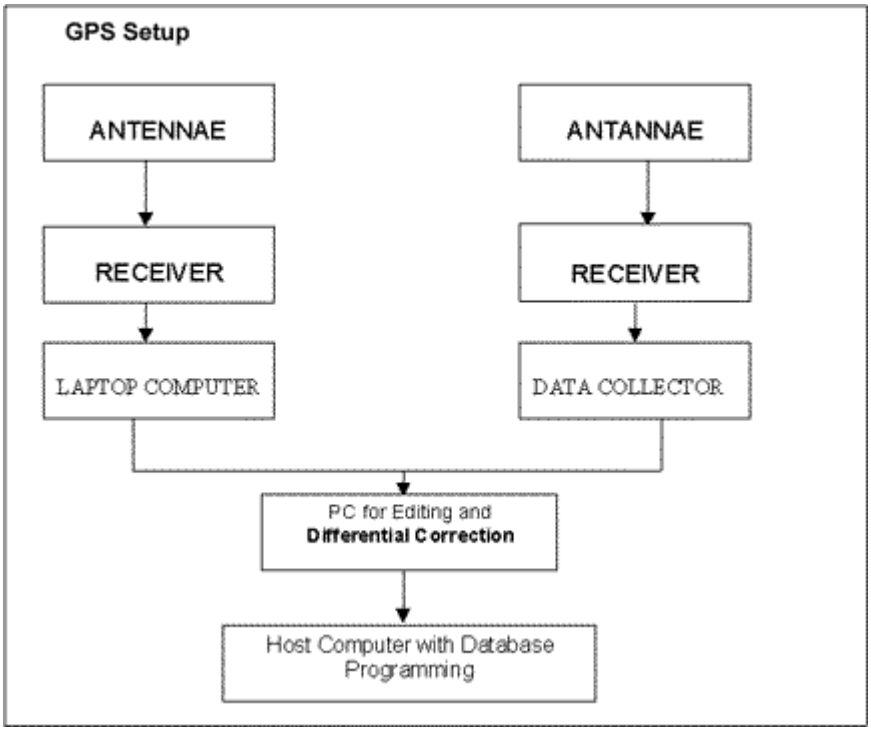


Figure 49: GPS setup

1-4.5 Accuracy of GPS

There are four basic levels of accuracy - or types of solutions - you can obtain with your real-time GPS mining system, shown in Table 8.

Table 8: Accuracy of GPS

Autonomous	Accuracy	15 - 100 meters
Differential GPS (DGPS)	Accuracy	0.5 - 5 meters
Real-Time Kinematic Float (RTK Float)	Accuracy	20cm - 1 meter
Real-Time Kinematic Fixed (RTK Fixed)	Accuracy	1cm - 5 cm

GPS satellites broadcast on three different frequencies, and each frequency (or career wave) has some information or codes on it. You can think of it as three different radio stations broadcasting several different programs. The table below (Table 9) lists the signals and the contents:

Table 9: GPS signals and contents:

L1 Career	L2 Career	L3 Career
19 cm wavelength	24 cm wavelength	Data not available
1575.42 M Hz	1227.6 M Hz	
C/A Code	P Code	
Navigation	Navigation Message	

- *P Code :* Reserved for direct use only by the military

- *C/A Code :* Used for rougher positioning
- For Single frequency use only L1 career is used
- For Double frequency, L1/L2/L3 career is used
- The navigation message (usually referred to as the ephemeris) tells us where the satellites are located, in a special coordinate system called WGS-84. If you know where the satellites are at any given time, then you can compute your location here on earth.

1-4.6 Different types of answers given by a GPS

Autonomous Positions
Uses…….. C/A code only
Requires….. Only one receiver
Data from at least four satellites
Provides…… An accuracy range of about 15 - 100 meters

This solution is designed for people who just need an approximate location on the earth, such as a boat at sea or a hiker in the mountains.

Real-Time Differential GPS (DGPS) Positions
Uses…….. C/A code only
Requires….. Two receivers
A radio link between the two receivers
- Reference receiver at a known location broadcasts RTCM (Radio Technical Commission for Maritime Services) corrections.
- Rover receiver applies corrections for improved GPS positions

Data from at least four satellites - the same four at both the references and rover (common satellites)

Provides…… An accuracy range of about 0.5 - 5 meters depending upon the quality of receiver and antennae used.

This solution gives much better results because here we have a known position at a reference receiver. However it must have a radio link between the reference receiver and the roving (moving) receiver.

Real Time Kinematic (RTK) Float Positions
Uses…….. C/A code and career waves.
Requires….. Two receivers
- Reference receiver at a known location tracks satellites and then broadcasts this satellite data over a radio link in a format called CMR. (CMR is a Trimble - defined format)
- Rover receiver receives data from both the satellites and the reference station.

A radio link between the two receivers.
Data from at least four common satellites.

Provides…… An accuracy range of about 20 cm to 1 meters.
This solution uses more of the satellite signal than the autonomous or DGPS solution. The CMR data is carrier phase data. The float solution is actually an intermediate step towards the most precise answer, which we'll discuss next.

Real Time Kinematic (RTK) Fixed Solutions
Uses…….. C/A code and career waves.

Requires….. Two receivers

- Reference receiver at a known location tracks satellites and then broadcasts CMR data over a radio link.
- Rover receiver receives data from both the satellites and the reference station.

A radio link between the two receivers.

Initialization, which is achieved most easily with dual-frequency receivers. Data from at least five common satellites to initialize on-the-fly (in motion) Tracking at least four common satellites after initializing.

Provides…… An accuracy range of about 1 - 5 cm.

We noticed that with each increasing level of precision, there are more requirements. The most important unique requirement for the RTK fixed solution is something called an initialization. Here it is not feasible to explain what's happening in an initialization, but it is relevant to mention that initialization is necessary to work at centi-meter level accuracy. Dual frequency receivers can perform this process automatically.

If the receiver looses the initialization - which can happen if it fails to track enough satellites - then your working accuracy will drop to the float solutions status temporarily. Remember, however both of these solutions require a radio link to your reference receiver. If, for any reason, you loose your radio link, you will drop back to the autonomous level - the least precise - until the radio link is regained.

1-4.7 Factors That Affect GPS

There are a number of potential error sources that affect either the GPS signal directly or your ability to produce optimal results:

- **Number of satellites - minimum number required:** You must track at least four common satellites - the same four satellites - at both the reference receiver and rover for either DGPS or RTK solutions. Also to achieve centimetre -level accuracy, remember you must have a fifth satellite for on-the fly RTK initialization. This extra satellite adds a check on the internal calculation. Any additional satellites beyond five provide even more checks, which is always useful.
- **Multipath - reflection of GPS signals near the antennae:** Multipath is simply reflection of signals similar to the phenomenon of ghosting on our television screen. GPS signals may be reflected by surfaces near the antennae, causing error in the travel time and therefore error in the GPS positions.
- **Ionosphere - change in the travel time of the signal:** Before GPS signals reach your antenna on the earth, they pass through a zone of charged particles called the ionosphere, which changes the speed of the signal. If your reference and rover receivers are relatively close together, the effect of ionosphere tends to be minimal. And if you are working with the lower range of GPS precisions, the ionosphere is not a major consideration. However if your rover is working too far from the reference station, you may experience problems, particularly with initializing your RTK fixed solution.

- **Troposphere - change in the travel time of the signal:** Troposphere is essentially the weather zone of our atmosphere, and droplets of water vapour in it can affect the speed of the signals. The vertical component of your GPS answer (your elevation) is particularly sensitive to the troposphere.
- **Satellite Geometry - general distribution of the satellites:** Satellite Geometry - or the distribution of satellites in the sky - effects the computation of your position. This is often referred to as Position Dilution of Precision (PDOP).
    - PDOP is expressed as a number, where lower numbers are preferable to higher numbers. The best results are obtained when PDOP is less than about 7.
    - PDOP is determined by your geographic location, the time of day you are working, and any site obstruction, which might block satellites. You can use planning software to help you determine when you'll have the most satellites in a particular area.
    - When satellites are spread out, PDOP is Low (good).
    - When satellites are closer together, PDOP is High (weak).

- **Satellite Health - Availability of Signal:** While the satellite system is robust and dependable, it is possible for the satellites to occasionally be unhealthy. A satellite broadcasts its health status, based on information from the U.S. Department of Defense. Your receivers have safeguards to protect against using data from unhealthy satellites.

- **Signal Strength - Quality of Signal:** The strength of the satellite signal depends on obstructions and the elevation of the satellites above the horizon. To the extent it is possible, obstructions between your GPS antennae and the sky should be avoided. Also watch out for satellites which are close to the horizon, because the signals are weaker.
- **Distance from the Reference Receiver:** The effective range of a rover from a reference station depends primarily on the type of accuracy you are trying to achieve. For the highest real time accuracy (RTK fixed), rovers should be within about 10-15 Km (about 6-9 miles) of the reference station. As the range exceeds this recommended limit, you may fail to initialize and be restricted to RTK float solutions (decimetre accuracy).
- **Radio Frequency (RF) Interference:** RF interference may sometimes be a problem both for your GPS reception and your radio system. Some sources of RF interference include:
    - Radio towers
    - Transmitters
    - Satellite dishes
    - Generators

  One should be particularly careful of sources, which transmit either near the GPS frequencies (1227 and 1575 MHz) or near harmonics (multiples) of these frequencies. One should also be aware of the RF generated by his own machines.

- **Loss of Radio Transmission from Base:** If, for any reason, there is an interruption in the radio link between a reference receiver and a rover, then your rover is left with an autonomous position. It is very important to set up a network of radios and repeaters, which can provide the uninterrupted radio link needed from Reference Station.

We have already discussed that there are different levels of accuracy in GPS positions. For any level, except autonomous (which can have a large amount of error in it); you must have a reference receiver, which is stationary, and a rover, which can be mobile or stationary.

The GPS reference station normally operates continuously, 24 hours a day. The coordinates of this station must be known before you can begin using GPS on any of your machines. First a proper site for the reference station is to be selected, and then a GPS survey is performed to obtain the known coordinates. This is usually done as part of the installation, either by the installation team or other qualified personnel.

Once it is installed, the GPS reference station can perform two functions simultaneously:

- Receive data from the satellites
- Broadcast GPS data to the rovers in the field.
- One reference station can support unlimited rovers. The primary constraint may be distance, because your accuracy may suffer if you're working too far from the reference station. This maximum distance will vary with your accuracy requirements and environment.

1-4.8 GPS Applications

One of the most significant and unique features of the Global Positioning Systems is the fact that the positioning signal is available to users in any position worldwide at any time. With a fully operational GPS system, it can be generated to a large community of likely to grow as there are multiple applications, ranging from surveying, mapping, and navigation to GIS data capture. The GPS will soon be a part of the overall utility of technology.

There are countless GPs applications; a few important ones are covered in the following passage.

A. Surveying and Mapping

The high precision of GPS carrier phase measurements, together with appropriate adjustment algorithms, provides an adequate tool for a variety of tasks for surveying and mapping. Using DGPs methods, accurate and timely mapping of almost anything can be carried out. The GPS is used to map cut blocks, road alignments, and environmental hazards such as landslides, forest fires, and oil spills. Applications, such as cadastral mapping, needing a high degree of accuracy also can be carried out using high grade GPS receivers. Continuous kinematic techniques can be used for topographic surveys and accurate linear mapping.

B. Navigation

Navigation using GPS can save countless hours in the field. Any feature, even if it is under water, can be located up to one hundred meters simply by scaling coordinates from a map, entering waypoints, and going directly to the site. Examples include road intersections, corner posts, accident sites, geological formations, and so on. GPS navigation in helicopters, in vehicles, or in a ship can provide an easy means of navigation with substantial savings.

C. Remote Sensing and GIS

It is also possible to integrate GPS positioning into remote-sensing methods such as photogrammetry and aerial scanning, magnetometry, and video technology. Using DGPS or kinematic techniques, depending upon the accuracy required, real time or post-

processing will provide positions for the sensor which can be projected to the ground, instead of having ground control projected to an image. GPS are becoming very effective tools for GIS data capture. The GIS user community benefits from the use of GPS for locational data capture in various GIS applications. The GPS can easily be linked to a laptop computer in the field, and, with appropriate software, users can also have all their data on a common base with every little distortion. Thus GPS can help in several aspects of construction of accurate and timely GIS databases.

D. **Geodesy**
Geodetic mapping and other control surveys can be carried out effectively using high-grade GPs equipment. Especially when helicopters were used or when the line of sight is not possible, GPS can set new standards of accuracy and productivity.

E. **Military**
The GPS was primarily developed for real time military positioning. Military applications include airborne, marine, and land navigation.

1-4.9 Future of GPS Technology

Barring significant new complications due to S/A (Selective Availability) from DOD, the GPS industry is likely to continue to develop in the civilian community. There are currently more than 50 manufacturers of GPs receivers, with the trend continuing to be towards smaller, less expensive, and more easily operated devices. While highly accurate, portable (hand-held) receivers are already available, current speculation envisions inexpensive and equally accurate 'wristwatch locators' and navigational guidance systems for automobiles. However, there is one future trend that will be very relevant to the GIS user community, namely, community base stations and regional receive networks, as GPS management and technological innovations that will make GPS surveying easier and more accurate.



Self-Assessment

4. What are the advantages and disadvantages of GPS surveying over the traditional surveying methods?
5. List and discuss four factors that affect GPS operation.
6. Write short notes on the components of GPS.

SESSION 2-4: COMPUTATION OF AREAS

The area of any surveyed figure can be calculated from either (1) the field notes or (2) the plotted plan.

2-4.1 Areas From Field Notes

Areas computed under this, may be straight-sided, irregular sided or a combination of both.

2-4.1.1 Straight sided areas

A. Areas from triangles

Plotted areas bounded by straight sides can be divided into a series of triangles, the areas of which can be computed by using one of the following formulae's.

i. Semi- perimeter method

The area contain within any one triangle ABC is found from the formulae

Area = √{s (s- a) (s-b) (s-c)}

Where s, is the semi- perimeter and a, b and c are the lengths of the sides of the triangle.

ii. **Area of triangle with height known**.

Area = ½ (base of triangle x height of triangle)

iii. **Area = ½ a b sine C**

Where C is the angle contained between side lengths a and b

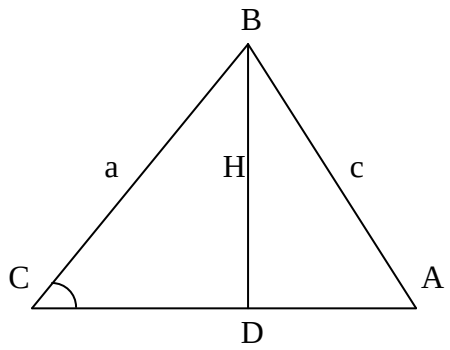


Figure 50: Triangle

Example1.

Compute the area of the triangle bounded by the sides a, b and c of length 3m, 4m and 5m respectively.

Solution

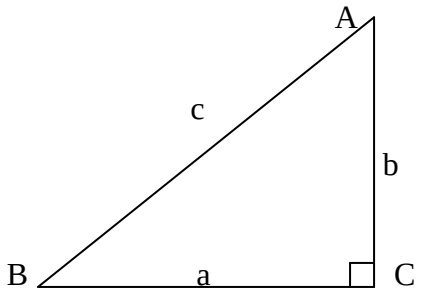


Figure 51: Right angle triangle

1. Area=½ (base x height) = ½ x4x 3= 6m²
2. Area = √{s (s- a) (s-b)(s-c)} = √6(6-3)(6-4)(6-5)= √36 =6m²
3. Area = ½ a b sine C =½ x3x4 sin 90° = 6m²

B. **Areas from trapezium**.

Area = ½ (a + b) h. Where 'a' and 'b' are parallel sides and 'h' is the perpendicular distance or height between the parallel sides.

Compute the area of a trapezium whose parallel sides are 4m and 6m and of height 5m.

Solution
Area = ½ x (4m+ 6m) x 5m = 25m².

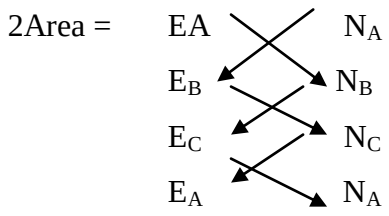
Areas from coordinates
There are two methods by which the area enclosed by a straight-sided figure is computed when the coordinates of the stations of the sides are known or given.

C. **Cross coordinate method**.
The figure 52 below is a triangular area of coordinates of stations.

Station	Easting (m)	Northing (m)
A	E_A	N_A
B	E_B	N_B
C	E_C	N_C

$$\text{Area} = \tfrac{1}{2}[N_A\{E_B-E_C\} + N_B\{E_C-E_A\} + N_C\{E_A-E_B\}]$$

This final result is easily remembered by writing the array in the following manner:



$$\text{Area} = \tfrac{1}{2}[\{E_AN_B+E_BN_C+E_CN_A\} - \{N_AE_B+N_BE_C+N_CE_A\}]$$

The following is the sequence of operations in the calculation of the area of any polygon:

i. Write the array of Eastings and Northings as shown above. **NOTE that E_A and N_A are repeated at the bottom of the columns.**
ii. Multiply the Easting of each station by the Northing of the preceding station and find the sum.
iii. Multiply the Easting of each station by the Northing of the following station and find the sum.
iv. Find the algebraic difference between (ii) and (iii) above.
v. Halve this figure to give the area of the polygon.

Example

Given that the coordinates of the traverse stations A, B, C and D are as follows:

Station	Easting (m)	Northing (m)
A	100.0	200.0
B	205.0	300.0
C	250.0	350.0
D	200.0	400.0

Calculate the area in hectares enclosed by the stations.

Solution

A = ½[{100.300} + {205.350} + {250.400} + {200.200}-{200.205}-{300.250}-{350.200} - {400.100}]

A = ½[15750] = 7,875m² = 0.7875Ha

2-4.2 Areas Enclosed By Irregular Sides

Areas could be obtained by (i) Graphical method (ii) Simpson's or trapezoidal rules (Mathematical) and (iii) Mechanical methods.

i. **Graphical method**: A piece of transparent graph paper is laid over the area, the squares are counted and the area calculated.

ii. **Mathematical methods:** The area is divided into a series of equidistant strips w, the ordinates (or offsets) y measured and the rules applied. There are two rules normally used.

2-4.2.1 Trapezoidal rule

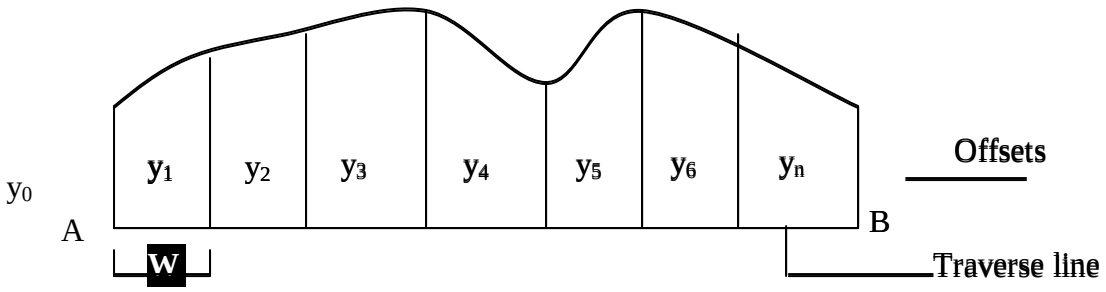


Figure 52: Irregular plots

Let y_0 and y_n be the end ordinates, y_1, y_2, y_3, etc., the intermediate ones, w the common distance between the ordinates and Y the sum of the intermediate ordinates. Then the area A by trapezoidal rule:

Area = strip width x [Average of 1st and last offsets + sum of others]

Area = $w\ [½(y_0+y_n) + y_1 + y_2 + y_3 + \ldots\ldots\ldots + y_{n-1}]$

Or

Area = $½w\ [y_0+y_n+ 2\{y_1 + y_2 + y_3 + \ldots\ldots\ldots + y_{n-1}\}]$

$A = ½w\ [y_0 + 2Y + y_n]$

Note:

i. If the first or last offset is zero, it must still be included in the equation.

ii. The number of offsets could be odd or even.

iii. The offsets must be at regular intervals.

This method is based on the assumption that the several offset figures are trapezoids and leads to results that are sufficiently accurate for the purposes. If, however, the boundaries are really curved to such an extent that appreciable error is likely to be introduced, the areas should be calculated by Simpson's rule.

2-4.2.2 Simpson's rule

In applying this rule, it is necessary to divide the area into an even number of strips of the same width w and an odd number of offsets as in figure 53.

$$\text{Area} = \tfrac{1}{3}w\,[y_0 + y_n + 4(y_2+y_4+\ldots) + 2(y_1+y_3+\ldots)]$$

Example 1

Calculate the area between the line RP and the irregular boundary line.

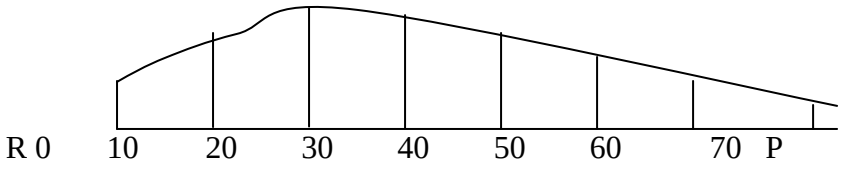


Figure 53: Example of irregular area

The offset distances from the station R to P are 3.0, 8.0, 10.0, 9.5, 9.2, 7.1, 4.5 and 2.5 respectively.

Solution

Since there is an even number of offsets between R and P at regular intervals of 10m, Trapezoidal rule could be used.

$A = 2x10[3+2.5 +2\{8+10+9.5+9.2+7.1+4.5\}]$

$A = 515.5m^2$

Example 2

The following offsets 10m apart were measured at right angles from a traverse line PQ to an irregular boundary. 4m, 4.5m, 5.1m, 6.5m, 6.3m, 5.1m and 4m respectively.

Solution

Using Simpson's rule:

$\text{Area} = \frac{1}{3} \times 10[4+4 + 2(5.1 +6.3) +4(4.5 +6.5 +5.1)]$

$= \underline{317.3m^2}$

2-4.3 Measurement of area by Planimeter

Measurement of areas by planimeter is also referred to as measurement of areas by mechanical integration. The Planimeter is thus a mechanical integrator used for the measurement of the areas (of all shapes) from plans. It consists essentially of

a) The pole block (which is fixed in position on the plan by a fine retaining needle).

b) The pole arm (which is pivoted about the pole block at one end, and carries the integrating unit at the other end).
c) The tracing arm, attached at one end to the integrating unit, and carrying at the other end the tracing point or optical tracer.
d) The measuring unit, consisting of a hardened steel integrating disc carried on pivots. A primary drum, divided into 100 parts, is directly connected to the disc spindle, and readings up to 1/1000th of a revolution of the integrating disc are obtained, either by estimation with reference to an index mark, or by vernier on an opposite drum. Another indicator gives the number of complete revolutions of the disc.

It is simple to use, and capable of a high degree of accuracy. To use it, place the pole block in a suitable position relative to the figure such that the tracing point can reach every part of the outline, (Figure 54). Put the tracing point on a known point on the outline and read both drums with reference to the index mark.

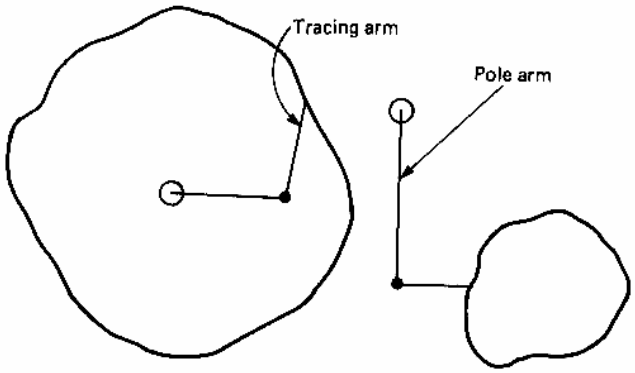


Figure 54: Area by planimeter

Note: Some models have a vernier, which should be read, in this case.

Carefully move the tracing point in a clockwise direction round the outline of the figure, back to the starting point. Take a second reading. The different between the two readings, multiplied by the scale factor, gives the required area. Repeat the operation until three consistent values are obtained. The mean of the three is then taken as the accepted value.

Note: the foregoing refers to conventional mechanical planimeters. There are now, on the market, Digital Planimeters which are more versatile, yet simple to use. Read about Digital Planimeters in Surveying by Bannister and Raymond.

2-4.4 Calculation of Volumes

Calculation of volumes of earth works, or capacities of reservoirs may be carried out by direct use of field measurements, or more generally, by measurement from plans.

Methods of calculation based on the field measurements are not of a precise nature, as they involve assumptions as to the geometrical figures whose volumes are required. The general methods of calculation are based on measurement from plans, and are classified according to the form of the solid as defined by (a) cross-sections, (b) spot levels and (c) contours.

The unit employed for the quantities of earthworks is the cubic yard or cubic metre, while, for the estimation of the capacity of a reservoir, the results are expressed in millions of gallons, or litres, as the basic measurements are in Imperial, or S.I., units.

2-4.4.1 Calculation from Cross-Sections

Calculation of volumes of earthworks from cross-sections is capable of general application, but it mostly used on long narrow works, such as roads, railways, canals, pipe excavations and others. The total volume is divided into a series of solids by planes of cross-sections, usually at intervals of 100ft or 25m, taken at right angles to some convenient line which runs longitudinally through the earthworks.

Note: Cross-sections are also taken at points of change from excavation to embankment if these are known and at places where there is a marked change of slope, either longitudinally or transversely.

The volumes of earthworks between successive cross sections are calculated from a consideration of the cross-sectional areas obtained by the general methods of area calculation discussed above, namely, by geometry, counting of squares or by measurement with a planimeter.

Calculation of cross-sectional areas by geometry may be applied to the following types of cross sections: (a) sections level across, (b) section with a crossfall, (c) sections part in cut and part in fill, and (d) sections of variable levels. As an illustration for the method of calculation by geometry, let us consider sections level across (Figure 55).

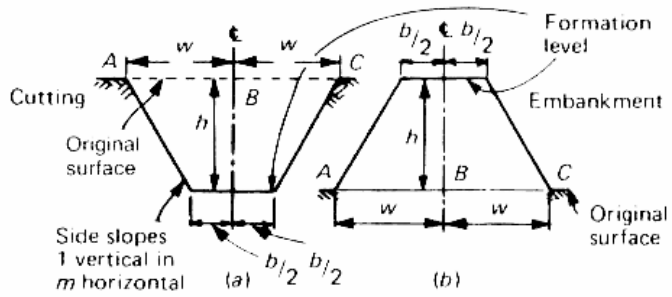


Figure 55: Volume calculation

Let

b = the constant formation or "sub-grade" breadth.
h = depth at the centre line (or, rise, in case of embankment)
w = side width
m = horizontal to 1 vertical = side slope.

The sloping side has to fall (or rise) h units from the original level to the final level, in such a way that m units of horizontal distance corresponds to a unit of vertical fall (or rise).

Therefore, for the fall (or rise) of h units, the corresponding horizontal distance is mh units. Hence, Side-width,

$$W = AB = BC = (\frac{B}{2} + mh)$$

and area of cross-section $= {}^1/_2$ (Sum of parallel sides) x h
$= {}^1/_2 \{2 ({}^b/_2 + mh) + b\}$ x h
$= h (b + mh)$

Having obtained the cross-sectional areas, the volume of earthworks may be computed by one of the following methods: (i) 'mean areas', (ii) 'end areas', and (iii) prismoidal formula.

2-4.4.2 Volume by 'Mean Areas'

Let the cross-sectional areas be A_1, A_2, $A_3 ... A_{n-1}$, A_n, and let the distance between the two extreme sections A_1 and A_n be L. Then the volume V of earthworks is given by $V = {}^1/n (A_1 + A_2 + A_3 {}^{+...+} A_{n-1} + A_n) L$, which is not too accurate.

A. **Volume by 'End Areas'**

The volume of solid between two consecutive cross-sections of sectional areas A_1 and A_2, D distance apart is given by

$$V = \frac{1}{2}(A_1 + A_2) \times D$$

Thus, for a series of cross-sections taken at regular intervals, the total volume

$$\Sigma V = \frac{1}{2}D(A_1 + A_2) + \frac{1}{2}D(A_2 + A_3) + \frac{1}{2}D(A_3 + A_4) + ... + \frac{1}{2}D(A_{n-2} + A_{n-1}) + \frac{1}{2}D(A_{n-1} + A_n)$$

$$= D[\frac{1}{2}(A_1 + A_n) + A_2 + A_3 + ... + A_{n-1]}$$

This formula is sometimes referred to as the trapezoidal rule for volumes. It is correct so long as the area of the section midway between, say, the first two cross-sections is actually the mean, i.e. $^1/_2 (A_1 + A_2)$, and this can be assumed so long as there is no wide variation between successive sections. If there are variations, then a correction must be applied, and this is a refinement taken care of by the prismoidal formula.

2-4.4.3 The Prismoidal Formula

The prismoid may be defined as a solid having two end faces which are parallel to each other, and made up of polygons not necessarily of the same number of sides, with the longitudinal faces, i.e. the sides, top and bottom, being formed by straight, continuous lines running between the end planes.

The volume V, of a prismoid is given by

where:

$$V = \frac{1}{6}D(A_1 + 4M + A_2)$$

A_1 and A_2 are the areas of the two end faces, distance D apart, and
M is the cross-sectional area midway between the end faces.

The theory assumes that in calculating M, each linear dimension is the average of the corresponding dimensions of the two end areas. It is wrong to simply assume that M is the mean of A_1 and A_2, since the formula then reduces to that of the 'end areas', discussed above.

There are many ways to getting a good estimate for the value of M for use in the prismoidal formula, and we shall look at a few of them:

i. We may treat each cross-section as the end area of a prismoid of length D, and estimate the dimensions of the mid-area at the $^{D}/_{2}$ points as the mean of the corresponding dimension of the two end areas. This approach is difficult when the sections are irregular.

ii. For irregular sections where the estimation of the mid-area is difficult, we can arrange for additional sections to be levelled at the mid-area points, as required. This approach increases the amount of field work and, hence, cost.

iii. We can also treat alternate sections as the end areas. In this case, the first and third sections become end areas of a prismoid 2D long, with the second section as the mid-area. In this way, we treat the even-numbered sections and the mid-areas.

Thus, with our usual notation,

$$\text{Volume of the first prismoid} = \frac{2D}{6}\left(A_1 + 4A_2 + A_3\right)$$

$$\text{Volume of the second prismoid} = \frac{2D}{6}\left(A_3 + 4A_4 + A_5\right)$$

and so on

Summing up, we get the total volume,

$$V = \frac{1}{3}D(A_{1+} 4A_2 + 2A_3 + 4A_4 + ... + 2A_{n+2} + 4_{n-1} + A_n$$

In words, the total volume is the product of one third the common distance apart and the sum of the first, and last end areas, four times each even-numbered cross-sectional area, and two times each odd-numbered cross-sectional area. (The formula requires that n, the total number of cross-sections, should be odd).

By analogy to Section 4.2.2.1 (B1) ii, we can say that the above formula is Simpson's Rule for volumes, which can be applied with greater accuracy where, for reasons of sections being irregular, mid-sections at half-way points are actually levelled. In this case each section is an end area and the prismoid considered is of length D; hence the "D" in the above formula should be halved.

We can also do away with the necessity of calculating the mid-section areas altogether by using the 'end areas' formula, and applying, for refinement, a correction known as the prismoidal correction. An example of the derivation of the prismoidal correction is given below:

Consider sections level across. If D is the distance between the two end faces of areas A_1 and A_2, then with our usual notation,

$A_1 = h_1(b + mh_1)$

$A_2 = h_2(b + mh_2)$

Hence, by the 'End Areas' formula, volume VEA of the solid

$$= \frac{1}{2}D(bh_1 + mh_1^{\ 2} + bh_2 + mh_2^{\ 2})$$

Now, assuming $\mathrm{hm} = \frac{1}{2}(\mathrm{h}_1 + \mathrm{h}_2)$, the difference in level at the mid area, between ground level and the formation level,

$$\mathrm{M} = \frac{(\mathrm{h}_1 + \mathrm{h}_2)}{2}\,\frac{(\mathrm{b} + \mathrm{mh}_1 + \mathrm{h}_2)}{2}$$

$$= \frac{\mathrm{b}}{2}(\mathrm{h}_1 + \mathrm{h}_2) + \frac{\mathrm{m}}{4}(\mathrm{h}_1 + \mathrm{h}_2)^2$$

Hence by the Prismoidal formula, volume $\mathrm{V_{PF}}$ of the solid

$$= \frac{1}{6}\mathrm{D}[\mathrm{bh}_1 + \mathrm{mh}_1{}^2 + \frac{4\mathrm{b}}{2(\mathrm{h}_1 + \mathrm{h}_2)} + \frac{\mathrm{m}}{4}(\mathrm{h}_1 + \mathrm{h}_2)^2 + \mathrm{bh}_2 + \mathrm{mh}_2{}^2]$$

$$= \frac{1}{6}D[bh_1 + mh_1{}^2 + 2bh_1 + 2bh_2 + mh_1{}^2 + 2mh_1h_2 + mh_2{}^2 + bh_2 + mh_2 + mh_2{}^2]$$

$$= \frac{1}{6}D[3bh_1 + 2mh_1{}^2 3bh_2 2mh_2{}^2 + 2mh_1h_2]$$

$$= \frac{1}{2}[bh_1 + \frac{2}{3}mh_1{}^2 + bh_2 + \frac{2}{3}mh_2{}^2 + \frac{2}{3}mh_1h_2]$$

Hence,

$$\mathrm{V_{EA}} - \mathrm{B_{PF}} = \frac{\mathrm{Dm}}{6}(\mathrm{h}_1 + \mathrm{h}_2)^2$$

which is the prismoidal correction for sections level across, which must always be deducted from the volume calculated with the 'end area' formula.

NOTE: Corrections can be similarly derived for the other types of cross-sections, starting from the general expression

$$V_{EA} - V_{PF} = \frac{1}{2}D(A_1 = A_2) - \frac{1}{6}D(A_1 + 4M + A_2)$$

$$= \frac{1}{6}D(3A_1 - 3A_2 - A_1 - 4M - A_2)$$

$$= \frac{1}{6}D(2A_1 - 4M + 2A_2)$$

$$= \frac{1}{3}D(A_1 - 2M + A_2)$$

2-4.4.4 Calculation from Contours

Calculation of volumes from contours is similar to that from cross-sections: Simply replace the areas of the end faces with the areas enclosed within each contour, and the distance D between the end faces with the contour interval C; and all the formula derived above apply; but first let us consider how to obtain the areas within each contour, as well as the contours themselves.

The area within each contour is best measured by a planimeter, and for best volume calculation results, the contour interval should be small so that the land can generally be assumed to slope uniformly from one contour to the other. Contour intervals of 1m or 2m, or from 2 to 5ft are usually employed. In principle, the method is ideal, but it is of limited application owing to the high cost of accurately contouring large areas. However where accurate contour plans are available, the method may be conveniently used to estimate, say the volume of water up to a certain level in a reservoir.

A. Simpson's rule

a. 'End Areas'

Let the areas enclosed within successive contours be A_1 and A_2 and let the contour interval be C. Then the volume of water, gravel, tipped material, etc between two contour lines is given by

$$V = \frac{1}{2}(A_1 + A_2) \times C$$

b. Primoidal Formula

Here the areas enclosed within alternate contours are regarded as mid-areas. Hence suppose the areas enclosed within three successive contours are A_1, A_2 and A_3, respectively, then the volume V enclosed within these contours is given by

$$V = \frac{1}{3}C(A_1 + 4A_2 + A_3)$$

Thus for the total volume of water contained in a reservoir, Simpson's rule for volumes apply.

NOTE: Where the contour interval is not small enough, it may be necessary to draw form lines by interpolation, which then should be used for the mid-areas.

2-4.4.5 Calculation from Spot levels

This method of volume calculation is especially useful in the determination of volumes of large open excavations for tanks, basements, borrow pits etc, and for ground levelling operations such as playing fields and building sites. It can also be applied to the determination of volumes of spoil heaps.

The field work consists in taking spot levels at points along straight lines so arranged that the points form a grid, square or rectangular, covering the area (see 1-2.3.2). The grid size depends upon the nature of the ground, but the corners should be sufficiently close for the ground surface between the lines to be considered as planes.

To calculate volume from spot levels, we first obtain the difference of each existing ground level, and the corresponding formation level. This gives us a series of heights from which the volume within each square, or rectangle, can be taken as the product of the plan area and the average height of the depth of excavation, or fill, at the four corners **(Figure 56).**

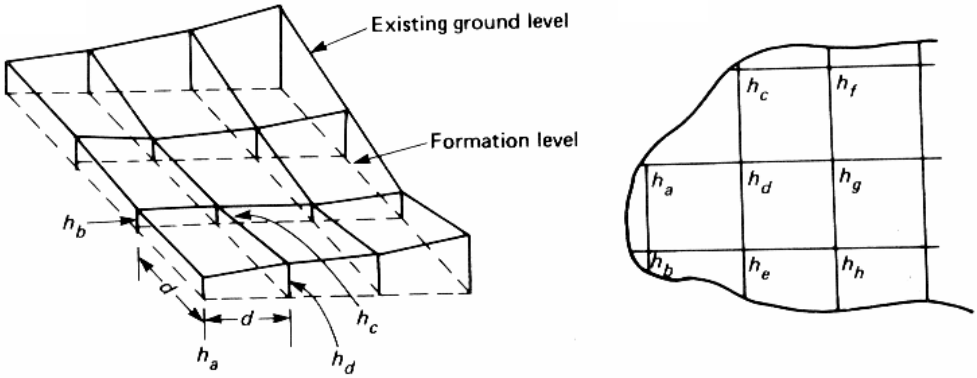


Figure 56: Spot levels

The volume of cut under the first square

$$d^2 \times \frac{1}{4}(h_a + h_b + h_c + h_d)$$

If the volumes within the other squares are similarly computed and added up, we shall find that one of the heights are used once, others twice, thrice or four times, and we can write an expression for the total volume V, as follows:

$$V = \frac{1}{4} d^2 (\Sigma h_1 + 2\Sigma h_2 + 3\Sigma h_3 + 4\Sigma h_4)$$

where h_n is the height of a corner used n times. (The formula does not cover cases where the base area is not a square or a rectangle, and you should calculate the volumes of irregular shapes separately).



Self-Assessment

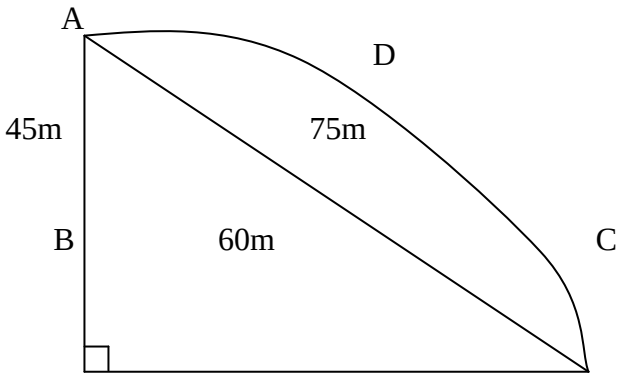
1. The coordinates listed below refer to a closed traverse PQRS

Station	Easting (m)	Northing (m)
P	+35.2	+46.1
Q	+162.9	+151.0
R	+14.9	+218.6
S	-69.2	-25.2

   Calculate the area in hectares enclosed by the stations.
2. The following offsets 10m apart were measured at right angles from a

traverse line to an irregular boundary
0m, 2.3m, 5.5m, 7.9m, 8.6m, 6.9, 7.3m, 6.2, 3.1m and 0m. Calculate the area between the traverse line and the irregular boundary.

3. a). Explain the difference between the Trapezoidal rule and Simpson's rule for determining areas of irregular parcels of land.
   b). The figure below represents a parcel of land consisting of a right-angled triangle ABC (with dimensions shown) and an irregular portion ADC.



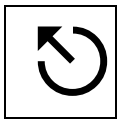
Offsets taken from AC to the irregular boundary were as follows:

Chainage(m)	0	15	30	45	60	75
Offset (m)	0	3.5	4.5	4.0	3.5	0

Compute the total area of the parcel in hectares.

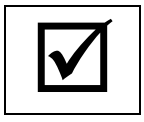
Learning Track Activities

Summary



Unit Summary

- Concept of GPS data capturing has been discussed
- Factors that affect the accuracy of GPS data have been analysed.
- Different GPS positioning systems have been discussed
- Calculation of areas from field notes, plans and coordinates has been carried out.
- Accuracies of different methods of area computation has been studied



Key terms

a. Control segment
b. Space segment
c. User segment
d. Absolute positioning
e. Differential positioning
f. Position Dilution of precision (PDOP)
g. Multipath
h. Receivers
i. WGS 84

Review Question

2.



Unit Assignment – 4

1. http:// www.garmin.com/aboutGPS
    a. go to the website above and find out more about GPS
    b. Kavanagh, B. F and Bird G. S. J. (1996) "Surveying: Principles and Applications"
    c. Irvine,, W. (1980). Surveying for Construction. Second Edition. McGraw-Hill Book Company (UK) Limited.
    d. Uren, J. and Price, W. F. (1994) Surveying for Engineers. Third edition. Macmillan Press ltd., London, UK.